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GRADE 5 MATHS

ADVANCED

BOOK 1

Teacher's edition

For Grade 5 students

ThinkAlgo — From Math Thinking to Code

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1 Natural Numbers

① Arithmetic sequences

- An **arithmetic sequence** is a list of numbers in which, from the second term on, each term is greater (or smaller) than the term before it by the same amount, called the **common difference**.
- Examples: 3, 7, 11, 15, ... increases with difference 4; 90, 80, 70, ... decreases with difference 10.
- When the first term, the last term and the difference are known:

$$\text{Number of terms} = (\text{Last} - \text{First}) : \text{Difference} + 1,$$

$$\text{Sum} = (\text{First} + \text{Last}) \times \text{Number of terms} : 2.$$

- The n-th term of an increasing sequence is $\text{First} + (n-1) \times \text{Difference}$; of a decreasing one, $\text{First} - (n-1) \times \text{Difference}$.

② Counting digits – Forming numbers from given digits

- There are 9 one-digit numbers (1 to 9), 90 two-digit numbers (10 to 99) and 900 three-digit numbers (100 to 999).
- **Multiplication rule:** to form a number we choose a digit for each place in turn. If the first place has m choices, the second n choices, the third p choices, ... then there are $m \times n \times p \times \ldots$ numbers in all.
- The digit 0 cannot be the leading digit. If the digits must be *different*, a digit already used cannot be chosen again.
- To add up all the numbers formed, count *how many times each digit appears in each place.*

③ Place value: writing an extra digit onto a number

Writing numbers by place value links the old number to the new one. For a two-digit number $\overline{ab}$:

- writing a digit c on the **right**: $\overline{abc} = \overline{ab} \times 10 + c$ (ten times the old number, plus c);

- writing a digit c on the **left**: $\overline{cab} = c \times 100 + \overline{ab}$;
- writing 0 **in the middle**: $\overline{a0b} = a \times 100 + b$.

Example 1

Consider the arithmetic sequence 3, 7, 11, 15, ..., 199.

a) How many terms does it have?

b) Find the sum of all its terms.

Guided solution.

The common difference is $7 - 3 = 4$.

a) Number of terms: $(199 - 3) : 4 + 1 = 50$.

b) Sum of the terms: $(3 + 199) \times 50 : 2 = 5\,050$.

Remember: pair the first term with the last, the second with the second-to-last,... every pair adds up to 202, and there are $50 : 2 = 25$ pairs.

Example 2

How many digits are needed to number the pages of a 234-page book, starting from page 1?

Guided solution.

Split the pages into three groups by the number of digits:

Pages 1 to 9: 9 pages, using $9 \times 1 = 9$ digits.

Pages 10 to 99: 90 pages, using $90 \times 2 = 180$ digits.

Pages 100 to 234: $234 - 100 + 1 = 135$ pages, using $135 \times 3 = 405$ digits.

Total: $9 + 180 + 405 = 594$ digits.

Example 3

Using the digits 0, 3, 5, 9, how many three-digit numbers can be formed? How many with three different digits?

Guided solution.

Let the number be $\overline{abc}$ ($a \neq 0$).

Digits may repeat: a has 3 choices $(3, 5, 9)$, b has 4, c has 4.

That gives $3 \times 4 \times 4 = 48$ numbers.

Different digits: a has 3 choices; b must differ from a, so 3 choices (now 0 is allowed); c must differ from both, so 2 choices.

That gives $3 \times 3 \times 2 = 18$ numbers.

Example 4

Find the sum of all three-digit numbers with different digits that can be formed from 2, 3, 5, 7.

Guided solution.

There are $4 \times 3 \times 2 = 24$ numbers.
The four digits play the same role, so in each place every digit appears $24 : 4 = 6$ times.
The digits in each place add up to $(2 + 3 + 5 + 7) \times 6 = 102$.
Hence the required sum is:

$$102 \times 100 + 102 \times 10 + 102 \times 1 = 102 \times 111 = 11\,322.$$

Example 5

Find a two-digit number such that writing the digit 5 on its right gives a new number that is 293 more than it.

Guided solution.

Let the number be $\overline{ab}$.
Writing 5 on the right gives $\overline{ab5} = \overline{ab} \times 10 + 5$.
By the problem: $\overline{ab} \times 10 + 5 = \overline{ab} + 293$.
Take $\overline{ab}$ from both sides: $\overline{ab} \times 9 + 5 = 293$, so $\overline{ab} \times 9 = 288$ and $\overline{ab} = 32$.
Check: $325 - 32 = 293$.
So the number is 32.

Practice

Problem 1

Consider the sequence 2, 5, 8, 11, 14, ...

a) State the rule and write the next three terms.

b) Find the 100th term.

c) Do 2 024 and 2 025 belong to the sequence? If so, which term is it?

Solution

a) Each term is 3 more than the one before.
Next three terms: 17, 20, 23.
b) The 100th term: $2 + (100 - 1) \times 3 = 299$.
c) Every term leaves remainder 2 when divided by 3.
Since $2\,024 : 3 = 674$ remainder 2, the number 2 024 is a term: the $(2\,024 - 2) : 3 + 1 = 675$th.
But 2 025 is divisible by 3, so it is not a term.

Problem 2

For the sequence $4,\ 9,\ 14,\ 19,\ \ldots,\ 499$, find the number of terms and their sum.

Solution

Difference: 5.
Number of terms: $(499 - 4) : 5 + 1 = 100$.
Sum: $(4 + 499) \times 100 : 2 = 25\,150$.
Answer: 100 terms; sum 25 150.

Problem 3

Find the sum of 80 consecutive natural numbers starting from 25.

Solution

The last number: $25 + (80 - 1) \times 1 = 104$.
Sum: $(25 + 104) \times 80 : 2 = 5\,160$.
Answer: 5 160.

Problem 4

Find the sum of the first 40 terms of the sequence $3,\ 7,\ 11,\ 15,\ \ldots$

Solution

The 40th term: $3 + 39 \times 4 = 159$.
Sum: $(3 + 159) \times 40 : 2 = 3\,240$.
Answer: 3 240.

Problem 5

Find the sum of all odd two-digit numbers.

Solution

The odd two-digit numbers form the sequence $11,\ 13,\ \ldots,\ 99$ (difference 2).
Number of terms: $(99 - 11) : 2 + 1 = 45$.
Sum: $(11 + 99) \times 45 : 2 = 2\,475$.
Answer: 2 475.

Problem 6

How many digits are needed to number the pages of a 208-page storybook, starting from page 1?

Solution

Pages 1–9: 9 digits; pages 10–99: $90 \times 2 = 180$ digits.
Pages 100–208: 109 pages, using $109 \times 3 = 327$ digits.
Total: $9 + 180 + 327 = 516$ digits.
Answer: 516 digits.

Problem 7

Exactly 435 digits were used to number the pages of a book, starting from page 1. How many pages does the book have?

Solution

The first 99 pages use $9 + 180 = 189$ digits.
Remaining digits: $435 - 189 = 246$, for three-digit pages: $246 : 3 = 82$ pages.
The book has $99 + 82 = 181$ pages.
Answer: 181 pages.

Problem 8

Writing the natural numbers one after another from 1 gives $A = 123456789101112\ldots$ What is the 150th digit of A, and which number does it belong to?

Solution

The numbers 1 to 9 take 9 digits; the other $150 - 9 = 141$ digits belong to two-digit numbers.
$141 : 2 = 70$ remainder 1: after the first 70 two-digit numbers (10 to 79) there is one more digit, the first digit of 80.
Answer: the digit 8, belonging to 80.

Problem 9

Using the digits 1, 4, 6, 9, how many natural numbers can be written that

a) have three digits? **b)** have three different digits? **c)** have three different digits and are even?

Solution

a) $4 \times 4 \times 4 = 64$. **b)** $4 \times 3 \times 2 = 24$.
c) The ones digit is 4 or 6: 2 choices; then the hundreds has 3 choices and the tens 2.
That gives $2 \times 3 \times 2 = 12$ numbers.

Problem 10

Using the digits 0, 2, 5, 7, how many three-digit numbers can be written? How many with three different digits?

Solution

The hundreds digit cannot be 0, so it has 3 choices.
Digits may repeat: $3 \times 4 \times 4 = 48$.
Different digits: $3 \times 3 \times 2 = 18$.
Answer: 48 and 18.

Problem 11

Using the digits 0, 1, 3, 6, 8, how many four-digit numbers with different digits can be written? How many of them are even?

Solution

All of them: the thousands digit has 4 choices (not 0), then 4, 3, 2 choices: $4 \times 4 \times 3 \times 2 = 96$.
Even numbers (ending in 0, 6 or 8):
– Ending in 0: $4 \times 3 \times 2 = 24$.
– Ending in 6 (or 8): the thousands digit has 3 choices (not 0, not the last digit), then 3 and 2: $3 \times 3 \times 2 = 18$.
Both cases give 36.
Even numbers: $24 + 36 = 60$.
Answer: 96 numbers, of which 60 are even.

Problem 12

Find the sum of all three-digit numbers with different digits formed from 1, 3, 5, 7.

Solution

There are $4 \times 3 \times 2 = 24$ numbers; in each place, each digit appears $24 : 4 = 6$ times.
Sum: $(1 + 3 + 5 + 7) \times 6 \times 111 = 16 \times 6 \times 111 = 10\,656$.
Answer: 10 656.

Problem 13

How many three-digit numbers contain exactly one digit 5?

Solution

– 5 in the hundreds: the tens and ones each have 9 choices (not 5): $9 \times 9 = 81$.
– 5 in the tens: the hundreds has 8 choices (not 0, not 5), the ones 9: $8 \times 9 = 72$.

– 5 in the ones: likewise $8 \times 9 = 72$.
Total: $81 + 72 + 72 = 225$.
Answer: 225 numbers.

Problem 14

How many four-digit numbers with different digits have 3 as the thousands digit?

Solution

The thousands digit is fixed as 3.
The hundreds has 9 choices (any digit except 3), the tens 8, the ones 7: $9 \times 8 \times 7 = 504$.
Answer: 504 numbers.

Problem 15

Find a two-digit number such that writing 4 on its right gives a number 571 more than it.

Solution

Let the number be $\overline{ab}$.
Then $\overline{ab} \times 10 + 4 = \overline{ab} + 571$, so $\overline{ab} \times 9 = 567$ and $\overline{ab} = 63$.
Check: $634 - 63 = 571$.
Answer: 63.

Problem 16

Find a three-digit number such that writing 9 on its right gives a number 1 350 more than it.

Solution

Let the number be x.
The new number is $x \times 10 + 9$, so $x \times 10 + 9 = x + 1\,350$, giving $x \times 9 = 1\,341$ and $x = 149$.
Check: $1\,499 - 149 = 1\,350$.
Answer: 149.

Problem 17

Find a three-digit number such that writing 4 on its left gives a number 17 times as large.

Solution

Let the number be x.
The new number is $4\,000 + x$, so $4\,000 + x = 17 \times x$, giving $16 \times x = 4\,000$ and $x = 250$.
Check: $4\,250 = 17 \times 250$.
Answer: 250.

Problem 18

Find a two-digit number such that writing 0 between its two digits gives a number 7 times as large.

Solution

Let the number be $\overline{ab}$.
Then $\overline{a0b} = 7 \times \overline{ab}$, i.e.
$a \times 100 + b = a \times 70 + b \times 7$.
Hence $a \times 30 = b \times 6$, i.e.
$b = a \times 5$.
As b is a digit, $a = 1$ and $b = 5$.
Check: $105 = 7 \times 15$.
Answer: 15.

Problem 19

Consider 5, 10, 15, ..., 500. How many terms are there? Find their sum.

Solution

Number of terms: $(500 - 5) : 5 + 1 = 100$.
Sum: $(5 + 500) \times 100 : 2 = 25\,250$.
Answer: 100 terms; 25 250.

Problem 20

Consider 1, 4, 7, 10, ... **a)** Find the 50th term. **b)** Is 2 026 a term? If so, which one?

Solution

a) The 50th term: $1 + (50 - 1) \times 3 = 148$.
b) Every term leaves remainder 1 when divided by 3; $2\,026 = 3 \times 675 + 1$, so it is a term.
It is term number $(2\,026 - 1) : 3 + 1 = 676$.
Answer: a) 148;
b) yes, term 676.

Problem 21

Find the sum of the even numbers from 2 to 200.

Solution

Number of terms: $(200 - 2) : 2 + 1 = 100$.
Sum: $(2 + 200) \times 100 : 2 = 10\,100$.

Answer: 10 100.

Problem 22

Find the sum of all three-digit multiples of 10.

Solution

The list $100,\ 110,\ \ldots,\ 990$ has $(990 - 100) : 10 + 1 = 90$ terms.

Sum: $(100 + 990) \times 90 : 2 = 49\,050$.

Answer: 49 050.

Problem 23

How many two-digit numbers are multiples of 3?

Solution

They are $12,\ 15,\ \ldots,\ 99$.

Count: $(99 - 12) : 3 + 1 = 30$.

Answer: 30 numbers.

Problem 24

How many digits are needed to number the pages of a 120-page book, starting from page 1?

Solution

Pages $1 - 9$: 9 digits.

Pages $10 - 99$: $90 \times 2 = 180$ digits.

Pages $100 - 120$: $21 \times 3 = 63$ digits.

In all: $9 + 180 + 63 = 252$ digits.

Answer: 252 digits.

Problem 25

A street has 75 houses numbered 1 to 75. How many digits are used to number them?

Solution

One-digit numbers: 9, using 9 digits.

Two-digit numbers: $75 - 9 = 66$, using $66 \times 2 = 132$ digits.

In all: $9 + 132 = 141$ digits.

Answer: 141 digits.

Problem 26

Using the digits 2, 4, 6, write every three-digit number with different digits, then add them up.

Solution

The numbers: 246, 264, 426, 462, 624, 642 (6 numbers).

Each digit appears in each place exactly 2 times.

Sum: $(2 + 4 + 6) \times 2 \times 111 = 2\,664$.

Answer: 2 664.

Problem 27

Using 1, 2, 3, 4, how many two-digit numbers with different digits can be written?

Solution

Tens digit: 4 choices.

Units digit: 3 choices (different from the tens).

Count: $4 \times 3 = 12$.

Answer: 12 numbers.

Problem 28

Find a two-digit number that increases by 362 when the digit 2 is written on its right.

Solution

The new number is 10 times the old one plus 2.

So 9 times the old number is $362 - 2 = 360$.

The number is $360 : 9 = 40$.

Answer: 40.

Problem 29

Find a three-digit number such that writing 1 on its left gives a number 9 times as large.

Solution

Writing 1 on the left adds 1 000.

The new number is 9 times as large, so 1 000 is $9 - 1 = 8$ times the number.

The number is $1\,000 : 8 = 125$.

Answer: 125.

Problem 30

A whole number ends in 7. Crossing out that 7 makes it 1 789 smaller. Find the number.

Solution

Call the number after crossing out A; the original is $10 \times A + 7$.
The difference is $9 \times A + 7 = 1\,789$, so $9 \times A = 1\,782$ and $A = 198$.
The number is 1 987.
Answer: 1 987.

Challenge – Extension

Problem 31 *

Using 1, 2, 5, 8, 9, how many four-digit numbers with different digits have the digits 2 and 5 next to each other?

Solution

Treat 2 and 5 as one "block", written (25) or (52): 2 ways.
Choose 2 more digits from 1, 8, 9: 3 ways (leave one out).
Arrange the block and the two digits (three "items"): $3 \times 2 \times 1 = 6$ ways.
Total: $2 \times 3 \times 6 = 36$.
Answer: 36 numbers.

Problem 32 *

A three-digit number has hundreds digit 7. Moving this 7 to the far right gives a number 324 less than the original. Find the number.

Solution

Let the number be $\overline{7bc} = 700 + \overline{bc}$.
The new number is $\overline{bc7} = \overline{bc} \times 10 + 7$.
So $700 + \overline{bc} - (\overline{bc} \times 10 + 7) = 324$, i.e.
$693 - \overline{bc} \times 9 = 324$.
Hence $\overline{bc} \times 9 = 369$ and $\overline{bc} = 41$.
The number is 741 (*check:* $741 - 417 = 324$).
Answer: 741.

Problem 33 *

Find a two-digit number such that writing 12 between its two digits gives a number 76 times as large.

Solution

Let the number be $\overline{ab}$.
The new number is $\overline{a12b} = a \times 1\,000 + 120 + b$.
So $a \times 1\,000 + 120 + b = 76 \times (a \times 10 + b) = a \times 760 + b \times 76$.
Hence $a \times 240 + 120 = b \times 75$; dividing by 15: $a \times 16 + 8 = b \times 5$.
The right side is at most 45, so $a \leq 2$; $a = 1$ gives 24, not a multiple of 5; $a = 2$ gives $40 = b \times 5$, $b = 8$.
Check: $2\,128 = 76 \times 28$.
Answer: 28.

Problem 34 *

Writing the even numbers one after another from 2 gives 2468101214... What is the 200th digit?

Solution

The one-digit evens $(2, 4, 6, 8)$ use 4 digits.
The two-digit evens (10 to 98) are 45 numbers, using 90 digits.
So far: 94 digits.
Remaining: $200 - 94 = 106$ digits; $106 : 3 = 35$ remainder 1.
After 35 three-digit evens (100 to 168) comes the first digit of 170.
Answer: the digit 1 (of 170).

Problem 35 *

An increasing arithmetic sequence has 30 terms, first term 9, and sum 2 445. Find its common difference and its last term.

Solution

From the sum formula: $(9 + \text{last}) \times 30 : 2 = 2\,445$, so $9 + \text{last} = 2\,445 \times 2 : 30 = 163$ and the last term is 154.
From the first to the last term there are 29 steps: $(154 - 9) : 29 = 5$.
Answer: difference 5, last term 154.

Problem 36 **

Find the largest natural number with all digits different whose digit sum is 20.

Solution

More digits means a larger number.
Seven different digits add up to at least $0 + 1 + \cdots + 6 = 21 > 20$, so the number has at most 6 digits.

To be largest, the leading digit should be as large as possible: take 9.
The other five different digits must add up to 11; the smallest five add up to $0+1+2+3+4=10$, so they can only be $0, 1, 2, 3, 5$.
Arrange the digits in decreasing order: 953 210.
Answer: 953 210.

Problem 37 *

An writes the natural numbers from 1 to 100 on the board. How many times does An write the digit 1?

Solution

– In the ones place: 1, 11, 21, ..., 91: 10 times.
– In the tens place: 10, 11, ..., 19: 10 times.
– In the hundreds place: the number 100: once.
Total: $10 + 10 + 1 = 21$ times.
Answer: 21 times.

Problem 38 *

Find the whole number n such that $1 + 2 + 3 + \cdots + n = 210$.

Solution

The sum is $(1 + n) \times n : 2$, so $n \times (n + 1) = 420$.
Two consecutive numbers multiply to $420 = 20 \times 21$.
So $n = 20$.
Answer: $n = 20$.

Problem 39 *

In 1, 3, 6, 10, 15, ... each term is the previous one plus its position number. Find the 20th term.

Solution

The nth term is $1 + 2 + \cdots + n$.
The 20th term: $1 + 2 + \cdots + 20 = (1 + 20) \times 20 : 2 = 210$.
Answer: 210.

Problem 40 *

Find $S = 1 + 2 + 4 + 8 + 16 + \cdots + 512$ (each term is double the one before).

Solution

$S \times 2 = 2 + 4 + 8 + \cdots + 512 + 1\,024$.
Subtracting S from $S \times 2$, the shared terms cancel: $S = 1\,024 - 1 = 1\,023$.
Answer: 1 023.

Problem 41 *

Writing the numbers 1 to 50 one after another gives a long number. **a)** How many digits does it have? **b)** What is its 50th digit from the left?

Solution

a) $9 + 41 \times 2 = 91$ digits.
b) The first 9 digits are 1 – 9; the other $50 - 9 = 41$ digits come from two-digit numbers.
$41 = 20 \times 2 + 1$, so 20 numbers (10 – 29) are complete and we reach the first digit of 30.
The 50th digit is 3.
Answer: a) 91;
b) 3.

Problem 42 *

How many three-digit numbers have digits adding up to 5?

Solution

Go by the hundreds digit:
Hundreds digit 1: the other two add to 4, giving 5 numbers (104, 113, 122, 131, 140).
Hundreds digit 2, 3, 4, 5: 4, 3, 2, 1 numbers.
In all: $5 + 4 + 3 + 2 + 1 = 15$.
Answer: 15 numbers.

Problem 43 *

How many three-digit numbers are multiples of 5?

Solution

They are 100, 105, ..., 995.
Count: $(995 - 100) : 5 + 1 = 180$.
Answer: 180 numbers.

Problem 44 *

Using 0, 1, 2, 3, 4, how many even three-digit numbers with different digits can be written?

Solution

Units digit 0*:* 4 choices for hundreds, 3 for tens: 12 numbers.

Units digit 2 *or* 4*:* 3 choices for hundreds (not 0, not the units digit), 3 for tens: $2 \times 3 \times 3 = 18$.

In all: $12 + 18 = 30$.

Answer: 30 numbers.

Problem 45 *

Find every two-digit number that is 4 times the sum of its digits.

Solution

$\overline{ab} = a \times 10 + b$ and $4 \times (a + b) = a \times 4 + b \times 4$.

They are equal when $a \times 6 = b \times 3$, i.e.

$b = a \times 2$.

The numbers: 12, 24, 36, 48.

Answer: 12; 24; 36; 48.

Problem 46 *

A two-digit number has digits adding up to 10. Swapping its digits gives a number 36 larger. Find it.

Solution

Swapping changes the number by 9 times the difference of the digits.

The difference is $36 : 9 = 4$, with the larger digit in the units.

Tens digit: $(10 - 4) : 2 = 3$; units digit: 7.

Answer: 37.

Problem 47 *

1 092 digits are used to number the pages of a book from page 1. How many pages are there?

Solution

Pages 1 – 99 use $9 + 180 = 189$ digits.

The remaining $1\,092 - 189 = 903$ digits number $903 : 3 = 301$ three-digit pages.

Pages: $99 + 301 = 400$.

Answer: 400 pages.

Problem 48 **

Writing the numbers 1 to 200, how many times is the digit 2 written?

Solution

Units place: 2, 12, ..., 192, 20 times.
Tens place: 20 – 29 and 120 – 129, 20 times.
Hundreds place: only 200, once.
In all: $20 + 20 + 1 = 41$ times.
Answer: 41 times.

Problem 49 ★★

Find the smallest whole number whose digits add up to 30.

Solution

To be smallest it needs as few digits as possible; each digit is at most 9, and $9 \times 3 = 27 < 30$, so it needs at least 4 digits.
The leading digit should be as small as possible with three 9s after it: $30 - 27 = 3$.
The number is 3 999.
Answer: 3 999.

Problem 50 ★★

Five consecutive whole numbers add up to 2 025. Find them.

Solution

The middle number is the average: $2\,025 : 5 = 405$.
The numbers: 403, 404, 405, 406, 407.
Answer: 403; 404; 405; 406; 407.

2 Fractions and Mixed Numbers

① Fractions – the basic property

- A fraction has the form $\frac{a}{b}$, where $a,\ b$ are natural numbers and $b \neq 0$; a is the **numerator**, b the **denominator**. The quotient $a : b$ is the value of the fraction $\frac{a}{b}$.
- **Basic property:** multiplying (or dividing) the numerator and the denominator by the same non-zero number gives an equal fraction:

$$\frac{a}{b} = \frac{a \times n}{b \times n} = \frac{a : m}{b : m} \qquad (n,\ m \neq 0).$$

- This lets us **simplify** a fraction (to lowest terms) and bring fractions to a **common denominator**.
- Two fractions are equal, $\frac{a}{b} = \frac{c}{d}$, exactly when $a \times d = b \times c$ (cross-multiplying).

② Mixed numbers

When the numerator of $\frac{a}{b}$ is larger than the denominator, divide a by b to get a quotient q and a remainder r $(0 \leq r < b)$. Then

$$\frac{a}{b} = q\frac{r}{b} \quad \text{(read “}q \text{ and } \tfrac{r}{b}\text{”)}, \qquad \text{for example } \frac{17}{5} = 3\frac{2}{5}.$$

$q\frac{r}{b}$ is a **mixed number**: a *whole part* q and a *fraction part* $\frac{r}{b}$ (less than 1). Conversely, $q\frac{r}{b} = \frac{q \times b + r}{b}$.

③ Properties of the operations – clever calculation

For fractions (or mixed numbers) $A,\ B,\ C$:

Commutative: $A + B = B + A$; $A \times B = B \times A$

Associative: $(A + B) + C = A + (B + C)$

Subtracting a sum: $A - (B + C) = A - B - C$

Distributive: $A \times (B \pm C) = A \times B \pm A \times C$

Subtracting a difference: $A - (B - C) = A - B + C$

Dividing by a product: $A : (B \times C) = A : B : C$

To calculate quickly, *group terms whose sum (or difference) is a whole number*, or *take out a common factor*.

④ Patterned sums of fractions

- **Splitting a fraction into a difference:** for $b > a$,

$$\frac{b-a}{a \times b} = \frac{1}{a} - \frac{1}{b}, \qquad \text{for example} \quad \frac{1}{2 \times 3} = \frac{1}{2} - \frac{1}{3}; \quad \frac{3}{4 \times 7} = \frac{1}{4} - \frac{1}{7}.$$

  When such fractions are added, the middle terms *cancel out*, leaving only the first and the last.

- **Sums with a constant ratio** (each term is the previous one times the same number), e.g. $\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \ldots$: multiply the sum by that number and subtract the original sum.

Example 1

Calculate in the most convenient way:

$$A = \frac{3}{5} + 2\frac{1}{4} + \frac{2}{5} + \frac{3}{4} \qquad B = 4\frac{2}{7} - \left(1\frac{3}{8} + \frac{2}{7}\right) \qquad C = \frac{7}{4} \times 3\frac{1}{3} - \frac{7}{4} \times 1\frac{1}{3}$$

Guided solution.

A) Group terms with whole-number sums: $A = \left(\frac{3}{5} + \frac{2}{5}\right) + \left(2\frac{1}{4} + \frac{3}{4}\right) = 1 + 3 = 4.$

B) Subtracting a sum: $B = 4\frac{2}{7} - \frac{2}{7} - 1\frac{3}{8} = 4 - 1\frac{3}{8} = 2\frac{5}{8}.$

C) Take out the common factor: $C = \frac{7}{4} \times \left(3\frac{1}{3} - 1\frac{1}{3}\right) = \frac{7}{4} \times 2 = \frac{7}{2}.$

Example 2

Calculate:

$$A = \frac{1}{1\times 2} + \frac{1}{2\times 3} + \frac{1}{3\times 4} + \cdots + \frac{1}{9\times 10} \qquad B = \frac{1}{6} + \frac{1}{12} + \frac{1}{20} + \cdots + \frac{1}{110}$$

Guided solution.

A) Split each fraction: $\frac{1}{1\times 2} = \frac{1}{1} - \frac{1}{2}$, $\frac{1}{2\times 3} = \frac{1}{2} - \frac{1}{3}$, $\cdots$

$$A = 1 - \frac{1}{2} + \frac{1}{2} - \frac{1}{3} + \cdots + \frac{1}{9} - \frac{1}{10} = 1 - \frac{1}{10} = \frac{9}{10}.$$

B) Notice: $6 = 2\times 3$, $12 = 3\times 4$, $20 = 4\times 5, \ldots, 110 = 10\times 11$.

Hence

$$B = \frac{1}{2} - \frac{1}{3} + \frac{1}{3} - \frac{1}{4} + \cdots + \frac{1}{10} - \frac{1}{11} = \frac{1}{2} - \frac{1}{11} = \frac{9}{22}.$$

Example 3

Calculate $D = \frac{1}{3} + \frac{1}{9} + \frac{1}{27} + \frac{1}{81} + \frac{1}{243}$.

Guided solution.

Each term is the previous one times $\frac{1}{3}$.

Multiply D by 3:

$$3\times D = 1 + \frac{1}{3} + \frac{1}{9} + \frac{1}{27} + \frac{1}{81}.$$

Subtracting, the common terms cancel: $3\times D - D = 1 - \frac{1}{243}$, so $2\times D = \frac{242}{243}$.

Therefore $D = \frac{242}{243} : 2 = \frac{121}{243}$.

Example 4

Calculate the product $P = \left(1 - \frac{1}{2}\right) \times \left(1 - \frac{1}{3}\right) \times \left(1 - \frac{1}{4}\right) \times \cdots \times \left(1 - \frac{1}{20}\right)$.

Guided solution.

Work out each factor, then cancel diagonally (each numerator cancels the previous denominator):

$$P = \frac{1}{2} \times \frac{2}{3} \times \frac{3}{4} \times \cdots \times \frac{19}{20} = \frac{1\times 2\times 3\times \cdots \times 19}{2\times 3\times 4\times \cdots \times 20} = \frac{1}{20}.$$

Practice

Problem 1

Simplify to lowest terms:

$$\frac{18}{24};\quad \frac{45}{60};\quad \frac{56}{98};\quad \frac{125}{1\,000}.$$

Solution

Divide the numerator and the denominator by their greatest common factor:

$$\frac{18}{24}=\frac{18:6}{24:6}=\frac{3}{4};\quad \frac{45}{60}=\frac{45:15}{60:15}=\frac{3}{4};\quad \frac{56}{98}=\frac{56:14}{98:14}=\frac{4}{7};\quad \frac{125}{1\,000}=\frac{125:125}{1\,000:125}=\frac{1}{8}.$$

Problem 2

Fill in $<$, $>$, $=$ and explain:

a) $\frac{5}{8}\ \square\ \frac{7}{12}$ **b)** $\frac{17}{19}\ \square\ \frac{17}{23}$ **c)** $\frac{2\,024}{2\,025}\ \square\ \frac{2\,025}{2\,026}$

Solution

a) Common denominator: $\frac{5}{8}=\frac{15}{24}$, $\frac{7}{12}=\frac{14}{24}$.

Since $15>14$, $\frac{5}{8}>\frac{7}{12}$.

b) Same numerator: the one with the smaller denominator is larger, so $\frac{17}{19}>\frac{17}{23}$.

c) Compare the "gaps" to 1: $1-\frac{2\,024}{2\,025}=\frac{1}{2\,025}$ is larger than $1-\frac{2\,025}{2\,026}=\frac{1}{2\,026}$.

The larger gap belongs to the smaller fraction, so $\frac{2\,024}{2\,025}<\frac{2\,025}{2\,026}$.

Problem 3

Calculate:

$$A=\frac{7}{9}-\frac{4}{15}+\frac{1}{6} \qquad B=\frac{38}{15}-\left(\frac{2}{3}+\frac{8}{15}\right) \qquad C=\frac{16}{11}-\frac{5}{6}+\frac{6}{11}-\frac{1}{6}$$

Solution

A) Common denominator 90: $A=\frac{70}{90}-\frac{24}{90}+\frac{15}{90}=\frac{61}{90}$.

B) $B=\frac{38}{15}-\frac{8}{15}-\frac{2}{3}=2-\frac{2}{3}=\frac{4}{3}$.

C) $C=\left(\frac{16}{11}+\frac{6}{11}\right)-\left(\frac{5}{6}+\frac{1}{6}\right)=2-1=1.$

Problem 4

Calculate:

$A = 2\frac{3}{5} - 1\frac{1}{3} + 1\frac{7}{15}$ $B = 5\frac{1}{4} + 2\frac{5}{6} - 3\frac{1}{12}$ $C = 6\frac{2}{7} - \left(2\frac{1}{3} + 1\frac{2}{7}\right)$

Solution

A) $A = \frac{39}{15} - \frac{20}{15} + \frac{22}{15} = \frac{41}{15} = 2\frac{11}{15}.$

B) $B = \frac{63}{12} + \frac{34}{12} - \frac{37}{12} = \frac{60}{12} = 5.$

C) $C = \left(6\frac{2}{7} - 1\frac{2}{7}\right) - 2\frac{1}{3} = 5 - 2\frac{1}{3} = 2\frac{2}{3}.$

Problem 5

Calculate in the most convenient way:

$A = \frac{5}{7} \times \frac{3}{8} + \frac{5}{7} \times \frac{5}{8}$ $B = \frac{3}{4} \times \frac{12}{5} \times \frac{5}{9} \times \frac{4}{3}$

$C = \frac{7}{9} \times \frac{5}{6} - \frac{7}{9} \times \frac{1}{3}$ $D = 2\frac{1}{3} : 1\frac{1}{6}$

Solution

A) $A = \frac{5}{7} \times \left(\frac{3}{8} + \frac{5}{8}\right) = \frac{5}{7} \times 1 = \frac{5}{7}.$

B) $B = \left(\frac{3}{4} \times \frac{4}{3}\right) \times \left(\frac{12}{5} \times \frac{5}{9}\right) = 1 \times \frac{4}{3} = \frac{4}{3}.$

C) $C = \frac{7}{9} \times \left(\frac{5}{6} - \frac{1}{3}\right) = \frac{7}{9} \times \frac{1}{2} = \frac{7}{18}.$

D) $D = \frac{7}{3} : \frac{7}{6} = \frac{7}{3} \times \frac{6}{7} = 2.$

Problem 6

Find y:

a) $y \times \frac{3}{5} = \frac{9}{10}$ **b)** $y + \frac{5}{6} = 2\frac{1}{4}$ **c)** $\frac{4}{7} : y = \frac{2}{3}$

d) $\left(y - \frac{1}{2}\right) \times \frac{4}{5} = \frac{2}{5}$ **e)** $y \times 1\frac{1}{4} + y \times \frac{3}{4} = 6$ **g)** $y : \frac{2}{3} + y : \frac{1}{3} = 18$

Solution

a) $y = \frac{9}{10} : \frac{3}{5} = \frac{9}{10} \times \frac{5}{3} = \frac{3}{2}$. **b)** $y = \frac{9}{4} - \frac{5}{6} = \frac{27}{12} - \frac{10}{12} = \frac{17}{12}$.

c) $y = \frac{4}{7} : \frac{2}{3} = \frac{4}{7} \times \frac{3}{2} = \frac{6}{7}$. **d)** $y - \frac{1}{2} = \frac{2}{5} : \frac{4}{5} = \frac{1}{2}$, $y = 1$.

e) $y \times \left(1\frac{1}{4} + \frac{3}{4}\right) = 6$, i.e.

$y \times 2 = 6$, $y = 3$.

g) $y \times \frac{3}{2} + y \times 3 = 18$, i.e.

$y \times \frac{9}{2} = 18$, $y = 4$.

Problem 7

Find the natural number y:

a) $\frac{y-1}{4} = \frac{15}{20}$ **b)** $\frac{8}{12} = \frac{10}{y+3}$

c) $\frac{1}{4} < \frac{2}{y} < \frac{1}{2}$ **d)** $\frac{3}{7} < \frac{y}{14} < \frac{5}{7}$

Solution

a) $\frac{15}{20} = \frac{3}{4}$, so $y - 1 = 3$, $y = 4$. **b)** $\frac{8}{12} = \frac{2}{3} = \frac{10}{15}$, so $y + 3 = 15$, $y = 12$.

c) Use the same numerator 2: $\frac{2}{8} < \frac{2}{y} < \frac{2}{4}$, so $4 < y < 8$: $y \in \{5; 6; 7\}$.

d) Use the same denominator 14: $\frac{6}{14} < \frac{y}{14} < \frac{10}{14}$, so $y \in \{7; 8; 9\}$.

Problem 8

A rectangular cloth is $\frac{7}{4}$ m long, and its width is $\frac{5}{12}$ m less than its length. Find its perimeter and area.

Solution

Width: $\frac{7}{4} - \frac{5}{12} = \frac{21}{12} - \frac{5}{12} = \frac{4}{3}$ (m).

Perimeter: $\left(\frac{7}{4} + \frac{4}{3}\right) \times 2 = \frac{37}{12} \times 2 = \frac{37}{6}$ (m). Area: $\frac{7}{4} \times \frac{4}{3} = \frac{7}{3}$ (m^2).

Answer: perimeter $\frac{37}{6}$ m; area $\frac{7}{3}$ m^2.

Problem 9

A rectangle has area $\frac{5}{8}$ m^2 and length $\frac{5}{4}$ m. Find its perimeter.

Solution

Width: $\frac{5}{8} : \frac{5}{4} = \frac{5}{8} \times \frac{4}{5} = \frac{1}{2}$ (m). Perimeter: $\left(\frac{5}{4} + \frac{1}{2}\right) \times 2 = \frac{7}{2}$ (m).

Answer: $\frac{7}{2}$ m.

Problem 10

From a 15 m rope, 6 pieces of $1\frac{3}{4}$ m each are cut. The rest is cut into 3 equal pieces. How long is each of these pieces?

Solution

The 6 pieces: $1\frac{3}{4} \times 6 = \frac{7}{4} \times 6 = \frac{21}{2}$ (m).

Left: $15 - \frac{21}{2} = \frac{9}{2}$ (m).

Each later piece: $\frac{9}{2} : 3 = \frac{3}{2}$ (m).

Answer: $1\frac{1}{2}$ m.

Problem 11

Calculate:

$$A = \frac{1}{2 \times 3} + \frac{1}{3 \times 4} + \cdots + \frac{1}{11 \times 12} \qquad B = \frac{3}{2 \times 5} + \frac{3}{5 \times 8} + \cdots + \frac{3}{17 \times 20}$$

Solution

A) $A = \frac{1}{2} - \frac{1}{3} + \frac{1}{3} - \frac{1}{4} + \cdots + \frac{1}{11} - \frac{1}{12} = \frac{1}{2} - \frac{1}{12} = \frac{5}{12}$.

B) Each denominator is a product of two numbers that differ by 3, the numerator: $B = \frac{1}{2} - \frac{1}{5} + \frac{1}{5} - \frac{1}{8} + \cdots + \frac{1}{17} - \frac{1}{20} = \frac{1}{2} - \frac{1}{20} = \frac{9}{20}$.

Problem 12

Calculate:

$C = \dfrac{2}{3\times 5} + \dfrac{2}{5\times 7} + \cdots + \dfrac{2}{23\times 25}$ $\qquad$ $D = \dfrac{1}{2} + \dfrac{1}{4} + \dfrac{1}{8} + \cdots + \dfrac{1}{512}$

Solution

C) $C = \dfrac{1}{3} - \dfrac{1}{5} + \dfrac{1}{5} - \dfrac{1}{7} + \cdots + \dfrac{1}{23} - \dfrac{1}{25} = \dfrac{1}{3} - \dfrac{1}{25} = \dfrac{22}{75}$.

D) $2\times D = 1 + \dfrac{1}{2} + \dfrac{1}{4} + \cdots + \dfrac{1}{256}$;

$2\times D - D = 1 - \dfrac{1}{512}$, so $D = \dfrac{511}{512}$.

Problem 13

Find a fraction equal to $\dfrac{3}{5}$ whose denominator is 26 more than its numerator.

Solution

The fraction is $\dfrac{3\times k}{5\times k}$.

The denominator minus the numerator is $2\times k = 26$, so $k = 13$.

The fraction is $\dfrac{39}{65}$.

Answer: $\dfrac{39}{65}$.

Problem 14

Write in simplest form: **a)** $\dfrac{18}{27}$; **b)** $\dfrac{56}{70}$; **c)** $\dfrac{125}{1\,000}$.

Solution

a) Divide top and bottom by 9: $\dfrac{2}{3}$.

b) Divide by 14: $\dfrac{4}{5}$.

c) Divide by 125: $\dfrac{1}{8}$.

Answer: $\dfrac{2}{3}$; $\dfrac{4}{5}$; $\dfrac{1}{8}$.

Problem 15

Rewrite $\frac{2}{3}$, $\frac{3}{4}$, $\frac{5}{6}$ with the lowest common denominator.

Solution

Lowest common denominator: 12.

$\frac{2}{3} = \frac{8}{12}$;

$\frac{3}{4} = \frac{9}{12}$;

$\frac{5}{6} = \frac{10}{12}$.

Answer: $\frac{8}{12}$; $\frac{9}{12}$; $\frac{10}{12}$.

Problem 16

Put in order from smallest to largest: $\frac{3}{5}$, $\frac{2}{3}$, $\frac{4}{7}$, $\frac{7}{10}$.

Solution

Common denominator 210: $\frac{126}{210}$, $\frac{140}{210}$, $\frac{120}{210}$, $\frac{147}{210}$.

Compare the numerators: $120 < 126 < 140 < 147$.

Answer: $\frac{4}{7} < \frac{3}{5} < \frac{2}{3} < \frac{7}{10}$.

Problem 17

Compare $\frac{5}{12}$ and $\frac{7}{13}$ using $\frac{1}{2}$ as a stepping stone.

Solution

$\frac{5}{12} < \frac{6}{12} = \frac{1}{2}$.

$\frac{7}{13} > \frac{6.5}{13} = \frac{1}{2}$.

So $\frac{5}{12} < \frac{1}{2} < \frac{7}{13}$.

Answer: $\frac{5}{12} < \frac{7}{13}$.

Problem 18

Compare $\frac{2\,023}{2\,024}$ and $\frac{2\,024}{2\,025}$.

Solution

Gap to 1: $1 - \frac{2\,023}{2\,024} = \frac{1}{2\,024}$;

$1 - \frac{2\,024}{2\,025} = \frac{1}{2\,025}$.

Since $\frac{1}{2\,024} > \frac{1}{2\,025}$, the first fraction is smaller.

Answer: $\frac{2\,023}{2\,024} < \frac{2\,024}{2\,025}$.

Problem 19

a) Write as fractions: $3\frac{2}{5}$; $5\frac{3}{8}$. **b)** Write as mixed numbers: $\frac{29}{6}$; $\frac{47}{9}$.

Solution

a) $3\frac{2}{5} = \frac{17}{5}$;

$5\frac{3}{8} = \frac{43}{8}$.

b) $29 = 6 \times 4 + 5$, so $\frac{29}{6} = 4\frac{5}{6}$;

$47 = 9 \times 5 + 2$, so $\frac{47}{9} = 5\frac{2}{9}$.

Answer: $\frac{17}{5}$; $\frac{43}{8}$; $4\frac{5}{6}$; $5\frac{2}{9}$.

Problem 20

Calculate: **a)** $3\frac{1}{4} + 2\frac{2}{3}$; **b)** $4\frac{1}{2} - 1\frac{5}{6}$.

Solution

a) $= \frac{13}{4} + \frac{8}{3} = \frac{39}{12} + \frac{32}{12} = \frac{71}{12} = 5\frac{11}{12}$.

b) $= \frac{27}{6} - \frac{11}{6} = \frac{16}{6} = \frac{8}{3} = 2\frac{2}{3}$.

Answer: $5\frac{11}{12}$; $2\frac{2}{3}$.

Problem 21

Calculate: **a)** $\frac{3}{4} \times 2\frac{2}{3}$; **b)** $1\frac{1}{2} : \frac{3}{4}$.

Solution

a) $= \frac{3}{4} \times \frac{8}{3} = \frac{24}{12} = 2.$

b) $= \frac{3}{2} \times \frac{4}{3} = \frac{12}{6} = 2.$

Answer: 2; 2.

Problem 22

Calculate cleverly: $\frac{3}{7} \times \frac{5}{9} + \frac{3}{7} \times \frac{4}{9}$.

Solution

$= \frac{3}{7} \times \left(\frac{5}{9} + \frac{4}{9}\right)$

$= \frac{3}{7} \times 1 = \frac{3}{7}.$

Answer: $\frac{3}{7}$.

Problem 23

Calculate: $\left(1 - \frac{1}{4}\right) \times \left(1 + \frac{1}{3}\right)$.

Solution

$= \frac{3}{4} \times \frac{4}{3}$

$= 1.$

Answer: 1.

Problem 24

Find x: **a)** $x \times \frac{3}{5} = \frac{9}{10}$; **b)** $x : \frac{2}{3} = \frac{9}{4}$.

Solution

a) $x = \frac{9}{10} : \frac{3}{5} = \frac{9}{10} \times \frac{5}{3} = \frac{3}{2}.$

b) $x = \frac{9}{4} \times \frac{2}{3} = \frac{3}{2}$.

Answer: $x = \frac{3}{2}$; $x = \frac{3}{2}$.

Problem 25

Find x: **a)** $x + \frac{1}{3} = \frac{5}{6}$; **b)** $\frac{7}{8} - x = \frac{1}{4}$.

Solution

a) $x = \frac{5}{6} - \frac{1}{3} = \frac{5}{6} - \frac{2}{6} = \frac{1}{2}$.

b) $x = \frac{7}{8} - \frac{1}{4} = \frac{7}{8} - \frac{2}{8} = \frac{5}{8}$.

Answer: $x = \frac{1}{2}$; $x = \frac{5}{8}$.

Problem 26

A tank is full. First $\frac{2}{5}$ of the tank is used, then $\frac{1}{3}$ of the tank. What fraction of the tank is left?

Solution

Used: $\frac{2}{5} + \frac{1}{3} = \frac{11}{15}$ of the tank.

Left: $1 - \frac{11}{15} = \frac{4}{15}$.

Answer: $\frac{4}{15}$ of the tank.

Problem 27

A rectangle is $\frac{5}{6}$ m long and $\frac{2}{3}$ m wide. Find its perimeter and area.

Solution

Perimeter: $\left(\frac{5}{6} + \frac{2}{3}\right) \times 2 = \frac{9}{6} \times 2 = 3$ m.

Area: $\frac{5}{6} \times \frac{2}{3} = \frac{10}{18} = \frac{5}{9}$ m^2.

Answer: 3 m; $\frac{5}{9}$ m^2.

Problem 28

Find a fraction equal to $\frac{3}{5}$ whose numerator and denominator add up to 40.

Solution

Numerator 3 parts, denominator 5 parts; 8 parts make 40.
One part: $40 : 8 = 5$.
Numerator 15, denominator 25.

Answer: $\frac{15}{25}$.

Problem 29

Find a fraction equal to $\frac{2}{7}$ whose numerator and denominator add up to 54.

Solution

$2 + 7 = 9$ parts; one part: $54 : 9 = 6$.
Numerator 12, denominator 42.

Answer: $\frac{12}{42}$.

Problem 30

A rope $7\frac{1}{2}$ m long is cut into pieces $\frac{3}{4}$ m long. How many pieces are there?

Solution

$7\frac{1}{2} = \frac{15}{2}$.
Pieces: $\frac{15}{2} : \frac{3}{4} = \frac{15}{2} \times \frac{4}{3} = 10$.

Answer: 10 pieces.

Challenge – Extension

Problem 31 *

Calculate:

$$E = \frac{1}{6} + \frac{1}{12} + \frac{1}{20} + \cdots + \frac{1}{420} \qquad F = \frac{5}{3 \times 8} + \frac{5}{8 \times 13} + \cdots + \frac{5}{48 \times 53}$$

Solution

E) $420 = 20 \times 21$, so $E = \dfrac{1}{2\times 3} + \cdots + \dfrac{1}{20\times 21} = \dfrac{1}{2} - \dfrac{1}{21} = \dfrac{19}{42}$.

F) $F = \dfrac{1}{3} - \dfrac{1}{8} + \dfrac{1}{8} - \dfrac{1}{13} + \cdots + \dfrac{1}{48} - \dfrac{1}{53} = \dfrac{1}{3} - \dfrac{1}{53} = \dfrac{50}{159}$.

Problem 32 *

Calculate $G = \dfrac{1}{4} + \dfrac{1}{28} + \dfrac{1}{70} + \dfrac{1}{130} + \dfrac{1}{208}$.

Solution

Notice: $4 = 1\times 4$, $28 = 4\times 7$, $70 = 7\times 10$, $130 = 10\times 13$, $208 = 13\times 16$ (factors differ by 3).
Multiply G by 3:

$$3 \times G = \frac{3}{1\times 4} + \frac{3}{4\times 7} + \cdots + \frac{3}{13\times 16} = 1 - \frac{1}{16} = \frac{15}{16}.$$

So $G = \dfrac{15}{16} : 3 = \dfrac{5}{16}$.

Problem 33 *

Calculate $H = \dfrac{2}{3} + \dfrac{2}{9} + \dfrac{2}{27} + \dfrac{2}{81} + \dfrac{2}{243} + \dfrac{2}{729}$.

Solution

$3 \times H = 2 + \dfrac{2}{3} + \dfrac{2}{9} + \cdots + \dfrac{2}{243}$.

Subtract H: $2 \times H = 2 - \dfrac{2}{729} = \dfrac{1\,456}{729}$, so $H = \dfrac{728}{729}$.

Problem 34 *

Calculate $K = \left(1 - \dfrac{1}{3}\right) \times \left(1 - \dfrac{1}{4}\right) \times \left(1 - \dfrac{1}{5}\right) \times \cdots \times \left(1 - \dfrac{1}{50}\right)$.

Solution

$$K = \frac{2}{3} \times \frac{3}{4} \times \frac{4}{5} \times \cdots \times \frac{49}{50} = \frac{2}{50} = \frac{1}{25}.$$

Problem 35

Calculate $M = \left(1 - \frac{1}{4}\right) \times \left(1 - \frac{1}{9}\right) \times \left(1 - \frac{1}{16}\right) \times \cdots \times \left(1 - \frac{1}{100}\right)$.

Solution

The denominators are 2×2, 3×3, …, 10×10, and $1 - \frac{1}{n \times n} = \frac{(n-1) \times (n+1)}{n \times n}$.

Hence

$$M = \frac{1 \times 3}{2 \times 2} \times \frac{2 \times 4}{3 \times 3} \times \frac{3 \times 5}{4 \times 4} \times \cdots \times \frac{9 \times 11}{10 \times 10}.$$

Split into two products: $\frac{1 \times 2 \times \cdots \times 9}{2 \times 3 \times \cdots \times 10} \times \frac{3 \times 4 \times \cdots \times 11}{2 \times 3 \times \cdots \times 10} = \frac{1}{10} \times \frac{11}{2} = \frac{11}{20}$.

Problem 36

Find the natural number y: $\frac{1}{5 \times 6} + \frac{1}{6 \times 7} + \frac{1}{7 \times 8} + \cdots + \frac{1}{y \times (y+1)} = \frac{1}{10}$.

Solution

The left side equals $\frac{1}{5} - \frac{1}{y+1}$.

So $\frac{1}{5} - \frac{1}{y+1} = \frac{1}{10}$, giving $\frac{1}{y+1} = \frac{1}{5} - \frac{1}{10} = \frac{1}{10}$.

Hence $y + 1 = 10$ and $y = 9$.

Answer: $y = 9$.

Problem 37

Without adding, compare $S = \frac{1}{11} + \frac{1}{12} + \frac{1}{13} + \cdots + \frac{1}{20}$ with $\frac{1}{2}$.

Solution

S has 10 terms.

Each of $\frac{1}{11}, \ldots, \frac{1}{19}$ is greater than $\frac{1}{20}$, and the last term equals $\frac{1}{20}$.

Therefore $S > \frac{1}{20} \times 10 = \frac{1}{2}$.

Answer: $S > \frac{1}{2}$.

Problem 38

Calculate: $\frac{1}{1\times 2}+\frac{1}{2\times 3}+\frac{1}{3\times 4}+\cdots+\frac{1}{99\times 100}$.

Solution

Each term $\frac{1}{n\times(n+1)}=\frac{1}{n}-\frac{1}{n+1}$.

Sum $=1-\frac{1}{2}+\frac{1}{2}-\frac{1}{3}+\cdots+\frac{1}{99}-\frac{1}{100}$.

$=1-\frac{1}{100}=\frac{99}{100}$.

Answer: $\frac{99}{100}$.

Problem 39

Calculate: $\frac{2}{1\times 3}+\frac{2}{3\times 5}+\frac{2}{5\times 7}+\cdots+\frac{2}{19\times 21}$.

Solution

$\frac{2}{n\times(n+2)}=\frac{1}{n}-\frac{1}{n+2}$.

Sum $=1-\frac{1}{3}+\frac{1}{3}-\frac{1}{5}+\cdots+\frac{1}{19}-\frac{1}{21}$.

$=1-\frac{1}{21}=\frac{20}{21}$.

Answer: $\frac{20}{21}$.

Problem 40

Calculate: $\frac{1}{2}+\frac{1}{4}+\frac{1}{8}+\cdots+\frac{1}{1\,024}$.

Solution

Each term is a difference: $\frac{1}{2}=1-\frac{1}{2}$;

$\frac{1}{4}=\frac{1}{2}-\frac{1}{4}$; …;

$\frac{1}{1\,024}=\frac{1}{512}-\frac{1}{1\,024}$.

Adding: $1-\frac{1}{1\,024}=\frac{1\,023}{1\,024}$.

Answer: $\frac{1\,023}{1\,024}$.

Problem 41

Calculate: $\left(1+\frac{1}{2}\right)\times\left(1+\frac{1}{3}\right)\times\left(1+\frac{1}{4}\right)\times\cdots\times\left(1+\frac{1}{99}\right)$.

Solution

$=\frac{3}{2}\times\frac{4}{3}\times\frac{5}{4}\times\cdots\times\frac{100}{99}$.

Each numerator cancels the next denominator, leaving $\frac{100}{2}=50$.

Answer: 50.

Problem 42

Compare $\frac{13}{27}$ and $\frac{27}{55}$ without a common denominator.

Solution

Both are less than $\frac{1}{2}$.

Look at how far short they fall:

$\frac{1}{2}-\frac{13}{27}=\frac{27-26}{54}=\frac{1}{54}$;

$\frac{1}{2}-\frac{27}{55}=\frac{55-54}{110}=\frac{1}{110}$.

Since $\frac{1}{54}>\frac{1}{110}$, $\frac{13}{27}$ is further below $\frac{1}{2}$, so it is smaller.

Answer: $\frac{13}{27}<\frac{27}{55}$.

Problem 43

Write three different fractions greater than $\frac{1}{3}$ and less than $\frac{1}{2}$.

Solution

With denominator 24: $\frac{1}{3}=\frac{8}{24}$ and $\frac{1}{2}=\frac{12}{24}$.

Three fractions: $\frac{9}{24}$, $\frac{10}{24}$, $\frac{11}{24}$.

Answer: For example: $\frac{9}{24}$; $\frac{10}{24}$; $\frac{11}{24}$.

Problem 44

Find every whole number x with $\frac{1}{3} < \frac{x}{12} < \frac{3}{4}$.

Solution

$\frac{1}{3} = \frac{4}{12}$;
$\frac{3}{4} = \frac{9}{12}$.
So $4 < x < 9$, giving $x = 5,\ 6,\ 7,\ 8$.
Answer: $x = 5;\ 6;\ 7;\ 8$.

Problem 45

Which whole number added to both the top and bottom of $\frac{5}{11}$ gives a fraction equal to $\frac{3}{5}$?

Solution

Adding the same number keeps the difference $11 - 5 = 6$.
The new fraction is $\frac{3}{5}$: the difference of 2 parts is 6, so one part is 3.
New numerator 9; the number added is $9 - 5 = 4$.
Answer: 4.

Problem 46

Which whole number, taken from the top of $\frac{23}{37}$ and added to the bottom, gives a fraction equal to $\frac{1}{3}$?

Solution

The sum stays $23 + 37 = 60$.
The new fraction is $\frac{1}{3}$: numerator 1 part out of 4; new numerator $60 : 4 = 15$.
The number is $23 - 15 = 8$.
Answer: 8.

Problem 47 *

Calculate: $\frac{1}{3}+\frac{1}{15}+\frac{1}{35}+\frac{1}{63}+\frac{1}{99}$.

Solution

Write the denominators as products: $\frac{1}{1\times 3}+\frac{1}{3\times 5}+\frac{1}{5\times 7}+\frac{1}{7\times 9}+\frac{1}{9\times 11}$.

Twice the sum is $1-\frac{1}{11}=\frac{10}{11}$.

So the sum is $\frac{10}{11}:2=\frac{5}{11}$.

Answer: $\frac{5}{11}$.

Problem 48 *

Write $\frac{7}{12}$ as the sum of two different fractions with numerator 1.

Solution

Split 7 into two factors of 12: $7=3+4$.

$\frac{7}{12}=\frac{3}{12}+\frac{4}{12}=\frac{1}{4}+\frac{1}{3}$.

Answer: $\frac{7}{12}=\frac{1}{3}+\frac{1}{4}$ (or $\frac{1}{2}+\frac{1}{12}$).

Problem 49 **

Calculate: $1+\frac{1}{3}+\frac{1}{6}+\frac{1}{10}+\frac{1}{15}+\cdots+\frac{1}{45}$ (the denominators are 1, 3, 6, 10, 15, ...).

Solution

Each term is $\frac{2}{n\times(n+1)}$: $1=\frac{2}{1\times 2}$, $\frac{1}{3}=\frac{2}{2\times 3}$, ..., $\frac{1}{45}=\frac{2}{9\times 10}$.

Sum $=2\times\left(1-\frac{1}{10}\right)=\frac{9}{5}$.

Answer: $\frac{9}{5}=1\frac{4}{5}$.

Problem 50 **

Find a fraction with denominator 13 such that adding 6 to its numerator (same denominator) makes it 3 times as large.

Solution

The denominator is fixed, so the new numerator is 3 times the old.

So 6 is $3 - 1 = 2$ times the old numerator, which is $6 : 2 = 3$.

The fraction is $\frac{3}{13}$.

Answer: $\frac{3}{13}$.

3 Decimals

① Decimal fractions – Decimals

- A **decimal fraction** has denominator 10, 100, 1 000, ... For example $\frac{4}{10}$, $\frac{45}{100}$, $\frac{358}{1\,000}$.
- Every decimal fraction can be written as a **decimal**: $\frac{4}{10} = 0.4$; $\frac{45}{10} = 4.5$; $\frac{358}{1\,000} = 0.358$.
- Many other fractions can be changed into decimal fractions and then into decimals: $\frac{1}{2} = \frac{5}{10} = 0.5$; $\frac{13}{5} = \frac{26}{10} = 2.6$.
- A decimal has a **whole-number part** (left of the decimal point) and a **decimal part** (right of it). The decimal places are: tenths, hundredths, thousandths, ...
- One unit of any place equals 10 units of the next place to the right. Writing (or removing) zeros at the right end of the decimal part does not change the value: $2.5 = 2.50 = 2.500$.

② Comparing two decimals

- Compare the **whole-number parts** first: the larger whole part gives the larger number.
- If the whole parts are equal, compare the decimal places in turn (tenths, hundredths, thousandths, ...). At the first place where the digits differ, the larger digit gives the larger number.

Example: $7.305 > 7.29$ because the whole parts are equal and in the tenths $3 > 2$.

③ Moving the decimal point

- Moving the point 1, 2, 3, ... places to the **right** makes the number 10, 100, 1 000, ... **times larger**: $3.75 \times 10 = 37.5$.
- Moving the point 1, 2, 3, ... places to the **left** makes it 10, 100, 1 000, ... **times smaller**: $52.1 : 100 = 0.521$.
- So when the point moves one place left, if the new number is 1 part, the old number is 10 such parts: the number has *decreased* by 9 parts.

Example 1

Write these decimals:

a) 4 tens, 7 ones, 5 tenths, 3 **b)** 5 ones and 38 hundredths; **c)** 0 ones and 7 thousandths. thousandths;

Guided solution.

Put each digit in its place; write 0 in any empty place:

a) 47.503; **b)** 5.38; **c)** 0.007.

Example 2

Write as fractions in lowest terms: 12.25 and 3.125.

Guided solution.

Write as a decimal fraction, then simplify:

$$12.25 = \frac{1\,225}{100} = \frac{49}{4}; \qquad 3.125 = \frac{3\,125}{1\,000} = \frac{25}{8}.$$

Example 3

Find all natural numbers x with $2.46 < x < \frac{31}{5}$.

Guided solution.

Change $\frac{31}{5} = \frac{62}{10} = 6.2$.

We need $2.46 < x < 6.2$, so $x \in \{3;\ 4;\ 5;\ 6\}$.

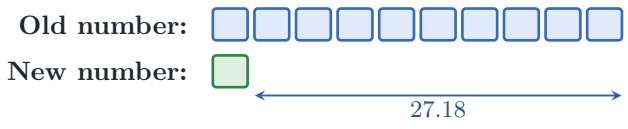
Example 4

When the decimal point of a number is moved one place to the left, the number decreases by 27.18. Find the number.

Guided solution.

Moving the point one place left makes the number 10 times smaller.

Bar model:



Difference in parts: $10 - 1 = 9$.

One part (the new number): $27.18 : 9 = 3.02$.

The number: $3.02 \times 10 = 30.2$.

Check: $30.2 - 3.02 = 27.18$.

Example 5

Calculate quickly: $S = 0.2 + 0.4 + 0.6 + \cdots + 3.8 + 4$.

Guided solution.

This is an arithmetic sequence with difference 0.2.

Number of terms: $(4 - 0.2) : 0.2 + 1 = 20$.

Sum: $S = (0.2 + 4) \times 20 : 2 = 42$.

Practice

Problem 1

Write as decimals:

a) $\frac{7}{10}$; $\frac{9}{1\,000}$; $\frac{63}{100}$; $\frac{1\,234}{100}$; $4\frac{3}{10}$; $12\frac{45}{1\,000}$ **b)** $\frac{2}{5}$; $\frac{7}{4}$; $3\frac{9}{20}$; $\frac{11}{8}$

Solution

a) 0.7; 0.009; 0.63; 12.34; 4.3; 12.045.

b) Change to decimal fractions: $\frac{2}{5} = \frac{4}{10} = 0.4$;

$\frac{7}{4} = \frac{175}{100} = 1.75$;

$3\frac{9}{20} = 3\frac{45}{100} = 3.45$;

$\frac{11}{8} = \frac{1\,375}{1\,000} = 1.375$.

Problem 2

Write as decimal fractions: 0.27; 0.048; 3.6; 15.0091.

Solution

$$0.27 = \frac{27}{100}; \quad 0.048 = \frac{48}{1\,000}; \quad 3.6 = \frac{36}{10}; \quad 15.0091 = \frac{150\,091}{10\,000}.$$

Problem 3

Write the decimal made of:

a) 6 ones, 3 tenths, 4 hundredths, 1 thousandth;

b) 42 ones, 5 tenths, 9 thousandths;

c) 3 hundreds, 7 ones, 8 hundredths;

d) 0 ones and 6 hundredths.

Solution

a) 6.341. **b)** 42.509. **c)** 307.08. **d)** 0.06.

Problem 4

Order from smallest to largest: 5.34; 5.043; 5.4; 5.304; 4.99.

Solution

4.99 has the smallest whole part.
The other four have whole part 5; compare tenths: $0 < 3 = 3 < 4$; for the two with 3 tenths, compare hundredths: $0 < 4$.
Answer: $4.99 < 5.043 < 5.304 < 5.34 < 5.4$.

Problem 5

Find the suitable digits:

a) $7.6\overline{x}2 < 7.612$ **b)** $3.4\overline{y}5 > 3.475$ **c)** $5.726 < 5.7\overline{a}7 < 5.755$

Solution

a) Both start 7.6; compare hundredths: we need $x < 1$ (if $x = 1$ we would need $2 < 2$).
So $x = 0$.
b) We need $y > 7$ ($y = 7$ makes them equal).
So $y \in \{8; 9\}$.
c) $5.7\overline{a}7 > 5.726$ when $a \geq 2$;
$5.7\overline{a}7 < 5.755$ when $a \leq 4$.
So $a \in \{2; 3; 4\}$.

Problem 6

Find all natural numbers n such that

a) $0.7 < n < 3.2$; **b)** $46.85 < n < 50.1$.

Solution

a) $n \in \{1;\ 2;\ 3\}$. **b)** $n \in \{47;\ 48;\ 49;\ 50\}$.

Problem 7

Find two consecutive natural numbers m and n such that

a) $m < 27.4 < n$; **b)** $m < 99.9 < n$.

Solution

a) $m = 27$, $n = 28$. **b)** $m = 99$, $n = 100$.

Problem 8

Find all decimals x with two decimal places such that $0.345 < x < 0.4$.

Solution

Write $0.4 = 0.40$.
The numbers with two decimal places greater than 0.345 and less than 0.40 are:

$$0.35;\quad 0.36;\quad 0.37;\quad 0.38;\quad 0.39.$$

Problem 9

Using the four digits 1, 3, 6, 8 once each, write the smallest and the largest decimals with two decimal places.

Solution

The whole part has two digits.
Smallest: digits in increasing order, 13.68.
Largest: in decreasing order, 86.31.
Answer: 13.68 and 86.31.

Problem 10

Calculate in the most convenient way:

$A = 0.01 + 0.02 + 0.03 + \cdots + 0.09 + 0.10$ $B = 1.3 + 1.5 + 1.7 + \cdots + 19.9 + 20.1$

Solution

A) 10 terms: $A = (0.01 + 0.10) \times 10 : 2 = 0.55$.

B) Difference 0.2; number of terms: $(20.1 - 1.3) : 0.2 + 1 = 95$.

$B = (1.3 + 20.1) \times 95 : 2 = 1\,016.5$.

Problem 11

Work out mentally:

3.75×10	0.068×100	$52.1 : 10$
$7.4 : 100$	$0.5 \times 1\,000$	$216 : 1\,000$

Solution

37.5; 6.8; 5.21; 0.074; 500; 0.216.

Problem 12

Fill in the decimals:

a) 3 m 45 cm = ... m **b)** 2 kg 50 g = ... kg **c)** 4 km 25 m = ... km

d) 905 g = ... kg **e)** 38 dm^2 = ... m^2 **g)** 7 m^2 5 dm^2 = ... m^2

Solution

a) 3.45 m. **b)** 2.05 kg. **c)** 4.025 km. **d)** 0.905 kg. **e)** 0.38 m^2. **g)** 7.05 m^2.

Note: 1 $m^2 = 100\ dm^2$, *so* 5 $dm^2 = 0.05\ m^2$.

Problem 13

The first rope is 3 m 15 cm long. The second is 8 dm longer than the first. The third is 65 cm shorter than the second. Find the total length of the three ropes in metres.

Solution

Convert: 3 m 15 cm = 3.15 m; 8 dm = 0.8 m; 65 cm = 0.65 m.

Second rope: $3.15 + 0.8 = 3.95$ (m).

Third rope: $3.95 - 0.65 = 3.3$ (m).

Total: $3.15 + 3.95 + 3.3 = 10.4$ (m).

Answer: 10.4 m.

Problem 14

Moving the decimal point of a number one place to the left makes it 16.74 smaller. Find the number.

Solution

If the new number is 1 part, the old one is 10 parts; the difference is 9 parts, which is 16.74.
New number: $16.74 : 9 = 1.86$.
The number: $1.86 \times 10 = 18.6$.
Answer: 18.6.

Problem 15

Moving the decimal point of a number one place to the right makes it 48.24 larger. Find the number.

Solution

If the old number is 1 part, the new one is 10 parts; the difference, 9 parts, is 48.24.
The number: $48.24 : 9 = 5.36$.
Check: $53.6 - 5.36 = 48.24$.
Answer: 5.36.

Problem 16

Write the decimal made of: **a)** five ones, four tenths, seven hundredths; **b)** zero ones, three thousandths; **c)** twelve ones, six hundredths.

Solution

a) 5.47.
b) 0.003.
c) 12.06 (the tenths digit is 0).
Answer: 5.47; 0.003; 12.06.

Problem 17

What is the value of the digit 7 in each number? **a)** 3.472; **b)** 70.25; **c)** 0.007.

Solution

a) Hundredths: $\frac{7}{100}$.
b) Tens: 70.
c) Thousandths: $\frac{7}{1\,000}$.

Answer: $\frac{7}{100}$; 70; $\frac{7}{1\,000}$.

Problem 18

Write as decimals: $\frac{3}{10}$; $\frac{47}{100}$; $\frac{9}{1\,000}$; $1\frac{3}{4}$.

Solution

$\frac{3}{10} = 0.3$;

$\frac{47}{100} = 0.47$;

$\frac{9}{1\,000} = 0.009$.

$1\frac{3}{4} = 1\frac{75}{100} = 1.75$.

Answer: 0.3; 0.47; 0.009; 1.75.

Problem 19

Write as decimals (change the denominator to 10, 100 or 1 000): $\frac{3}{5}$; $\frac{7}{20}$; $\frac{9}{25}$; $\frac{3}{8}$.

Solution

$\frac{3}{5} = \frac{6}{10} = 0.6$;

$\frac{7}{20} = \frac{35}{100} = 0.35$.

$\frac{9}{25} = \frac{36}{100} = 0.36$;

$\frac{3}{8} = \frac{375}{1\,000} = 0.375$.

Answer: 0.6; 0.35; 0.36; 0.375.

Problem 20

Fill in $>$, $<$, $=$: **a)** $4.7 \square 4.70$; **b)** $0.09 \square 0.1$; **c)** $12.305 \square 12.35$.

Solution

a) A zero added at the end of the decimal part changes nothing: $=$.

b) Tenths: $0 < 1$, so $<$.

c) Hundredths: $0 < 5$, so $<$.

Answer: =; <; <.

Problem 21

Order from largest to smallest: 0.8, 0.08, 0.88, 0.808.

Solution

With three decimal places: 0.800, 0.080, 0.880, 0.808.
Compare: $880 > 808 > 800 > 80$.
Answer: $0.88 > 0.808 > 0.8 > 0.08$.

Problem 22

Write as decimals: **a)** 5 m 7 cm = ... m; **b)** 450 g = ... kg; **c)** 3 quintals 5 kg = ... quintals (1 quintal = 100 kg); **d)** 2 km 30 m = ... km.

Solution

a) $7 \text{ cm} = \frac{7}{100}$ m: 5.07 m.

b) $450 \text{ g} = \frac{450}{1\,000}$ kg: 0.45 kg.

c) $5 \text{ kg} = \frac{5}{100}$ quintal: 3.05 quintals.

d) $30 \text{ m} = \frac{30}{1\,000}$ km: 2.03 km.

Answer: 5.07; 0.45; 3.05; 2.03.

Problem 23

Write as decimals: **a)** $3 \text{ m}^2\ 25 \text{ dm}^2 = \ldots \text{ m}^2$; **b)** $7 \text{ dm}^2\ 5 \text{ cm}^2 = \ldots \text{ dm}^2$; **c)** $4\,500 \text{ m}^2 = \ldots$ ha.

Solution

Each area unit is 100 times the next smaller one.
a) 3.25 m^2.

b) 7.05 dm^2.

c) 1 ha = 10 000 m^2, so 4 500 m^2 = 0.45 ha.

Answer: 3.25; 7.05; 0.45.

Problem 24

Work out mentally: 3.75×10; 0.42×100; $56.3 : 10$; $7.8 : 100$.

Solution

Multiplying by 10, 100 moves the point 1, 2 places right: 37.5;
42.
Dividing by 10, 100 moves it 1, 2 places left: 5.63;
0.078.
Answer: 37.5; 42; 5.63; 0.078.

Problem 25

Calculate: **a)** $12.5 + 3.75$; **b)** $20 - 7.36$; **c)** $4.5 + 2.35 - 1.8$.

Solution

a) $12.50 + 3.75 = 16.25$.

b) $20.00 - 7.36 = 12.64$.

c) $6.85 - 1.8 = 5.05$.

Answer: 16.25; 12.64; 5.05.

Problem 26

Find x: **a)** $x + 2.75 = 6.1$; **b)** $x - 4.6 = 3.85$.

Solution

a) $x = 6.1 - 2.75 = 3.35$.

b) $x = 3.85 + 4.6 = 8.45$.

Answer: $x = 3.35$; $x = 8.45$.

Problem 27

Calculate cleverly: $3.25 + 4.8 + 6.75 + 5.2$.

Solution

$= (3.25 + 6.75) + (4.8 + 5.2)$
$= 10 + 10 = 20$.
Answer: 20.

Problem 28

A shop sells sugar on three days: 12.5 kg on day one, 3.75 kg more on day two, and 2.4 kg less than day two on day three. How much sugar does it sell in all?

Solution

Day two: $12.5 + 3.75 = 16.25$ kg.
Day three: $16.25 - 2.4 = 13.85$ kg.
In all: $12.5 + 16.25 + 13.85 = 42.6$ kg.
Answer: 42.6 kg.

Problem 29

Find every decimal x with one decimal place such that $3.4 < x < 3.9$.

Solution

The whole part is 3 and the tenths digit is more than 4 and less than 9.
$x = 3.5,\ 3.6,\ 3.7,\ 3.8$.
Answer: 3.5; 3.6; 3.7; 3.8.

Problem 30

Write the largest and the smallest decimal with three different digits, two decimal places and a non-zero whole part.

Solution

Largest: the biggest digits, in decreasing order: 9.87.
Smallest: whole part 1, then 0 and 2: 1.02.
Answer: 9.87 and 1.02.

Challenge – Extension

Problem 31 *

Moving the decimal point of a number two places to the left makes it 40.887 smaller. Find the number.

Solution

Moving two places left divides by 100: new number 1 part, old number 100 parts, difference 99 parts.
New number: $40.887 : 99 = 0.413$.
The number: $0.413 \times 100 = 41.3$.

Answer: 41.3.

Problem 32

Two decimals add up to 151.5. Moving the decimal point of the first number two places left gives the second number. Find both numbers.

Solution

The first number is 100 times the second: second 1 part, first 100 parts, total 101 parts.
Second number: $151.5 : 101 = 1.5$.
First number: $1.5 \times 100 = 150$.
Answer: 150 and 1.5.

Problem 33

A decimal, with its point moved one place to the right, is added to the original number; the result is 75.9. Find the number.

Solution

If the number is 1 part, the shifted number is 10 parts; the sum is 11 parts.
The number: $75.9 : 11 = 6.9$.
Check: $69 + 6.9 = 75.9$.
Answer: 6.9.

Problem 34

Replace a, b with suitable digits: $\overline{1a.b} + \overline{a.b} = 20.8$.

Solution

Note that $\overline{1a.b} = 10 + \overline{a.b}$.
So $10 + \overline{a.b} + \overline{a.b} = 20.8$, i.e.
$\overline{a.b} \times 2 = 10.8$ and $\overline{a.b} = 5.4$.
So $a = 5$, $b = 4$.
Check: $15.4 + 5.4 = 20.8$.

Problem 35

Consider the sequence $0.5,\ 1,\ 1.5,\ 2,\ \ldots,\ 49.5,\ 50$.

a) How many terms are there?

b) What is the 37th term?

c) Find the sum of all terms.

d) Find the sum of the first 40 terms.

Solution

a) $(50 - 0.5) : 0.5 + 1 = 100$ terms. **b)** $0.5 + 36 \times 0.5 = 18.5$.

c) $(0.5 + 50) \times 100 : 2 = 2\,525$. **d)** The 40th term is 20; sum: $(0.5 + 20) \times 40 : 2 = 410$.

Problem 36 *

Using 0, 4, 6, 8 once each, how many decimals can be written whose whole-number part and decimal part both have two digits?

Solution

The number looks like $\overline{ab.cd}$ with $a \neq 0$: a has 3 choices, b has 3, c has 2, d has 1: $3 \times 3 \times 2 \times 1 = 18$.

Answer: 18 numbers.

Problem 37 **

Find the smallest decimal with all digits different whose digit sum is 10.

Solution

To be smallest, the whole part must be 0, and the decimal digits must be as small as possible, in increasing order.

0 is already used in the whole part, so take 1, 2, 3 in turn; what is left is $10 - 1 - 2 - 3 = 4$ (unused).

The number is 0.1234.

Answer: 0.1234.

Problem 38 **

A decimal has two decimal places. Removing its decimal point gives a number 1 221.66 greater than the original. Find the decimal.

Solution

Removing the point of a number with two decimal places multiplies it by 100: old 1 part, new 100 parts, difference 99 parts.

The number: $1\,221.66 : 99 = 12.34$.

Answer: 12.34.

Problem 39

Moving the decimal point of a number two places to the right makes it 207.9 larger. Find the number.

Solution

The new number is 100 times the old: old 1 part, new 100 parts, difference 99 parts.
The number is $207.9 : 99 = 2.1$.
Answer: 2.1.

Problem 40

Two decimals add up to 22.11. Moving the point of the smaller one place right gives the larger. Find them.

Solution

The larger is 10 times the smaller; 11 parts in all.
Smaller: $22.11 : 11 = 2.01$.
Larger: 20.1.
Answer: 2.01 and 20.1.

Problem 41

Two numbers differ by 49.5. Moving the point of the larger one place left gives the smaller. Find them.

Solution

The larger is 10 times the smaller; the difference is 9 parts.
Smaller: $49.5 : 9 = 5.5$.
Larger: 55.
Answer: 5.5 and 55.

Problem 42

Consider $0.1,\ 0.3,\ 0.5,\ \ldots,\ 9.9$. How many terms are there? Find their sum.

Solution

Step: 0.2.
Number of terms: $(9.9 - 0.1) : 0.2 + 1 = 50$.
Sum: $(0.1 + 9.9) \times 50 : 2 = 250$.
Answer: 50 terms; 250.

Problem 43

How many decimals with two decimal places are greater than 2 and less than 3?

Solution

They are 2.01, 2.02, ..., 2.99.

The decimal part runs from 01 to 99: 99 numbers.

Answer: 99 numbers.

Problem 44

Find every decimal x with one decimal place such that $5.15 < x < 5.52$.

Solution

$5.15 < 5.2$ and $5.5 < 5.52$.

$x = 5.2,\ 5.3,\ 5.4,\ 5.5$.

Answer: 5.2; 5.3; 5.4; 5.5.

Problem 45

Calculate quickly: $1.2 + 2.3 + 3.4 + 4.5 + 5.6 + 6.7 + 7.8 + 8.9$.

Solution

This list goes up by 1.1 and has 8 terms.

Sum: $(1.2 + 8.9) \times 8 : 2 = 10.1 \times 4 = 40.4$.

Answer: 40.4.

Problem 46

Find digits a and b with $\overline{a.b} + \overline{b.a} = 7.7$ and $a - b = 3$.

Solution

The sum is $(a + b)$ ones plus $(a + b)$ tenths, i.e.

$(a + b) \times 1.1 = 7.7$.

So $a + b = 7$, and with $a - b = 3$ we get $a = 5$, $b = 2$.

Check: $5.2 + 2.5 = 7.7$.

Answer: $a = 5$, $b = 2$.

Problem 47

Moving the point of a decimal one place right and then subtracting the original number gives 31.5. Find the number.

Solution

The new number is 10 times the old; the difference is 9 times the old.
The number is $31.5 : 9 = 3.5$.
Answer: 3.5.

Problem 48 *

Three decimals add up to 11.1. The first is 1.5 more than the second, which is 1.2 more than the third. Find them.

Solution

The second is 1.2 more than the third; the first is $1.5 + 1.2 = 2.7$ more than the third.
Three times the third: $11.1 - 1.2 - 2.7 = 7.2$, so the third is 2.4.
Second: 3.6; first: 5.1.
Answer: 5.1; 3.6; 2.4.

Problem 49 **

Using 2, 5, 7 once each, write every decimal with one decimal place and add them up.

Solution

There are 6: 25.7, 27.5, 52.7, 57.2, 72.5, 75.2.
Each digit is in each place 2 times; the digits add to 14.
Sum: $14 \times 2 \times (10 + 1 + 0.1) = 28 \times 11.1 = 310.8$.
Answer: 310.8.

Problem 50 **

Writing the block 142857 over and over after the point gives $0.142857142857\ldots$ What is the 100th digit after the point?

Solution

Each block has 6 digits; $100 = 6 \times 16 + 4$.
After 16 blocks, the 100th digit is the 4th digit of a block: 8.
Answer: 8.

4 Three Basic Fraction Problems

① The three basic problems

- **Problem 1 – A fraction of a number.** To find $\frac{m}{n}$ of A, compute $A \times \frac{m}{n}$ (or $A : n \times m$). Example: $\frac{2}{5}$ of 60 is $60 \times \frac{2}{5} = 24$.
- **Problem 2 – A number from the value of its fraction.** If $\frac{a}{b}$ of a number is X, the number is $X : \frac{a}{b}$ (or $X : a \times b$). Example: if $\frac{2}{5}$ of a number is 24, the number is $24 : \frac{2}{5} = 60$.
- **Problem 3 – The ratio of two numbers.** The ratio of a to b ($b \neq 0$) is $a : b$ or $\frac{a}{b}$, usually in lowest terms. Both quantities must be in *the same unit*. Example: the ratio of 45 to 20 is $\frac{45}{20} = \frac{9}{4}$.

② A fraction of what is left

When a problem says "$\frac{a}{b}$ *of what is left*", first find the remaining part (as a fraction of the whole), then take the fraction of it. For instance, using $\frac{1}{3}$ of a journey leaves $\frac{2}{3}$; going on for $\frac{3}{5}$ *of the rest* covers $\frac{3}{5} \times \frac{2}{3} = \frac{2}{5}$ of the journey.

A **bar model** shows clearly which part matches the known number.

Example 1

A garden has an area of 720 m^2. $\frac{3}{8}$ of it grows vegetables, $\frac{5}{12}$ grows flowers, and the rest is paths. Find the area of the paths.

Guided solution.

Way 1. Vegetables: $720 \times \frac{3}{8} = 270$ (m^2).

Flowers: $720 \times \frac{5}{12} = 300$ (m^2).

Paths: $720 - (270 + 300) = 150$ (m^2).

Way 2. The paths take $1 - \frac{3}{8} - \frac{5}{12} = \frac{24-9-10}{24} = \frac{5}{24}$ of the garden, i.e.

$720 \times \frac{5}{24} = 150$ (m^2).

Example 2

In the school chess club, $\frac{3}{5}$ of the members play chess, $\frac{1}{4}$ play Chinese chess, and the remaining 12 play tic-tac-toe. How many members does the club have?

Guided solution.

The 12 tic-tac-toe players are $1 - \frac{3}{5} - \frac{1}{4} = \frac{20-12-5}{20} = \frac{3}{20}$ of the members.

Number of members: $12 : \frac{3}{20} = 80$.

Example 3

A cyclist rides for three days. On day one she covers $\frac{1}{3}$ of the route, on day two $\frac{3}{5}$ of the remaining distance, and on day three the last 24 km. How long is the route?

Guided solution.



After day one, $1 - \frac{1}{3} = \frac{2}{3}$ of the route is left.

Day two covers $\frac{3}{5}$ of that, so day three covers $1 - \frac{3}{5} = \frac{2}{5}$ of the remainder, i.e.

$\frac{2}{5} \times \frac{2}{3} = \frac{4}{15}$ of the route.

The route is $24 : \frac{4}{15} = 90$ (km).

Example 4

Three numbers add up to 170. The first is $\frac{1}{2}$ of the second, and the second is $\frac{3}{4}$ of the third. Find the three numbers.

Guided solution.

Use a common "part": the second number is 2 parts compared with the first and 3 parts compared with the third.

Take it as 6 parts:

$$\text{first} = 3 \text{ parts}; \quad \text{second} = 6 \text{ parts}; \quad \text{third} = 8 \text{ parts}.$$

Total: $3 + 6 + 8 = 17$ parts; one part is $170 : 17 = 10$.
The numbers are $30,\ 60,\ 80$.

Example 5

A number plus 45 equals $\frac{8}{5}$ of the number. Find the number.

Guided solution.

The number is $\frac{5}{5}$ of itself; adding 45 makes it $\frac{8}{5}$.

So 45 is $\frac{8}{5} - \frac{5}{5} = \frac{3}{5}$ of the number.

The number is $45 : \frac{3}{5} = 75$.

Practice

Problem 1

A farm keeps 240 birds: chickens, ducks and geese. Chickens are $\frac{3}{8}$ of all the birds and ducks are $\frac{1}{3}$. How many geese are there?

Solution

Chickens: $240 \times \frac{3}{8} = 90$.

Ducks: $240 \times \frac{1}{3} = 80$.

Geese: $240 - 90 - 80 = 70$.
Answer: 70 geese.

Problem 2

A school has 1 200 pupils. Grade 5 makes up $\frac{1}{5}$ of them, and girls are $\frac{7}{12}$ of Grade 5. How many boys and how many girls are in Grade 5?

Solution

Grade 5: $1\,200 \times \frac{1}{5} = 240$ pupils.

Girls: $240 \times \frac{7}{12} = 140$.

Boys: $240 - 140 = 100$.

Answer: 100 boys and 140 girls.

Problem 3

A runner covers 30 km in three hours. In the first hour he runs $\frac{2}{5}$ of the distance, in the second hour $\frac{1}{2}$ of the remaining distance. How far does he run in the third hour?

Solution

First hour: $30 \times \frac{2}{5} = 12$ (km); left: $30 - 12 = 18$ (km).

Second hour: $18 \times \frac{1}{2} = 9$ (km).

Third hour: $18 - 9 = 9$ (km).

Answer: 9 km.

Problem 4

Mum buys 12 kg of rice. First she uses $\frac{1}{3}$ of it plus 1 kg more. Next she uses $\frac{1}{2}$ of what is left plus $\frac{1}{2}$ kg more. How much rice is left?

Solution

First time: $12 \times \frac{1}{3} + 1 = 5$ (kg); left: $12 - 5 = 7$ (kg).

Second time: $7 \times \frac{1}{2} + \frac{1}{2} = 4$ (kg); left: $7 - 4 = 3$ (kg).

Answer: 3 kg.

Problem 5

$\frac{2}{3}$ of the pupils in class 5B is 22 pupils. How many pupils are in class 5B?

Solution

Pupils in class 5B: $22 : \frac{2}{3} = 33$.

Answer: 33 pupils.

Problem 6

After 18 litres are taken from a can of oil, $\frac{5}{8}$ of the original amount is left. How much oil was there at first?

Solution

The 18 litres are $1 - \frac{5}{8} = \frac{3}{8}$ of the original.

At first: $18 : \frac{3}{8} = 48$ litres.

Answer: 48 litres.

Problem 7

A number minus 28 equals $\frac{3}{5}$ of the number. Find the number.

Solution

28 is $1 - \frac{3}{5} = \frac{2}{5}$ of the number, so the number is $28 : \frac{2}{5} = 70$.

Answer: 70.

Problem 8

This season a field's rice harvest grew by $\frac{1}{4}$ compared with last season, reaching 15 tonnes. How much was harvested last season?

Solution

If last season is 1 (that is $\frac{4}{4}$), this season is $1 + \frac{1}{4} = \frac{5}{4}$.

Last season: $15 : \frac{5}{4} = 12$ tonnes.

Answer: 12 tonnes.

Problem 9

After selling $\frac{3}{7}$ of its boxes of cakes plus 4 more boxes, a shop has 28 boxes left. How many boxes did it have at first?

Solution

Selling only $\frac{3}{7}$ would leave $28 + 4 = 32$ boxes, which is $1 - \frac{3}{7} = \frac{4}{7}$ of the boxes.

At first: $32 : \frac{4}{7} = 56$ boxes.

Answer: 56 boxes.

Problem 10

A racing cyclist covers $\frac{4}{15}$ of a route on day one, $\frac{2}{5}$ of the route on day two, and the last 100 km on day three. How long is the route?

Solution

100 km is $1 - \frac{4}{15} - \frac{6}{15} = \frac{5}{15} = \frac{1}{3}$ of the route.

Route: $100 : \frac{1}{3} = 300$ km.

Answer: 300 km.

Problem 11

Two barns hold equal amounts of rice. Taking $\frac{2}{5}$ from the first barn and $\frac{3}{10}$ from the second leaves the first with 12 tonnes less than the second. How much did each barn hold at first?

Solution

The first barn keeps $\frac{3}{5} = \frac{6}{10}$ and the second keeps $\frac{7}{10}$ of its original amount.

12 tonnes is $\frac{7}{10} - \frac{6}{10} = \frac{1}{10}$, so each barn held $12 : \frac{1}{10} = 120$ tonnes.

Answer: 120 tonnes each.

Problem 12

Nam has some marbles. Blue ones are $\frac{1}{3}$ of all, yellow ones are $\frac{3}{4}$ of the blue ones, and the other 30 are red. How many marbles does Nam have?

Solution

Yellow: $\frac{3}{4} \times \frac{1}{3} = \frac{1}{4}$ of all.

Red: $1 - \frac{1}{3} - \frac{1}{4} = \frac{5}{12}$ of all.

All marbles: $30 : \frac{5}{12} = 72$.

Answer: 72 marbles.

Problem 13

The first customer buys $\frac{1}{6}$ of the potatoes, the second buys $\frac{3}{5}$ of the rest, and the seller has 18 kg left. How much did each customer buy?

Solution

After the first customer $\frac{5}{6}$ is left.

Finally $\frac{2}{5}$ of that remains: $\frac{2}{5} \times \frac{5}{6} = \frac{1}{3}$ of the potatoes, which is 18 kg.

At first: $18 : \frac{1}{3} = 54$ kg.

First customer: $54 : 6 = 9$ kg; second: $(54 - 9) \times \frac{3}{5} = 27$ kg.

Answer: 9 kg and 27 kg.

Problem 14

Find the ratio of:

a) 45 to 20; **b)** 1.2 to 3.6; **c)** $\frac{2}{3}$ m to 75 cm.

Solution

a) $\frac{45}{20} = \frac{9}{4}$. **b)** $\frac{1.2}{3.6} = \frac{12}{36} = \frac{1}{3}$.

c) Same unit first: $\frac{2}{3}$ m $= \frac{200}{3}$ cm.

Ratio: $\frac{200}{3} : 75 = \frac{200}{225} = \frac{8}{9}$.

Problem 15

Find $\frac{3}{4}$ of 56 kg.

Solution

$56 \times \frac{3}{4} = 42$ kg.

Answer: 42 kg.

Problem 16

How many minutes are $\frac{2}{5}$ of an hour?

Solution

1 hour = 60 minutes.

$60 \times \frac{2}{5} = 24$ minutes.

Answer: 24 minutes.

Problem 17

Class 5C has 40 pupils, and $\frac{3}{8}$ of them earn top marks. How many is that?

Solution

$40 \times \frac{3}{8} = 15$ pupils.

Answer: 15 pupils.

Problem 18

A roll of cloth is 24 m long. The shop sells $\frac{5}{8}$ of it. How many metres are left?

Solution

Sold: $24 \times \frac{5}{8} = 15$ m.

Left: $24 - 15 = 9$ m.

Answer: 9 m.

Problem 19

Mr Tu's trip to town is 72 km. He has covered $\frac{4}{9}$ of it. How far is left?

Solution

Remaining: $1 - \frac{4}{9} = \frac{5}{9}$ of the trip.

Distance left: $72 \times \frac{5}{9} = 40$ km.

Answer: 40 km.

Problem 20

$\frac{3}{7}$ of Hung's money is 42 000 dong. How much does he have?

Solution

$42\,000 : \frac{3}{7} = 42\,000 \times \frac{7}{3} = 98\,000$ dong.

Answer: 98 000 dong.

Problem 21

Find a number given that $\frac{5}{6}$ of it is 35.

Solution

$35 : \frac{5}{6} = 42$.

Answer: 42.

Problem 22

An has read $\frac{2}{5}$ of a book and has 90 pages still to read. How many pages does the book have?

Solution

90 pages are $1 - \frac{2}{5} = \frac{3}{5}$ of the book.

Pages: $90 : \frac{3}{5} = 150$.

Answer: 150 pages.

Problem 23

Find the ratio of: **a)** 15 to 25; **b)** 2 hours to 40 minutes; **c)** 1.2 m to 30 cm.

Solution

a) $\frac{15}{25} = \frac{3}{5}$.

b) Same unit: 2 hours = 120 minutes; $\frac{120}{40} = 3$.

c) 1.2 m = 120 cm; $\frac{120}{30} = 4$.

Answer: $\frac{3}{5}$; 3; 4.

Problem 24

Class 5D has 32 pupils, 14 of them girls. Find the ratio of girls to boys, and of girls to the whole class.

Solution

Boys: $32 - 14 = 18$.

Girls to boys: $\frac{14}{18} = \frac{7}{9}$.

Girls to class: $\frac{14}{32} = \frac{7}{16}$.

Answer: $\frac{7}{9}$; $\frac{7}{16}$.

Problem 25

A shop has 360 kg of rice. On day one it sells $\frac{1}{4}$ of it; on day two, $\frac{2}{5}$ of what is left. How much rice remains?

Solution

Day one: $360 \times \frac{1}{4} = 90$ kg; 270 kg left.

Day two: $270 \times \frac{2}{5} = 108$ kg.

Remaining: $270 - 108 = 162$ kg.

Answer: 162 kg.

Problem 26

Mai has 450 000 dong. She spends $\frac{2}{9}$ of it on books and $\frac{1}{3}$ of it on notebooks. How much is left?

Solution

Books: $450\,000 \times \frac{2}{9} = 100\,000$ dong.

Notebooks: $450\,000 \times \frac{1}{3} = 150\,000$ dong.

Left: $450\,000 - 100\,000 - 150\,000 = 200\,000$ dong.

Answer: 200 000 dong.

Problem 27

After $\frac{3}{8}$ of the water in a tank is used, 750 litres remain. How much water was there at first?

Solution

750 litres are $1 - \frac{3}{8} = \frac{5}{8}$ of the water.

At first: $750 : \frac{5}{8} = 1\,200$ litres.

Answer: 1 200 litres.

Problem 28

A number plus $\frac{1}{4}$ of itself is 60. Find the number.

Solution

The number plus a quarter of it is $\frac{5}{4}$ of the number.

The number: $60 : \frac{5}{4} = 48$.

Answer: 48.

Problem 29

A number minus $\frac{2}{7}$ of itself is 35. Find the number.

Solution

What is left is $1 - \frac{2}{7} = \frac{5}{7}$ of the number.

The number: $35 : \frac{5}{7} = 49$.

Answer: 49.

Problem 30

An, Binh and Cuong have 84 marbles. An has $\frac{1}{3}$ of them and Binh $\frac{3}{7}$ of them. How many does each have?

Solution

An: $84 \times \frac{1}{3} = 28$.

Binh: $84 \times \frac{3}{7} = 36$.

Cuong: $84 - 28 - 36 = 20$.

Answer: An 28, Binh 36, Cuong 20 marbles.

Challenge – Extension

Problem 31 *

Classes 5A and 5B have 68 pupils in total, and $\frac{3}{4}$ of class 5A equals $\frac{2}{3}$ of class 5B. How many pupils are in each class?

Solution

Make the numerators equal: $\frac{3}{4} = \frac{6}{8}$, $\frac{2}{3} = \frac{6}{9}$.

So $\frac{1}{8}$ of 5A equals $\frac{1}{9}$ of 5B.

5A is 8 equal parts and 5B is 9; 17 parts are 68 pupils, so a part is 4.

5A: 32 pupils; 5B: 36 pupils. (*Check:* $\frac{3}{4} \times 32 = 24 = \frac{2}{3} \times 36$.)

Problem 32 *

In a music group, the number of boys is $\frac{2}{3}$ of the number of girls. After 3 more boys join, the boys are $\frac{5}{6}$ of the girls. How many boys and girls were there at first?

Solution

The number of girls does not change, so compare everything with it.

The 3 new boys are $\frac{5}{6} - \frac{2}{3} = \frac{1}{6}$ of the girls.

Girls: $3 : \frac{1}{6} = 18$.

Boys at first: $18 \times \frac{2}{3} = 12$.

Answer: 12 boys and 18 girls.

Problem 33 ⋆

Mr An and Mrs Ha go to the market with 3 000 000 VND in total. Mr An spends $\frac{5}{9}$ of his money and Mrs Ha spends $\frac{3}{4}$ of hers; then they have equal amounts left. How much did each bring?

Solution

Mr An keeps $\frac{4}{9}$ and Mrs Ha keeps $\frac{1}{4}$ of their money, and these are equal.

With equal numerators: $\frac{4}{9}$ and $\frac{1}{4} = \frac{4}{16}$.

So Mr An's money is 9 equal parts and Mrs Ha's is 16; the 25 parts are 3 000 000 VND, so a part is 120 000 VND.

Mr An: 1 080 000 VND; Mrs Ha: 1 920 000 VND. (*Check:* each keeps 480 000 VND.)

Problem 34 ⋆⋆

Three teams plant trees. Team One plants $\frac{1}{3}$ of all the trees, Team Two plants $\frac{2}{5}$ of the rest, and Team Three plants 30 trees more than Team Two. How many trees are planted in all?

Solution

After Team One, $\frac{2}{3}$ of the trees are left.

Team Two plants $\frac{2}{5} \times \frac{2}{3} = \frac{4}{15}$; Team Three plants $\frac{2}{3} - \frac{4}{15} = \frac{6}{15}$ of all.

30 trees are $\frac{6}{15} - \frac{4}{15} = \frac{2}{15}$ of all.

All trees: $30 : \frac{2}{15} = 225$.

Answer: 225 trees.

Problem 35 ⋆⋆

A shop sells all its oranges in 4 days. Day one sells $\frac{1}{3}$ of what the next three days sell together. Day two sells $\frac{2}{3}$ of day one. Day three sells half of day four. Day four sells 15 kg more than day one. How many kilograms of oranges were there at first?

Solution

If day one is 1 part, the other three days are 3 parts, 4 in all: day one sells $\frac{1}{4}$ of the oranges.

Day two: $\frac{2}{3} \times \frac{1}{4} = \frac{1}{6}$.

Days three and four: $1 - \frac{1}{4} - \frac{1}{6} = \frac{7}{12}$, split $1 : 2$, so day four sells $\frac{7}{12} \times \frac{2}{3} = \frac{7}{18}$.

15 kg is $\frac{7}{18} - \frac{1}{4} = \frac{14-9}{36} = \frac{5}{36}$ of the oranges.

At first: $15 : \frac{5}{36} = 108$ kg.

Check: the four days sell 27, 18, 21, 42 *kg.*

Answer: 108 kg.

Problem 36 ⋆

Mum shares sweets: An gets $\frac{1}{3}$ of them, Binh gets $\frac{1}{4}$, and the remaining 15 go to little Hoa. How many sweets were there?

Solution

An and Binh get $\frac{1}{3} + \frac{1}{4} = \frac{7}{12}$ of the sweets.

So 15 is $1 - \frac{7}{12} = \frac{5}{12}$ of them.

Sweets: $15 : \frac{5}{12} = 36$.

Answer: 36 sweets.

Problem 37 ⋆

A seller sells $\frac{2}{5}$ of her oranges, then $\frac{1}{3}$ of the rest, then the last 24. How many oranges did she have?

Solution

After the first sale $\frac{3}{5}$ are left.

The second sale is $\frac{1}{3} \times \frac{3}{5} = \frac{1}{5}$ of all the oranges.

So 24 is $\frac{3}{5} - \frac{1}{5} = \frac{2}{5}$ of them: $24 : \frac{2}{5} = 60$.

Answer: 60 oranges.

Problem 38

Two classes have 76 pupils. $\frac{3}{5}$ of class A equals $\frac{2}{3}$ of class B. How many pupils are in each class?

Solution

Make the numerators equal: $\frac{3}{5} = \frac{6}{10}$, $\frac{2}{3} = \frac{6}{9}$.

$\frac{6}{10}$ of A equals $\frac{6}{9}$ of B, so A is 10 parts and B is 9 parts.

One part: $76 : 19 = 4$; A: 40; B: 36.

Answer: 40 and 36 pupils.

Problem 39

A walker covers $\frac{1}{4}$ of a route on day one, $\frac{2}{3}$ of the rest on day two and the last 15 km on day three. How long is the route?

Solution

After day one, $\frac{3}{4}$ is left.

Day three is $\frac{1}{3} \times \frac{3}{4} = \frac{1}{4}$ of the route.

Route: $15 : \frac{1}{4} = 60$ km.

Answer: 60 km.

Problem 40

Three numbers add up to 140. The first is $\frac{2}{3}$ of the second, and the second is $\frac{4}{5}$ of the third. Find them.

Solution

Let the third be 15 parts; the second is $15 \times \frac{4}{5} = 12$ parts and the first $12 \times \frac{2}{3} = 8$ parts.

35 parts in all; one part: $140 : 35 = 4$.

The numbers: 32, 48, 60.

Answer: 32; 48; 60.

Problem 41

A drum holds oil. After $\frac{1}{3}$ of the oil is taken out, the drum weighs 32 kg. After another $\frac{1}{4}$ of the

original oil is taken out, it weighs 23 kg. How heavy is the empty drum?

Solution

$\frac{1}{4}$ of the oil weighs $32 - 23 = 9$ kg, so the oil was 36 kg.

After the first step $36 \times \frac{2}{3} = 24$ kg remain.

Drum: $32 - 24 = 8$ kg.

Answer: 8 kg.

Problem 42 *

In a class $\frac{3}{4}$ of the pupils like Maths and $\frac{2}{3}$ like Vietnamese; 10 like both, and everyone likes at least one. How many pupils are there?

Solution

Adding the two groups counts the 10 twice: $\frac{3}{4} + \frac{2}{3} = \frac{17}{12}$ of the class.

The extra $\frac{17}{12} - 1 = \frac{5}{12}$ of the class is 10 pupils.

Class: $10 : \frac{5}{12} = 24$.

Answer: 24 pupils.

Problem 43 *

A number is increased by $\frac{1}{5}$ of itself; the new number is then decreased by $\frac{1}{6}$ of itself. How does the result compare with the original?

Solution

After the increase the number is $\frac{6}{5}$ of the original.

After the decrease it is $\frac{5}{6}$ of that, i.e.

$\frac{5}{6} \times \frac{6}{5} = 1$ times the original.

The result equals the original.

Answer: It is equal to the original.

Problem 44

Two barns hold 380 tonnes of paddy. Taking $\frac{2}{5}$ from barn A and $\frac{1}{3}$ from barn B leaves equal amounts. How much did each hold?

Solution

What is left: $\frac{3}{5}$ of A equals $\frac{2}{3}$ of B, i.e.

$\frac{6}{10}$ of A equals $\frac{6}{9}$ of B.

A is 10 parts, B is 9; one part: $380 : 19 = 20$.

A: 200 t; B: 180 t.

Answer: 200 t and 180 t.

Problem 45

A shop sells $\frac{1}{3}$ of a roll of cloth and 2 m more, then half of the rest and 3 m more, leaving 7 m. How long was the roll?

Solution

Before day two: $(7 + 3) \times 2 = 20$ m.

$20 + 2 = 22$ m is $\frac{2}{3}$ of the roll.

Roll: $22 : \frac{2}{3} = 33$ m.

Answer: 33 m.

Problem 46

Two people have the same amount of money. One spends $\frac{3}{5}$ of it and the other $\frac{2}{3}$; together they have 330 000 dong left. How much did each have?

Solution

They keep $\frac{2}{5}$ and $\frac{1}{3}$ of their money.

Together that is $\frac{2}{5} + \frac{1}{3} = \frac{11}{15}$ of one person's money, i.e.

330 000 dong.

Each had $330\,000 : \frac{11}{15} = 450\,000$ dong.

Answer: 450 000 dong.

Problem 47 ★★

An gives Binh $\frac{1}{4}$ of his marbles and 3 more, then gives Cuong half of the rest and 1 more, keeping 8. How many did An have?

Solution

Before giving to Cuong: $(8+1)\times 2 = 18$.

$18+3=21$ is $\frac{3}{4}$ of the original.

At first: $21 : \frac{3}{4} = 28$.

Answer: 28 marbles.

Problem 48 ★★

Lan should find $\frac{3}{4}$ of a number but finds $\frac{4}{3}$ of it instead, getting 14 too much. Find the number and the correct answer.

Solution

The wrong result exceeds the right one by $\frac{4}{3}-\frac{3}{4}=\frac{7}{12}$ of the number.

The number: $14 : \frac{7}{12} = 24$.

Correct answer: $24\times\frac{3}{4}=18$.

Answer: 24; 18.

Problem 49 ★★

The top shelf holds $\frac{2}{3}$ as many books as the bottom shelf. Moving 10 books up makes the shelves equal. How many books are on each?

Solution

The total is fixed.

The top shelf is $\frac{2}{5}$ of it before and $\frac{1}{2}$ after.

So 10 books are $\frac{1}{2}-\frac{2}{5}=\frac{1}{10}$ of the total, which is 100.

Top: 40; bottom: 60.

Answer: 40 and 60 books.

Problem 50 ★★

An egg seller sells $\frac{2}{5}$ of the eggs in the morning and $\frac{3}{4}$ of the rest in the afternoon, with 30 left at the end. How many eggs were there?

Solution

After the morning $\frac{3}{5}$ are left.

At the end $\frac{1}{4} \times \frac{3}{5} = \frac{3}{20}$ are left, i.e.

30 eggs.

Eggs: $30 : \frac{3}{20} = 200$.

Answer: 200 eggs.

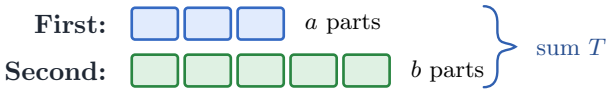
5 Sum–Ratio and Difference–Ratio Problems

① The ratio of two numbers

- The ratio of a to b is $a : b$ or $\frac{a}{b}$ ($b \neq 0$; a and b in the same unit), usually in lowest terms.
- A ratio may be given as a decimal or as "times": $0.75 = \frac{3}{4}$; $1.5 = \frac{3}{2}$; "a is 3 times b" means $\frac{a}{b} = \frac{3}{1}$.
- "$\frac{1}{3}$ of the first number equals $\frac{1}{6}$ of the second" means the first is 3 equal parts and the second 6 such parts: the ratio is $\frac{3}{6} = \frac{1}{2}$.

② Two problem types, solved with bar models

Sum–ratio: find two numbers given their sum T and their ratio $\frac{a}{b}$.



Step 1: total number of equal parts $a + b$. Step 2: one part $= T : (a + b)$. Step 3: first number = one part $\times a$; second number = $T -$ first.

Difference–ratio: find two numbers given their difference H and ratio $\frac{a}{b}$ ($a > b$). Work the same way, but H matches the *difference in parts* $a - b$; one part is $H : (a - b)$.

Note: draw equal parts with equal lengths, and read carefully to see *whether the sum or the difference stays the same* when the numbers change (for instance, the age gap between two people never changes).

Example 1

A rectangular stadium has perimeter 400 m, and its width is $\frac{3}{5}$ of its length. Find its area.

Guided solution.

Half the perimeter (length + width): $400 : 2 = 200$ (m).



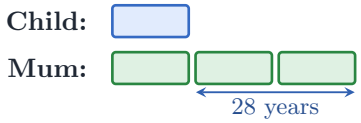
Total parts: $3 + 5 = 8$; one part: $200 : 8 = 25$ (m).
Width: $25 \times 3 = 75$ (m); length: $200 - 75 = 125$ (m).
Area: $125 \times 75 = 9\,375$ (m^2).

Example 2

Mum is 28 years older than her child. In 4 years she will be 3 times as old as the child. How old is each of them now?

Guided solution.

The age gap never changes; in 4 years Mum is still 28 years older.
Ages in 4 years:



Difference in parts: $3 - 1 = 2$.
The child's age in 4 years: $28 : 2 = 14$.
The child now: $14 - 4 = 10$; Mum now: $10 + 28 = 38$.

Example 3

The average of two natural numbers is 135, and $\frac{1}{3}$ of the first equals $\frac{1}{6}$ of the second. Find the two numbers.

Guided solution.

Sum: $135 \times 2 = 270$.
As $\frac{1}{3}$ of the first equals $\frac{1}{6}$ of the second, the first is 3 parts and the second 6 such parts.
Total parts: $3 + 6 = 9$; one part: $270 : 9 = 30$.
First: $30 \times 3 = 90$; second: $270 - 90 = 180$.

Example 4

Two numbers add up to 84. The second is 0.75 of the first. Find them.

Guided solution.

Change $0.75 = \frac{3}{4}$: the second is 3 parts, the first 4 parts; 7 parts in all; one part: $84 : 7 = 12$.
First: $12 \times 4 = 48$; second: $84 - 48 = 36$.

Example 5

Find two numbers whose difference is 24 and whose ratio (smaller to larger) is 0.7.

Guided solution.

$0.7 = \frac{7}{10}$: smaller 7 parts, larger 10 parts; difference 3 parts $= 24$, so one part is 8.
Smaller: $8 \times 7 = 56$; larger: $56 + 24 = 80$.

Practice

Problem 1

The average of two numbers is 168, and their ratio is $\frac{5}{9}$. Find them.

Solution

Sum: $168 \times 2 = 336$.
Parts: $5 + 9 = 14$; one part: $336 : 14 = 24$.
Numbers: $24 \times 5 = 120$ and $336 - 120 = 216$.
Answer: 120 and 216.

Problem 2

Find two numbers with sum 252 such that 8 times the smaller is the larger.

Solution

Smaller 1 part, larger 8 parts; 9 parts in all.
Smaller: $252 : 9 = 28$; larger: $28 \times 8 = 224$.
Answer: 28 and 224.

Problem 3

In an exact division, the dividend plus the divisor is 342 and the quotient is 8. Find the dividend and the divisor.

Solution

The quotient is 8, so the dividend is 8 times the divisor: divisor 1 part, dividend 8 parts, 9 in all.
Divisor: $342 : 9 = 38$; dividend: $38 \times 8 = 304$.

Answer: dividend 304, divisor 38.

Problem 4

Two numbers add up to 286. Writing 0 on the right of the first number gives the second. Find them.

Solution

Writing 0 on the right multiplies by 10: first 1 part, second 10 parts, 11 in all.
First: $286 : 11 = 26$; second: 260.
Answer: 26 and 260.

Problem 5

Barn A holds 180 tonnes of rice more than barn B. If barn B receives 30 more tonnes, then barn A holds $\frac{6}{5}$ of barn B. How much did each barn hold at first?

Solution

After barn B receives the rice, A has $180 - 30 = 150$ tonnes more, matching $6 - 5 = 1$ part.
Barn B then: $150 \times 5 = 750$ t; at first: $750 - 30 = 720$ t; barn A: $720 + 180 = 900$ t.
Answer: A 900 t, B 720 t.

Problem 6

Mum and her child are 46 years old in total. In 4 years the child's age will be $\frac{2}{7}$ of Mum's. How old is each now?

Solution

In 4 years each is 4 years older; the total is $46 + 4 \times 2 = 54$.
Parts: $2 + 7 = 9$; one part: $54 : 9 = 6$.
Then the child is 12 and Mum 42.
Now: child $12 - 4 = 8$, Mum $42 - 4 = 38$.
Answer: child 8, Mum 38.

Problem 7

What natural number must be added to both the numerator and the denominator of $\frac{13}{29}$ to get a fraction equal to $\frac{3}{5}$?

Solution

Adding the same number to both keeps the *difference* between denominator and numerator: $29 - 13 = 16$.
The new fraction has numerator 3 parts and denominator 5; the 2-part difference is 16, so a part is 8: the new fraction is $\frac{24}{40}$.
The number to add: $24 - 13 = 11$.
Answer: 11.

Problem 8

Subtract a natural number from the numerator of $\frac{11}{25}$ and add the same number to its denominator, to get a fraction equal to $\frac{1}{5}$. Find the number.

Solution

Moving the same amount from numerator to denominator keeps the *sum*: $11 + 25 = 36$.
New fraction: numerator 1 part, denominator 5; 6 parts $= 36$, so a part is 6: the new fraction is $\frac{6}{30}$.
The number: $11 - 6 = 5$.
Answer: 5.

Problem 9

Two sacks hold 160 kg of rice together. Moving 6 kg from the first sack to the second makes the first $\frac{3}{5}$ of the second. How much was in each sack at first?

Solution

Moving rice keeps the total (160 kg).
Afterwards: $3 + 5 = 8$ parts; one part: $160 : 8 = 20$ kg.
First sack afterwards 60 kg, so at first $60 + 6 = 66$ kg; second sack at first: $160 - 66 = 94$ kg.
Answer: 66 kg and 94 kg.

Problem 10

A rectangle has perimeter 180 cm. Adding 6 cm to its length and taking 6 cm off its width gives a rectangle whose length is twice its width. Find the area of the original rectangle.

Solution

Half the perimeter: 90 cm, and it stays 90 cm after the change.
New rectangle: width 1 part, length 2 parts; a part is $90 : 3 = 30$ cm: width 30 cm, length 60 cm.
Original: width $30 + 6 = 36$ cm, length $60 - 6 = 54$ cm; area $54 \times 36 = 1\,944$ cm^2.
Answer: $1\,944$ cm^2.

Problem 11

Grade 5 of a school has 315 pupils, and $\frac{1}{4}$ of the girls equals $\frac{1}{5}$ of the boys. How many girls and boys are there?

Solution

Girls are 4 equal parts and boys 5; 9 parts in all; one part: $315 : 9 = 35$.
Girls: $35 \times 4 = 140$; boys: $35 \times 5 = 175$.
Answer: 140 girls, 175 boys.

Problem 12

Find two numbers whose difference is the largest two-digit number and whose ratio is $\frac{4}{7}$.

Solution

The difference is 99.
Difference in parts: $7 - 4 = 3$; one part: $99 : 3 = 33$.
Numbers: $33 \times 4 = 132$ and $33 \times 7 = 231$.
Answer: 132 and 231.

Problem 13

Dad is 32 years older than his son. This year Dad is 5 times as old as the son. How old is the son?

Solution

Difference in parts: $5 - 1 = 4$; the son: $32 : 4 = 8$.
Answer: 8 years old.

Problem 14

Find two numbers whose difference is 516, given that the larger divided by the smaller gives quotient 7 and remainder 12.

Solution

The larger is 7 times the smaller plus 12.
Taking 12 off it makes it exactly 7 times the smaller, and the difference becomes $516 - 12 = 504$.
Difference in parts $7 - 1 = 6$; smaller: $504 : 6 = 84$; larger: $84 + 516 = 600$.
Answer: 84 and 600.

Problem 15

Find a number such that deleting the 0 at its right end makes it 1 845 smaller.

Solution

Deleting the final 0 divides the number by 10: new 1 part, old 10 parts; the 9-part difference is 1 845.
New number: $1\,845 : 9 = 205$; the number is 2 050.
Answer: 2 050.

Problem 16

The difference of two numbers is $\frac{1}{4}$ of the smaller number. Their sum is 162. Find them.

Solution

If the smaller is 4 parts, the difference is 1 part and the larger 5 parts.
Sum 9 parts; one part: $162 : 9 = 18$.
Smaller: 72; larger: 90.
Answer: 72 and 90.

Problem 17

Find two numbers with sum 12.4, the first being $\frac{2}{3}$ of the second.

Solution

Parts: $2 + 3 = 5$; one part: $12.4 : 5 = 2.48$.
First: $2.48 \times 2 = 4.96$; second: $12.4 - 4.96 = 7.44$.
Answer: 4.96 and 7.44.

Problem 18

Grade 5 has 288 pupils. The ratio of girls to boys is 0.6. How many girls and boys are there?

Solution

$0.6 = \frac{3}{5}$.

Parts: $3 + 5 = 8$; one part: $288 : 8 = 36$.

Girls: $36 \times 3 = 108$; boys: $36 \times 5 = 180$.

Answer: 108 girls, 180 boys.

Problem 19

A rectangular plot has perimeter 168 m; its length is 1.8 times its width. Find its area.

Solution

$1.8 = \frac{9}{5}$: width 5 parts, length 9 parts.

Half the perimeter, 84 m, is 14 parts; one part: 6 m.

Width 30 m, length 54 m; area $54 \times 30 = 1\,620$ m^2.

Answer: $1\,620$ m^2.

Problem 20

Two numbers add up to 150 and are in the ratio $\frac{2}{3}$. Find them.

Solution

$2 + 3 = 5$ parts.

One part: $150 : 5 = 30$.

The numbers: $30 \times 2 = 60$ and $30 \times 3 = 90$.

Answer: 60 and 90.

Problem 21

Two numbers differ by 45 and are in the ratio $\frac{3}{8}$. Find them.

Solution

$8 - 3 = 5$ parts.

One part: $45 : 5 = 9$.

Smaller: 27; larger: 72.

Answer: 27 and 72.

Problem 22

Two tanks hold 120 litres; the first holds $\frac{3}{5}$ as much as the second. How much does each hold?

Solution

8 parts; one part: $120 : 8 = 15$ litres.
First: 45 l; second: 75 l.
Answer: 45 l and 75 l.

Problem 23

A rectangular garden has perimeter 96 m, and its width is $\frac{1}{3}$ of its length. Find its area.

Solution

Half the perimeter: 48 m.
Width 1 part, length 3 parts: width $48 : 4 = 12$ m, length 36 m.
Area: $36 \times 12 = 432$ m^2.
Answer: 432 m^2.

Problem 24

This year a child's age is $\frac{2}{7}$ of the mother's, and she is 25 years older. Find both ages.

Solution

$7 - 2 = 5$ parts; one part: $25 : 5 = 5$ years.
Child: 10; mother: 35.
Answer: Child 10, mother 35 years.

Problem 25

The larger number is 5 times the smaller, and they differ by 72. Find them.

Solution

$5 - 1 = 4$ parts; smaller: $72 : 4 = 18$.
Larger: $18 \times 5 = 90$.
Answer: 18 and 90.

Problem 26

Two numbers add up to 7.2; the first is 3 times the second. Find them.

Solution

4 parts; second: $7.2 : 4 = 1.8$.
First: $1.8 \times 3 = 5.4$.
Answer: 5.4 and 1.8.

Problem 27

Class 5E has 35 pupils, with boys to girls in the ratio $\frac{3}{4}$. How many boys and girls are there?

Solution

7 parts; one part: $35 : 7 = 5$.
Boys: 15; girls: 20.
Answer: 15 boys, 20 girls.

Problem 28

Two numbers have average 60; the smaller is $\frac{1}{4}$ of the larger. Find them.

Solution

Sum: $60 \times 2 = 120$.
5 parts; smaller: $120 : 5 = 24$; larger: 96.
Answer: 24 and 96.

Problem 29

Two numbers differ by 30; the larger is 2.5 times the smaller. Find them.

Solution

The smaller is 1 part and the larger 2.5 parts; the difference is 1.5 parts.
Smaller: $30 : 1.5 = 20$; larger: 50.
Answer: 20 and 50.

Problem 30

A rectangular garden is 15 m longer than it is wide, and its width is $\frac{2}{5}$ of its length. Find its area.

Solution

3 parts; one part: $15 : 3 = 5$ m.
Width: 10 m; length: 25 m.

Area: 250 m^2.
Answer: 250 m^2.

Challenge – Extension

Problem 31 *

Two cans hold 160.5 litres of oil. After 9.5 litres are poured from the first can into the second, the second holds 1.5 times as much as the first. How much did each can hold at first?

Solution

The total stays the same.

Afterwards: first can 2 parts, second 3 parts ($1.5 = \frac{3}{2}$); one part: $160.5 : 5 = 32.1$ litres.

First can afterwards: 64.2 l; at first $64.2 + 9.5 = 73.7$ l.

Second can at first: $160.5 - 73.7 = 86.8$ l.

Answer: 73.7 l and 86.8 l.

Problem 32 *

The first sack has 20.4 kg more rice than the second. If 1.8 kg is added to each sack, the first will hold 3 times as much as the second. How much did each hold at first?

Solution

Adding the same amount to each keeps the difference (20.4 kg).

Afterwards the difference is $3 - 1 = 2$ parts; the second sack then holds $20.4 : 2 = 10.2$ kg.

Second sack at first: $10.2 - 1.8 = 8.4$ kg; first sack at first: $8.4 + 20.4 = 28.8$ kg.

Answer: 28.8 kg and 8.4 kg.

Problem 33 *

The son's age is $\frac{1}{4}$ of his dad's. In 10 years it will be 0.4 of his dad's. How old is each now?

Solution

The age gap stays the same.

Now: son 1 part, dad 4, gap 3 parts.

In 10 years ($0.4 = \frac{2}{5}$): son 2 parts, dad 5, gap also 3 parts.

The gaps are equal, so the parts are the same size: the son grows from 1 part to 2 parts, so 1 part = 10 years.

Now: son 10, dad 40. (*Check:* $20 : 50 = 0.4$.)

Problem 34 ★★

In spring tree planting, Grade 4 plants $\frac{3}{5}$ as many trees as Grade 5. If each grade planted 20 more, Grade 4 would have $\frac{2}{3}$ as many as Grade 5. How many trees did each grade plant?

Solution

The difference stays the same.

At first: Grade 4 has 3 parts, Grade 5 has 5, difference 2.

Afterwards: $\frac{2}{3} = \frac{4}{6}$, so 4 and 6 parts, difference still 2 parts of the same size.

Grade 4 goes from 3 parts to 4, so 1 part = 20 trees: Grade 4 planted 60 and Grade 5 planted 100.

Answer: Grade 4: 60; Grade 5: 100.

Problem 35 ★★

The ratio of books on the top shelf to the bottom shelf is 0.5. Moving 9 books from the bottom to the top makes the ratio 0.8. How many books were on each shelf at first?

Solution

The total stays the same.

At first the top shelf holds $\frac{1}{1+2} = \frac{1}{3}$ of all books; afterwards ($0.8 = \frac{4}{5}$) it holds $\frac{4}{4+5} = \frac{4}{9}$.

The 9 books are $\frac{4}{9} - \frac{1}{3} = \frac{1}{9}$ of the total, so the total is $9 : \frac{1}{9} = 81$.

Top shelf at first: $81 : 3 = 27$; bottom: 54.

Answer: top 27, bottom 54.

Problem 36 ★

Three numbers add up to 210. The first is $\frac{1}{2}$ of the second, and the third is twice the second. Find them.

Solution

First 1 part, second 2 parts, third 4 parts; 7 parts in all.

One part: $210 : 7 = 30$.

The numbers: 30, 60, 120.

Answer: 30; 60; 120.

Problem 37

Two numbers add up to 357. Crossing out the units digit 5 of the larger gives the smaller. Find them.

Solution

The larger is 10 times the smaller plus 5.
So 11 times the smaller plus 5 is 357; 11 times the smaller is 352.
Smaller: 32; larger: 325.
Answer: 32 and 325.

Problem 38

Two numbers differ by 33.3. Moving the point of the smaller one place right gives the larger. Find them.

Solution

The larger is 10 times the smaller; difference 9 parts.
Smaller: $33.3 : 9 = 3.7$; larger: 37.
Answer: 3.7 and 37.

Problem 39

A father is 30 years older than his son. In 5 years he will be 3 times as old. How old is each now?

Solution

The age gap stays 30.
In 5 years: $3 - 1 = 2$ parts are 30 years.
The son will be 15, so he is 10 now.
The father is 40.
Answer: Son 10, father 40 years.

Problem 40

A grandfather and grandchild have ages totalling 78. The child's age in months equals the grandfather's age in years. Find both ages.

Solution

A year has 12 months, so the grandfather is 12 times as old.
13 parts; child: $78 : 13 = 6$.
Grandfather: 72.
Answer: Child 6, grandfather 72 years.

Problem 41

An has 3 times as many marbles as Binh. If An gives Binh 12, they will have the same. How many does each have?

Solution

For giving 12 to make them equal, An must have 24 more.
2 parts are 24; Binh: 12; An: 36.
Answer: An 36, Binh 12 marbles.

Problem 42

Barn A holds 3 times as much paddy as barn B. Moving 20 tonnes from A to B makes A hold twice as much as B. How much does each hold?

Solution

The total is fixed.
A is $\frac{3}{4}$ of it before and $\frac{2}{3}$ after.
So 20 t is $\frac{3}{4} - \frac{2}{3} = \frac{1}{12}$ of the total, which is 240 t.
A: 180 t; B: 60 t.
Answer: 180 t and 60 t.

Problem 43

Two numbers add up to 216. Dividing the larger by the smaller gives 4 remainder 6. Find them.

Solution

The larger is 4 times the smaller plus 6, so the sum is 5 times the smaller plus 6.
Smaller: $(216 - 6) : 5 = 42$.
Larger: $216 - 42 = 174$.
Answer: 42 and 174.

Problem 44

Two numbers differ by 87. Dividing the larger by the smaller gives 3 remainder 5. Find them.

Solution

The larger is 3 times the smaller plus 5, so the difference is 2 times the smaller plus 5.
Smaller: $(87 - 5) : 2 = 41$.
Larger: $41 + 87 = 128$.
Answer: 41 and 128.

Problem 45 *

A fraction's numerator and denominator add up to 120. Moving 10 from the denominator to the numerator gives a fraction equal to 1. Find it.

Solution

The sum stays 120; afterwards top and bottom are equal, 60 each.

At first: numerator $60 - 10 = 50$, denominator $60 + 10 = 70$.

Answer: $\frac{50}{70}$.

Problem 46 *

Adding 5 m to the width of a rectangle makes it a square and adds 60 m^2. Find the rectangle's area.

Solution

The added strip is 5 m wide and as long as the rectangle: length $60 : 5 = 12$ m.

Width: $12 - 5 = 7$ m.

Area: $12 \times 7 = 84$ m^2.

Answer: 84 m^2.

Problem 47 **

A mother is 36 and her child 8. In how many years will she be 3 times as old?

Solution

The gap is always 28.

When she is 3 times as old, the gap is 2 parts.

The child will be $28 : 2 = 14$.

Years: $14 - 8 = 6$.

Answer: 6 years.

Problem 48 **

In an exact division the dividend, divisor and quotient add up to 195, and the quotient is 6. Find the dividend and divisor.

Solution

Dividend plus divisor: $195 - 6 = 189$.

The dividend is 6 times the divisor: 7 parts.

Divisor: $189 : 7 = 27$; dividend: 162.

Answer: 162 and 27.

Problem 49

A shop's sticky rice is $\frac{2}{5}$ of its plain rice. After 30 kg of plain rice is sold, the sticky rice is $\frac{1}{2}$ of the plain rice left. How much of each was there?

Solution

The sticky rice is fixed; call it 2 parts.
Plain rice: 5 parts before, 4 parts after.
So 30 kg is 1 part: sticky 60 kg, plain 150 kg.
Answer: Sticky 60 kg, plain 150 kg.

Problem 50

Two whole numbers add up to 1 001. The larger ends in 0, and crossing out that 0 gives the smaller. Find them.

Solution

The larger is 10 times the smaller; 11 parts in all.
Smaller: $1\,001 : 11 = 91$; larger: 910.
Answer: 91 and 910.

6 (*) Problems with Two Ratios

① Look for the quantity that does not change

In many problems two numbers change and so does their ratio (the problem gives *two ratios*: before and after). The key is to find a quantity that **does not change** and use it as the "unit" of comparison:

- The **sum is constant** when an amount moves from one number to the other. Write each number as a *fraction of the sum*.
- The **difference is constant** when the same amount is added to (or taken from) both; age gaps are an example. Write each number as a *fraction of the difference*.
- **One number is constant** when only the other changes. Compare everything with the unchanged number.

The known change then matches the *difference of two fractions*, which gives the unchanged quantity.

Example 1

Find two numbers in the ratio $\frac{3}{7}$ such that adding 6 to the first and taking 6 from the second gives two numbers in the ratio $\frac{2}{3}$.

Guided solution.

Whatever is added to one is taken from the other, so **the sum is constant**.

At first the first number is $\frac{3}{3+7} = \frac{3}{10}$ of the sum; afterwards $\frac{2}{2+3} = \frac{2}{5} = \frac{4}{10}$ of the sum.

The first number grows by 6, which is $\frac{4}{10} - \frac{3}{10} = \frac{1}{10}$ of the sum.

Sum: $6 : \frac{1}{10} = 60$.

First: $60 \times \frac{3}{10} = 18$; second: $60 - 18 = 42$. (*Check:* $24 : 36 = \frac{2}{3}$.)

Example 2

The first number is $\frac{3}{8}$ of the second. Taking 5 from each makes the second 3 times the first. Find the numbers.

Guided solution.

The same amount is taken from both, so **the difference is constant**.

At first: first 3 parts, second 8 parts, difference 5 parts; the first is $\frac{3}{5}$ of the difference.

Afterwards: first 1 part, second 3 parts, difference 2 parts; the first is $\frac{1}{2}$ of the difference.

The 5 is $\frac{3}{5} - \frac{1}{2} = \frac{1}{10}$ of the difference.

Difference: $5 : \frac{1}{10} = 50$.

First: $50 \times \frac{3}{5} = 30$; second: 80.

Example 3

On a bus there are 3 times as many men as women. At the first stop 3 men get on and 3 women get off; now there are 5 times as many men as women. How many men and women were there at first?

Guided solution.

3 get on and 3 get off, so **the total is constant**.

Women were $\frac{1}{1+3} = \frac{1}{4}$ of the total, and later $\frac{1}{1+5} = \frac{1}{6}$.

The 3 women who left are $\frac{1}{4} - \frac{1}{6} = \frac{1}{12}$ of the total.

Total: $3 : \frac{1}{12} = 36$.

At first: 9 women and 27 men.

Example 4

If Nam gives Giang 3 sweets, Nam will have 2 times as many as Giang. If instead Giang gives Nam 3 sweets, Nam will have 5 times as many. How many sweets does each have?

Guided solution.

Giving sweets keeps **the total T constant**.

– Case 1: after giving, Nam has $\frac{2}{2+1} = \frac{2}{3}T$; so Nam has $\frac{2}{3}T + 3$.

– Case 2: after receiving, Nam has $\frac{5}{5+1} = \frac{5}{6}T$; so Nam has $\frac{5}{6}T - 3$.

Comparing: $\frac{5}{6}T - \frac{2}{3}T = 3 + 3$, i.e.

$\frac{1}{6}T = 6$ and $T = 36$.

Nam has $\frac{2}{3} \times 36 + 3 = 27$; Giang has $36 - 27 = 9$.

Practice

Problem 1

Find two numbers in the ratio $\frac{2}{3}$ such that adding 4 to the first and taking 4 from the second gives the ratio $\frac{3}{2}$.

Solution

The sum is constant.

The first number: $\frac{2}{5}$ of the sum before, $\frac{3}{5}$ after.

So 4 is $\frac{1}{5}$ of the sum, and the sum is 20.

The numbers: 8 and 12.

Answer: 8 and 12.

Problem 2

A fraction equals $\frac{2}{5}$. Adding 9 to its numerator and taking 9 from its denominator gives a fraction equal to $\frac{5}{2}$. Find the original fraction.

Solution

The numerator plus denominator is constant.

The numerator is $\frac{2}{7}$ of it before, $\frac{5}{7}$ after; so 9 is $\frac{3}{7}$ of the sum, which is 21.

The original fraction: $\frac{6}{15}$.

Answer: $\frac{6}{15}$.

Problem 3

The ratio of books on shelf I to shelf II is $\frac{5}{7}$. Moving 10 books from shelf I to shelf II makes shelf II hold 3 times as many as shelf I. How many books were on each shelf at first?

Solution

The total is constant.

Shelf I is $\frac{5}{12}$ of it before and $\frac{1}{4} = \frac{3}{12}$ after.

The 10 books are $\frac{2}{12} = \frac{1}{6}$ of the total, which is 60.
Shelf I: 25; shelf II: 35.
Answer: 25 and 35 books.

Problem 4

Class 5A has twice as many boys as girls. If 3 girls join and 3 boys leave, the girls make up $\frac{5}{12}$ of the class. How many pupils are in class 5A?

Solution

The class size stays the same.
Girls are $\frac{1}{3} = \frac{4}{12}$ of it before and $\frac{5}{12}$ after, so 3 pupils are $\frac{1}{12}$ of the class.
Class size: $3 \times 12 = 36$.
Answer: 36 pupils.

Problem 5

The first number is $\frac{2}{7}$ of the second. Adding 12 to each makes the first $\frac{1}{2}$ of the second. Find the numbers.

Solution

The difference is constant.
The first number is $\frac{2}{7-2} = \frac{2}{5}$ of the difference before and $\frac{1}{2-1} = 1$ difference after.
So 12 is $1 - \frac{2}{5} = \frac{3}{5}$ of the difference, which is 20.
The numbers: 8 and 28.
Answer: 8 and 28.

Problem 6

This year the son's age is $\frac{1}{3}$ of the age gap between him and his dad. Six years ago it was $\frac{1}{5}$ of the gap. How much older is the dad?

Solution

The age gap is constant.
The 6 years are $\frac{1}{3} - \frac{1}{5} = \frac{2}{15}$ of the gap.

The gap: $6 : \frac{2}{15} = 45$ years. (The son is 15 and the dad 60.)

Answer: 45 years.

Problem 7

The masses of corn in two sacks are in the ratio $\frac{3}{8}$. Adding 4 kg to each makes the ratio $\frac{2}{5}$. How much corn was in each sack at first?

Solution

The difference is constant.

The smaller sack is $\frac{3}{5}$ of it before and $\frac{2}{3}$ after; 4 kg is $\frac{2}{3} - \frac{3}{5} = \frac{1}{15}$ of the difference, which is 60 kg.

Smaller sack: $60 \times \frac{3}{5} = 36$ kg; larger: $36 + 60 = 96$ kg.

Answer: 36 kg and 96 kg.

Problem 8

Mum is now 5 times as old as her child. In six years she will be 3 times as old. How old is each now?

Solution

The gap is constant.

The child is $\frac{1}{4}$ of it now and $\frac{1}{2}$ of it in six years; 6 years is $\frac{1}{4}$ of the gap, which is 24 years.

Child: $24 : 4 = 6$; Mum: 30.

Answer: child 6, Mum 30.

Problem 9

A fraction equals $\frac{3}{4}$. Taking 6 from both its numerator and denominator gives a fraction equal to $\frac{5}{7}$. Find the original fraction.

Solution

Denominator minus numerator is constant.

The numerator is 3 times it before and $\frac{5}{2}$ of it after; 6 is $3 - \frac{5}{2} = \frac{1}{2}$ of it, so it is 12.

Numerator 36, denominator 48: the fraction $\frac{36}{48}$.

Answer: $\frac{36}{48}$.

Problem 10

In class 5A the boys are $\frac{3}{4}$ of the girls. If 3 boys and 3 girls leave, the boys make up $\frac{5}{12}$ of the class. How many pupils were there at first?

Solution

Both lose 3, so (girls − boys) is constant.

At first boys 3 parts, girls 4: the boys are 3 times the difference.

Afterwards boys 5 parts, girls 7: the boys are $\frac{5}{2}$ of the difference.

3 pupils are $\frac{1}{2}$ of it, so it is 6.

At first: 18 boys, 24 girls, 42 pupils.

Answer: 42 pupils.

Problem 11

There are two barns. If 3 tonnes are added to barn A and 3 tonnes removed from barn B, the ratio of rice is $\frac{3}{7}$. If instead 5 tonnes are removed from A and 5 added to B, the ratio is $\frac{1}{4}$. How much does each barn hold?

Solution

In both cases the total T is unchanged.

Barn A then holds $\frac{3}{10}\,T$ (case 1) or $\frac{1}{5}\,T = \frac{2}{10}\,T$ (case 2).

The two cases differ by $3 + 5 = 8$ tonnes, which is $\frac{1}{10}\,T$; so $T = 80$ tonnes.

Barn A: $80 \times \frac{3}{10} - 3 = 21$ t; barn B: $80 - 21 = 59$ t.

Answer: A 21 t, B 59 t.

Problem 12

A fraction equals $\frac{5}{8}$. Adding 6 to both its numerator and denominator gives a fraction equal to $\frac{2}{3}$. Find the original fraction.

Solution

The difference is constant.

The numerator is $\frac{5}{3}$ of it before and 2 times it after; 6 is $\frac{1}{3}$ of it, so it is 18.

The original fraction: $\frac{30}{48}$.

Answer: $\frac{30}{48}$.

Problem 13

Two numbers are in the ratio $\frac{1}{2}$. Taking 5 from the second and adding it to the first gives the ratio $\frac{4}{5}$. Find them.

Solution

The sum is fixed.

The first is $\frac{1}{3} = \frac{3}{9}$ of it before and $\frac{4}{9}$ after.

So 5 is $\frac{1}{9}$ of the sum, which is 45.

The numbers: 15 and 30.

Answer: 15 and 30.

Problem 14

Drum I holds $\frac{3}{5}$ as much oil as drum II. Pouring 6 litres from II into I makes them equal. How much does each hold?

Solution

The total is fixed.

Drum I is $\frac{3}{8}$ of it before and $\frac{1}{2} = \frac{4}{8}$ after.

So 6 l is $\frac{1}{8}$ of the total, which is 48 l.

Drum I: 18 l; drum II: 30 l.

Answer: 18 l and 30 l.

Problem 15

A fraction equals $\frac{3}{7}$. Moving 4 from the denominator to the numerator gives a fraction equal to 1. Find it.

Solution

Top plus bottom is fixed.

The numerator is $\frac{3}{10}$ of it before and $\frac{1}{2} = \frac{5}{10}$ after.

So 4 is $\frac{2}{10}$ of the sum, which is 20.

The fraction: $\frac{6}{14}$.

Answer: $\frac{6}{14}$.

Problem 16

Barns I and II hold paddy in the ratio $\frac{7}{9}$. Moving 10 tonnes from I to II gives the ratio $\frac{3}{5}$. How much did each hold?

Solution

The total is fixed.

Barn I is $\frac{7}{16}$ of it before and $\frac{3}{8} = \frac{6}{16}$ after.

So 10 t is $\frac{1}{16}$ of the total, which is 160 t.

Barn I: 70 t; barn II: 90 t.

Answer: 70 t and 90 t.

Problem 17

A fraction equals $\frac{4}{5}$. Taking 6 from the numerator and adding 6 to the denominator gives $\frac{1}{2}$. Find it.

Solution

The sum is fixed.

The numerator is $\frac{4}{9}$ of it before and $\frac{1}{3} = \frac{3}{9}$ after.

So 6 is $\frac{1}{9}$ of the sum, which is 54.

The fraction: $\frac{24}{30}$.

Answer: $\frac{24}{30}$.

Problem 18

The first number is $\frac{2}{5}$ of the second. Adding 9 to each makes the first $\frac{1}{2}$ of the second. Find them.

Solution

The difference is fixed.

The first is $\frac{2}{3}$ of it before and 1 times it after.

So 9 is $\frac{1}{3}$ of the difference, which is 27.

The numbers: 18 and 45.

Answer: 18 and 45.

Problem 19

Two numbers are in the ratio $\frac{3}{4}$. Taking 10 from each gives the ratio $\frac{1}{2}$. Find them.

Solution

The difference is fixed.

The first is 3 times it before and 1 times it after.

So 10 is 2 times the difference, which is 5.

The numbers: 15 and 20.

Answer: 15 and 20.

Problem 20

A child is $\frac{1}{4}$ of the mother's age now and will be $\frac{2}{5}$ of it in 8 years. Find both ages now.

Solution

The age gap is fixed.

The child is $\frac{1}{3}$ of it now and $\frac{2}{3}$ of it in 8 years.

So 8 years is $\frac{1}{3}$ of the gap, which is 24.

Child: 8; mother: 32.

Answer: Child 8, mother 32 years.

Problem 21

Four years ago the elder brother was 3 times as old as the younger; now he is 2 times as old. Find both ages now.

Solution

The gap is fixed.

The younger was $\frac{1}{2}$ of it four years ago and is 1 times it now.

So 4 years is $\frac{1}{2}$ of the gap, which is 8.

Younger: 8; elder: 16.

Answer: Younger 8, elder 16 years.

Problem 22

A fraction equals $\frac{2}{3}$. Adding 5 to its top and bottom gives $\frac{3}{4}$. Find it.

Solution

The difference is fixed.
The numerator is 2 times it before and 3 times it after.
So 5 equals the difference.

The fraction: $\frac{10}{15}$.

Answer: $\frac{10}{15}$.

Problem 23

A fraction equals $\frac{5}{7}$. Taking 3 from its top and bottom gives $\frac{2}{3}$. Find it.

Solution

The difference is fixed.

The numerator is $\frac{5}{2}$ of it before and 2 times it after.

So 3 is $\frac{1}{2}$ of the difference, which is 6.

The fraction: $\frac{15}{21}$.

Answer: $\frac{15}{21}$.

Problem 24

Mr Ba's hens are $\frac{3}{5}$ as many as his ducks. After buying 12 more hens, the hens are $\frac{9}{10}$ as many as the ducks. How many of each did he have?

Solution

The ducks do not change.

The 12 hens are $\frac{9}{10} - \frac{3}{5} = \frac{3}{10}$ of the ducks.

Ducks: $12 : \frac{3}{10} = 40$.

Hens: $40 \times \frac{3}{5} = 24$.

Answer: 24 hens, 40 ducks.

Problem 25

The top shelf holds $\frac{4}{7}$ as many books as the bottom shelf. Adding 15 books to the top makes them equal. How many were on each?

Solution

The bottom shelf does not change.

The 15 books are $1 - \frac{4}{7} = \frac{3}{7}$ of it.

Bottom: $15 : \frac{3}{7} = 35$; top: 20.

Answer: Top 20, bottom 35 books.

Problem 26

A fraction equals $\frac{3}{4}$. Keeping the numerator and adding 6 to the denominator gives $\frac{3}{5}$. Find it.

Solution

The numerator is fixed; in both fractions it is 3 parts.

The denominator grows from 4 parts to 5 parts, so 1 part is 6.

The fraction: $\frac{18}{24}$.

Answer: $\frac{18}{24}$.

Problem 27

On a coach the women are $\frac{2}{3}$ as many as the men. At a stop 4 men get off and 4 women get on, and then there are equal numbers. How many men and women were there at first?

Solution

The total is fixed.

Women are $\frac{2}{5}$ of it before and $\frac{1}{2}$ after.

So 4 people are $\frac{1}{2} - \frac{2}{5} = \frac{1}{10}$ of the total, which is 40.

Women: 16; men: 24.

Answer: 24 men, 16 women.

Problem 28

An has $\frac{5}{7}$ as many marbles as Binh. If Binh gives An 4, they have the same. How many does each have?

Solution

The total is fixed.

An has $\frac{5}{12}$ of it before and $\frac{6}{12}$ after.

So 4 is $\frac{1}{12}$ of the total, which is 48.

An: 20; Binh: 28.

Answer: An 20, Binh 28 marbles.

Problem 29

A father is 4 times as old as his son now and will be 2 times as old in 20 years. Find both ages now.

Solution

The gap is fixed.

The son is $\frac{1}{3}$ of it now and 1 times it in 20 years.

So 20 years is $\frac{2}{3}$ of the gap, which is 30.

Son: 10; father: 40.

Answer: Son 10, father 40 years.

Problem 30

A shop has sugar equal to $\frac{3}{8}$ of its rice. After selling 40 kg of rice, the sugar is $\frac{1}{2}$ of the rice left. How much of each was there?

Solution

The sugar is fixed; call it 3 parts.
Rice: 8 parts before, 6 parts after.
So 40 kg is 2 parts; one part: 20 kg.
Sugar: 60 kg; rice: 160 kg.
Answer: Sugar 60 kg, rice 160 kg.

Challenge – Extension

Problem 31 *

Three years ago a dad was 7 times as old as his son. In five years he will be 3 times as old. How old is each now?

Solution

The age gap is constant.
The son was $\frac{1}{6}$ of it three years ago, and will be $\frac{1}{2}$ of it in five years.
From three years ago to five years ahead is 8 years, which is $\frac{1}{2} - \frac{1}{6} = \frac{1}{3}$ of the gap; the gap is 24 years.
Three years ago the son was 4; now he is 7 and his dad is 31.
Answer: son 7, dad 31.

Problem 32 *

If An gives Binh 5 marbles they will have the same number. If Binh gives An 5 marbles, An will have 3 times as many as Binh. How many marbles does each have?

Solution

Giving 5 makes them equal, so An has 10 more than Binh.
If Binh gives An 5, An has $10 + 5 + 5 = 20$ more; the difference is $3 - 1 = 2$ parts, so Binh then has $20 : 2 = 10$.
Binh has $10 + 5 = 15$; An has $15 + 10 = 25$.
Answer: An 25, Binh 15.

Problem 33 *

The top shelf holds $\frac{2}{5}$ as many books as the bottom shelf. Moving 6 books from the bottom to the top makes the top hold $\frac{1}{2}$ as many as the bottom. How many books are there in all?

Solution

The total is constant.

The top shelf is $\frac{2}{7}$ of it before and $\frac{1}{3}$ after; 6 books are $\frac{1}{3} - \frac{2}{7} = \frac{1}{21}$ of the total.

Total: $6 \times 21 = 126$ books.

Answer: 126 books.

Problem 34 ★★

Adding 3 to the numerator of a fraction gives a fraction equal to 1. Adding 3 to its denominator instead gives a fraction equal to $\frac{1}{2}$. Find the fraction.

Solution

Adding 3 to the numerator gives 1, so the denominator is 3 more than the numerator.

Adding 3 to the denominator makes it $3+3 = 6$ more than the numerator, and twice the numerator; so the numerator is 6 and the denominator 9.

Answer: $\frac{6}{9}$.

Problem 35 ★★

A shop has 4 times as much plain rice as sticky rice. After selling 30 kg of plain rice and 5 kg of sticky rice, the plain rice left is 3 times the sticky rice left. How much of each was there at first?

Solution

After selling, plain rice left $= 3\times$(sticky -5) $= 3\times$ sticky -15.

Also, plain rice left $= 4\times$ sticky -30.

Comparing the two: sticky rice $= 30 - 15 = 15$ kg.

Sticky: 15 kg; plain: 60 kg. (*Check:* $30 = 3 \times 10$.)

Answer: 60 kg plain, 15 kg sticky.

Problem 36 ★

If An gives Binh 6 marbles, Binh has twice as many as An. If Binh gives An 6, they have the same. How many does each have?

Solution

The total T is fixed.

First case: after giving, An has $\frac{1}{3}T$, so An has $\frac{1}{3}T + 6$.

Second case: after receiving, An has $\frac{1}{2}T$, so An has $\frac{1}{2}T - 6$.

Comparing: $\frac{1}{2}T - \frac{1}{3}T = 12$, i.e.

$\frac{1}{6}T = 12$ and $T = 72$.

An: 30; Binh: 42.

Answer: An 30, Binh 42 marbles.

Problem 37 *

A mother is 4 times as old as her child now; five years ago she was 7 times as old. Find both ages now.

Solution

The gap is fixed.

The child is $\frac{1}{3}$ of it now and was $\frac{1}{6}$ of it five years ago.

So 5 years is $\frac{1}{3} - \frac{1}{6} = \frac{1}{6}$ of the gap, which is 30.

Child: 10; mother: 40.

Answer: Child 10, mother 40 years.

Problem 38 *

In class 5G the boys are $\frac{5}{6}$ as many as the girls. If 3 boys leave, the boys are $\frac{3}{4}$ as many as the girls. How many boys and girls were there?

Solution

The girls do not change.

The 3 boys are $\frac{5}{6} - \frac{3}{4} = \frac{1}{12}$ of the girls.

Girls: 36; boys: $36 \times \frac{5}{6} = 30$.

Answer: 30 boys, 36 girls.

Problem 39 *

A fraction equals $\frac{3}{8}$ and its denominator is 10 more than its numerator. Adding the same whole number to top and bottom gives $\frac{1}{2}$. Find the fraction and the number added.

Solution

The difference 10 is $8 - 3 = 5$ parts; one part is 2.

The fraction is $\frac{6}{16}$.

The difference stays 10; the new fraction is $\frac{1}{2}$, so the new numerator equals the difference, 10.

Number added: $10 - 6 = 4$.

Answer: $\frac{6}{16}$; 4.

Problem 40 *

In class 5H the top pupils are $\frac{1}{4}$ as many as the rest. By the end of the year 3 more reach the top level, and the top pupils are $\frac{2}{5}$ as many as the rest. How many pupils are in the class?

Solution

The class size is fixed.

Top pupils are $\frac{1}{5}$ of it before and $\frac{2}{7}$ after.

So 3 pupils are $\frac{2}{7} - \frac{1}{5} = \frac{3}{35}$ of the class.

Class: $3 : \frac{3}{35} = 35$.

Answer: 35 pupils.

Problem 41 *

The younger brother is $\frac{2}{3}$ of the elder's age now and was $\frac{1}{2}$ of it five years ago. Find both ages now.

Solution

The gap is fixed.

The younger is 2 times it now and was 1 times it five years ago.

So the gap is 5 years.

Younger: 10; elder: 15.

Answer: Younger 10, elder 15 years.

Problem 42 *

Two tanks hold water in the ratio $\frac{2}{3}$. After 40 litres are used from each, the ratio is $\frac{1}{2}$. How much

did each hold?

Solution

The difference is fixed.
The first tank is 2 times it before and 1 times it after.
So the difference is 40 l.
First: 80 l; second: 120 l.
Answer: 80 l and 120 l.

Problem 43 *

Classes 5A and 5B are in the ratio $\frac{7}{8}$. Moving 3 pupils from 5B to 5A makes 5A 3 larger than 5B. How many pupils are in each class?

Solution

At first 5B has 1 part more than 5A.
Moving 3 pupils changes the difference by 6.
From "5B ahead by 1 part" to "5A ahead by 3": 1 part $+3 = 6$, so 1 part $= 3$.
5A: 21; 5B: 24.
Answer: 5A 21, 5B 24 pupils.

Problem 44 *

An has $\frac{3}{5}$ as many marbles as Binh. An buys 8 more and Binh gives 8 away; then they have the same. How many did each have?

Solution

An gains 8 and Binh loses 8, so the total is fixed.
An is $\frac{3}{8}$ of it before and $\frac{4}{8}$ after; 8 is $\frac{1}{8}$ of the total.
Total: 64; An: 24; Binh: 40.
Answer: An 24, Binh 40 marbles.

Problem 45 *

Barn A holds 3 times as much paddy as barn B. If B takes in 30 more tonnes, A holds $\frac{3}{2}$ as much as B. How much did each hold?

Solution

Barn A is fixed.

B is $\frac{1}{3}$ of A before and $\frac{2}{3}$ of A after.

So 30 t is $\frac{1}{3}$ of A, which is 90 t.

B: 30 t.

Answer: A 90 t, B 30 t.

Problem 46 ★★

A father is 3 times as old as his son now and was 5 times as old ten years ago. How old was he when his son was born?

Solution

The gap is fixed.

The son is $\frac{1}{2}$ of it now and was $\frac{1}{4}$ of it ten years ago.

So 10 years is $\frac{1}{4}$ of the gap, which is 40.

When the son was born, the father's age was exactly the gap: 40.

Answer: 40 years old.

Problem 47 ★★

Adding 2 to the numerator of a fraction gives $\frac{1}{2}$; taking 2 from the numerator gives $\frac{1}{4}$ (same denominator). Find the fraction.

Solution

The denominator is fixed.

The two new numerators differ by 4, which is $\frac{1}{2} - \frac{1}{4} = \frac{1}{4}$ of the denominator.

Denominator: 16.

Numerator: $16 \times \frac{1}{2} - 2 = 6$.

The fraction: $\frac{6}{16}$.

Answer: $\frac{6}{16}$.

Problem 48 ⋆⋆

If An gives Binh 4 sweets, An has 3 times as many as Binh. If Binh gives An 4, An has 7 times as many. How many does each have?

Solution

The total T is fixed.

First case: An keeps $\frac{3}{4}T$, so An has $\frac{3}{4}T + 4$.

Second case: An has $\frac{7}{8}T$ after receiving, so An has $\frac{7}{8}T - 4$.

Comparing: $\frac{1}{8}T = 8$, so $T = 64$.

An: 52; Binh: 12.

Answer: An 52, Binh 12 sweets.

Problem 49 ⋆⋆

Find a fraction such that adding 1 to top and bottom gives $\frac{1}{2}$, and taking 1 from top and bottom gives $\frac{1}{3}$.

Solution

The difference d between bottom and top is fixed.

For $\frac{1}{2}$ the new numerator is d; for $\frac{1}{3}$ it is $\frac{d}{2}$.

The two new numerators differ by 2, which is $\frac{d}{2}$; so $d = 4$.

Numerator: $4 - 1 = 3$; denominator: 7.

The fraction is $\frac{3}{7}$.

Answer: $\frac{3}{7}$.

Problem 50 ⋆⋆

This year An is $\frac{1}{3}$ of his father's age. In three years their ages will total 54. How old is each now?

Solution

Total now: $54 - 3 \times 2 = 48$.

An 1 part, father 3 parts; one part: $48 : 4 = 12$.

An: 12; father: 36.

Answer: An 12, father 36 years.

7 Direct and Inverse Proportion

① Two proportional quantities

Direct proportion. Two quantities are directly proportional if, when one is multiplied (or divided) by some number, the other is multiplied (or divided) by the same number. *Example:* metres of cloth and its cost (at a fixed price).

Inverse proportion. Two quantities are inversely proportional if, when one is multiplied by some number, the other is divided by that number; their *product* stays the same. *Example:* the number of workers and the number of days needed for the same job.

Two common methods:

- **The unitary method:** find the value for 1 unit (1 metre, 1 worker, 1 day,...), then multiply or divide.
- **The ratio method:** find how many times the first quantity grows (or shrinks), then change the second one accordingly.

For "workers – days" problems, think of the job as *(workers)* × *(days)* "worker-days"; the number of worker-days needed does not change.

Example 1

5 m of cloth cost 450 000 dong. How much do 15 m of the same cloth cost?

Guided solution.

Method 1 (unitary). 1 m costs $450\,000 : 5 = 90\,000$ dong, so 15 m cost $90\,000 \times 15 = 1\,350\,000$ dong.

Method 2 (ratio). 15 m is $15 : 5 = 3$ times 5 m, so the cost is 3 times as much: $450\,000 \times 3 = 1\,350\,000$ dong.

Answer: 1 350 000 dong.

Example 2

A team of 12 workers builds a stretch of road in 10 days. How many workers are needed to finish it in 8 days? (Everyone works at the same rate.)

Guided solution.

Workers and days are inversely proportional.

One worker alone would need $10 \times 12 = 120$ days, so the job takes 120 worker-days.

To finish in 8 days we need $120 : 8 = 15$ workers.

Answer: 15 workers.

Example 3

A kitchen has enough rice for 120 people for 20 days. After 5 days, 30 more people arrive. For how many more days will the remaining rice last? (Everyone eats the same amount.)

Guided solution.

After 5 days, the rice left would feed 120 people for $20 - 5 = 15$ days.
That rice would feed 1 person for $15 \times 120 = 1\,800$ days.
There are now $120 + 30 = 150$ people, so the rice lasts $1\,800 : 150 = 12$ days.
Answer: 12 days.

Example 4

10 workers repair 400 m of road in 5 days. How many metres can 12 workers repair in 6 days? (Everyone works at the same rate.)

Guided solution.

There are three quantities; reduce to "1 worker in 1 day".
In 5 days 1 worker repairs $400 : 10 = 40$ m; in 1 day, $40 : 5 = 8$ m.
12 workers in 6 days repair $8 \times 12 \times 6 = 576$ m.
Answer: 576 m.

Practice

Problem 1

6 identical boxes hold 72 pens altogether. How many pens do 15 such boxes hold?

Solution

Each box holds $72 : 6 = 12$ pens, so 15 boxes hold $12 \times 15 = 180$ pens.
Answer: 180 pens.

Problem 2

A car uses 9 litres of petrol for 120 km. How much petrol does it use for 200 km?

Solution

40 km uses $9 : 3 = 3$ litres (because $120 = 40 \times 3$).
200 km is $200 : 40 = 5$ times 40 km, so it uses $3 \times 5 = 15$ litres.
Answer: 15 litres.

Problem 3

15 outfits use 45 m of cloth. How many such outfits can be made from 72 m?

Solution

Each outfit needs $45 : 15 = 3$ m, so 72 m make $72 : 3 = 24$ outfits.
Answer: 24 outfits.

Problem 4

With enough money for 24 notebooks at 6 000 dong each, how many notebooks at 8 000 dong each can you buy?

Solution

The money is $6\,000 \times 24 = 144\,000$ dong, which buys $144\,000 : 8\,000 = 18$ notebooks.
Answer: 18 notebooks.

Problem 5

15 workers finish a job in 12 days. How long would 20 workers take? (Everyone works at the same rate.)

Solution

The job takes $15 \times 12 = 180$ worker-days, so 20 workers need $180 : 20 = 9$ days.
Answer: 9 days.

Problem 6

A kitchen has rice for 80 people for 15 days. How long would it last 100 people?

Solution

The rice would feed 1 person for $15 \times 80 = 1\,200$ days, so 100 people for $1\,200 : 100 = 12$ days.
Answer: 12 days.

Problem 7

A summer camp of 60 pupils has food for 30 days. After 10 days, 20 pupils go home early. For how many more days will the food last those who stay?

Solution

The food left would feed 60 pupils for 20 days, i.e.
$60 \times 20 = 1\,200$ pupil-days.

Now 40 pupils remain, so it lasts $1\,200 : 40 = 30$ days.
Answer: 30 days.

Problem 8

8 people plant 96 trees in 3 hours. How many trees do 10 people plant in 4 hours? (Everyone works at the same rate.)

Solution

1 person plants $96 : 8 : 3 = 4$ trees per hour.
10 people in 4 hours: $4 \times 10 \times 4 = 160$ trees.
Answer: 160 trees.

Problem 9

3 workers finish a job in 8 days. How many more workers are needed to finish it 2 days sooner? (Everyone works at the same rate.)

Solution

The job is $3 \times 8 = 24$ worker-days.
In $8 - 2 = 6$ days we need $24 : 6 = 4$ workers, i.e.
$4 - 3 = 1$ more.
Answer: 1 worker.

Problem 10

Every 100 kg of paddy gives 62 kg of rice. How many kilograms of rice does 2 500 kg of paddy give?

Solution

2 500 kg is 25 times 100 kg, so the rice is $62 \times 25 = 1\,550$ kg.
Answer: 1 550 kg.

Problem 11

5 identical pumps move 360 m^3 of water in 4 hours. How much do 8 such pumps move in 3 hours?

Solution

1 pump in 1 hour: $360 : 5 : 4 = 18$ m^3.
8 pumps in 3 hours: $18 \times 8 \times 3 = 432$ m^3.

Answer: 432 m^3.

Problem 12

A workshop of 45 workers plans to finish an order in 20 days. After 4 days, 15 more workers join. How many days early is the order finished?

Solution

The rest of the work is 45 workers for 16 days, i.e.
$45 \times 16 = 720$ worker-days.
With 60 workers it takes $720 : 60 = 12$ days.
In total $4 + 12 = 16$ days, which is $20 - 16 = 4$ days early.
Answer: 4 days.

Problem 13

A kitchen has rice for 60 people for 24 days. After 4 days more people arrive, and the rice left lasts only 16 more days. How many people arrived?

Solution

The rice left is $60 \times 20 = 1\,200$ person-days.
Over 16 days that feeds $1\,200 : 16 = 75$ people.
So $75 - 60 = 15$ people arrived.
Answer: 15 people.

Problem 14

4 kg of apples cost 120 000 dong. How much do 7 kg cost?

Solution

1 kg: $120\,000 : 4 = 30\,000$ dong.
7 kg: $30\,000 \times 7 = 210\,000$ dong.
Answer: 210 000 dong.

Problem 15

12 notebooks cost 96 000 dong. How much do 20 cost?

Solution

One: $96\,000 : 12 = 8\,000$ dong.
20: $8\,000 \times 20 = 160\,000$ dong.
Answer: 160 000 dong.

Problem 16

A car covers 135 km in 3 hours. How far does it go in 5 hours? (Same distance every hour.)

Solution

One hour: $135 : 3 = 45$ km.
5 hours: $45 \times 5 = 225$ km.
Answer: 225 km.

Problem 17

8 workers finish a job in 6 days. How long would 12 workers take? (Same rate.)

Solution

The job is $8 \times 6 = 48$ worker-days.
12 workers need $48 : 12 = 4$ days.
Answer: 4 days.

Problem 18

Reading 15 pages a day, Ha finishes a book in 12 days. How long would she take at 20 pages a day?

Solution

The book has $15 \times 12 = 180$ pages.
Days: $180 : 20 = 9$.
Answer: 9 days.

Problem 19

A kitchen has rice for 50 people for 18 days. How long would it last 60 people?

Solution

The rice feeds 1 person for $18 \times 50 = 900$ days.
60 people: $900 : 60 = 15$ days.
Answer: 15 days.

Problem 20

6 identical sewing machines make 600 shirts in 5 days. How many do 10 machines make in 3 days?

Solution

1 machine in 1 day: $600 : 6 : 5 = 20$ shirts.
10 machines in 3 days: $20 \times 10 \times 3 = 600$ shirts.
Answer: 600 shirts.

Problem 21

6 m of cable cost 54 000 dong. How many metres can 360 000 dong buy?

Solution

1 m costs $54\,000 : 6 = 9\,000$ dong.
$360\,000 : 9\,000 = 40$ m.
Answer: 40 m.

Problem 22

A tap delivers 60 litres in 5 minutes. How much does it deliver in 1 hour?

Solution

1 hour = 60 minutes, which is $60 : 5 = 12$ times 5 minutes.
Water: $60 \times 12 = 720$ litres.
Answer: 720 litres.

Problem 23

6 identical lorries carry 72 tonnes. How much do 15 such lorries carry?

Solution

Each lorry: $72 : 6 = 12$ t.
15 lorries: $12 \times 15 = 180$ t.
Answer: 180 t.

Problem 24

A team of 20 plans to finish a job in 15 days. How many more workers are needed to finish in 12 days? (Same rate.)

Solution

The job is $20 \times 15 = 300$ worker-days.
In 12 days we need $300 : 12 = 25$ workers.
So $25 - 20 = 5$ more.
Answer: 5 workers.

Problem 25

9 people plan to finish a job in 12 days. After 4 days, 3 of them stop. How many days late is the job finished?

Solution

Work left: $9 \times (12 - 4) = 72$ worker-days.
6 people need $72 : 6 = 12$ days.
Total: $4 + 12 = 16$ days, which is $16 - 12 = 4$ days late.
Answer: 4 days.

Problem 26

Every 30 kg of paddy gives 18 kg of rice. How much rice does 1 tonne of paddy give?

Solution

$1\text{ t} = 1\,000$ kg.
1 kg of paddy gives $18 : 30 = 0.6$ kg of rice.
$1\,000$ kg give $0.6 \times 1\,000 = 600$ kg.
Answer: 600 kg.

Problem 27

5 identical sacks of rice weigh 250 kg. How much do 12 sacks weigh?

Solution

One sack: $250 : 5 = 50$ kg.
12 sacks: $50 \times 12 = 600$ kg.
Answer: 600 kg.

Problem 28

A school pays 9 000 000 dong for 15 desk sets. How much more money is needed to buy 25 sets?

Solution

One set: $9\,000\,000 : 15 = 600\,000$ dong.
25 sets: $600\,000 \times 25 = 15\,000\,000$ dong.
Extra: $15\,000\,000 - 9\,000\,000 = 6\,000\,000$ dong.
Answer: 6 000 000 dong.

Problem 29

Carrying 4 tonnes a trip, a lorry needs 15 trips. How many trips are needed at 6 tonnes a trip?

Solution

Load: $4 \times 15 = 60$ t.
Trips: $60 : 6 = 10$.
Answer: 10 trips.

Problem 30

14 people eat 21 kg of rice in 3 days. How much do 20 people eat in 6 days? (Same amount each.)

Solution

1 person for 1 day: $21 : 14 : 3 = 0.5$ kg.
20 people for 6 days: $0.5 \times 20 \times 6 = 60$ kg.
Answer: 60 kg.

Challenge – Extension

Problem 31 ★

A unit has rice for 30 days. After 10 days, 40 more people join, and the rice left lasts only 15 more days. How many people were in the unit at first?

Solution

The rice left feeds the original group for 20 days, or the original group plus 40 for 15 days.
Since $20 : 15 = \frac{4}{3}$, the new group is $\frac{4}{3}$ of the original; so 40 people are $\frac{1}{3}$ of the original group.
Original group: $40 \times 3 = 120$ people.
Answer: 120 people.

Problem 32 *

A team plans to finish a project in 15 days. After 5 days, 6 workers leave, so the team needs 2 days more than planned. How many workers were there at first?

Solution

The rest of the work was planned for 10 days but took 12.

The days are $\frac{12}{10} = \frac{6}{5}$ times as many, so the workers left are $\frac{5}{6}$ of the original team.

So 6 workers are $1 - \frac{5}{6} = \frac{1}{6}$ of the team, which had $6 \times 6 = 36$ workers.

Answer: 36 workers.

Problem 33 *

12 people working 8 hours a day finish a job in 10 days. How many days do 16 people working 6 hours a day need? (Everyone works at the same rate.)

Solution

The job is $12 \times 8 \times 10 = 960$ worker-hours.

Each day 16 people put in $16 \times 6 = 96$ worker-hours.

Days needed: $960 : 96 = 10$.

Answer: 10 days.

Problem 34 *

Two gear wheels mesh together. The big one has 48 teeth and the small one 18. When the big wheel turns 15 times, how many times does the small one turn?

Solution

The wheels mesh, so the same number of teeth pass the contact point on both: $48 \times 15 = 720$.

The small wheel turns $720 : 18 = 40$ times.

Number of teeth and number of turns are inversely proportional.

Answer: 40 turns.

Problem 35 **

"A hen and a half lays an egg and a half in a day and a half." How many eggs do 6 such hens lay in 6 days?

Solution

1.5 hens lay 1.5 eggs in 1.5 days, so 1.5 hens lay 1 egg in 1 day.

6 hens are 4 times 1.5 hens, so they lay 4 eggs a day, and $4 \times 6 = 24$ eggs in 6 days.

Answer: 24 eggs.

Problem 36 ★★

With a sum of money, Ms Lan can buy 20 kg of type I rice or 30 kg of type II rice. If she spends it on equal amounts of both types, how many kilograms of each does she get?

Solution

1 kg of type I costs $\frac{1}{20}$ of the money, 1 kg of type II costs $\frac{1}{30}$.

One kilogram of each costs $\frac{1}{20} + \frac{1}{30} = \frac{1}{12}$ of the money.

So she gets 12 kg of each.

Answer: Each type: 12 kg.

Problem 37 ★

A team plans to finish a project in 20 days. With 10 more workers it would finish 4 days sooner. How many workers are in the team? (Same rate.)

Solution

The days drop from 20 to 16, i.e.

$\frac{16}{20} = \frac{4}{5}$ of the time, so the team must grow by $\frac{5}{4}$.

So 10 workers are $\frac{1}{4}$ of the team.

Team: $10 \times 4 = 40$.

Answer: 40 workers.

Problem 38 ★

A kitchen has rice for 100 people for 30 days. After 6 days, 20 people leave. How long will the rest of the rice last?

Solution

The rice left is $100 \times 24 = 2\,400$ person-days.

80 people remain: $2\,400 : 80 = 30$ days.

Answer: 30 days.

Problem 39

A kitchen has rice for 24 days. After 4 days, 20 more people arrive and the rest lasts only 15 days. How many people were there at first?

Solution

The rice left feeds the original group for 20 days, or the group plus 20 for 15 days.

$20 : 15 = \frac{4}{3}$, so the new group is $\frac{4}{3}$ of the old; the 20 newcomers are $\frac{1}{3}$ of it.

At first: 60 people.

Answer: 60 people.

Problem 40

6 identical taps fill a tank in 4 hours. All 6 run for 1 hour, then 2 are turned off. How much longer do the others need?

Solution

Filling takes $6 \times 4 = 24$ tap-hours.

The first hour gives 6; 18 remain.

4 taps need $18 : 4 = 4.5$ hours $= 4$ h 30 min.

Answer: 4 h 30 min.

Problem 41

Gears A, B, C mesh in a chain (A with B, B with C). A has 30 teeth, B 20 and C 45. When A turns 12 times, how many times does C turn?

Solution

The same number of teeth pass each contact: $30 \times 12 = 360$.

B turns $360 : 20 = 18$ times; C also has 360 teeth pass, so it turns $360 : 45 = 8$ times.

Answer: 8 turns.

Problem 42

A clock gains 3 minutes a day. After how many days will it be 1 hour fast?

Solution

1 hour = 60 minutes.

Days: $60 : 3 = 20$.

Answer: 20 days.

Problem 43 *

Walking at 4 km an hour, Mai takes 30 minutes to get to school. How long would she take cycling at 12 km an hour?

Solution

The distance is fixed; going $12 : 4 = 3$ times as fast takes a third of the time.
Time: $30 : 3 = 10$ minutes.
Answer: 10 minutes.

Problem 44 *

4 people working 8 hours a day finish a job in 6 days. How long would 6 people working 4 hours a day take?

Solution

The job is $4 \times 8 \times 6 = 192$ worker-hours.
Each day 6 people give $6 \times 4 = 24$.
Days: $192 : 24 = 8$.
Answer: 8 days.

Problem 45 *

18 lorries each carry 45 sacks. If only 15 lorries carry the whole load, how many extra sacks must each take?

Solution

Sacks: $45 \times 18 = 810$.
Each lorry: $810 : 15 = 54$; that is $54 - 45 = 9$ more.
Answer: 9 sacks.

Problem 46 **

"Three cats catch three mice in three minutes." How long do 100 cats take to catch 100 mice? (All cats are equally quick.)

Solution

Three cats catch three mice in 3 minutes, so each cat catches 1 mouse in 3 minutes.
100 cats each catching 1 mouse also take 3 minutes.
Answer: 3 minutes.

Problem 47 ★★

A sum of money buys 15 kg of apples or 10 kg of pears. After buying 6 kg of apples, how many kilograms of pears can the rest buy?

Solution

1 kg of apples costs $\frac{1}{15}$ of the money; 6 kg cost $\frac{6}{15} = \frac{2}{5}$.

The remaining $\frac{3}{5}$ buys $10 \times \frac{3}{5} = 6$ kg of pears.

Answer: 6 kg of pears.

Problem 48 ★★

3 kg of oranges and 2 kg of apples cost 175 000 dong, and apples cost twice as much per kilogram. Find each price.

Solution

2 kg of apples cost the same as 4 kg of oranges, so the bill is the price of $3 + 4 = 7$ kg of oranges.

Oranges: $175\,000 : 7 = 25\,000$ dong/kg; apples: 50 000 dong/kg.

Answer: Oranges 25 000, apples 50 000 dong/kg.

Problem 49 ★★

A team of 12 plans to build 360 m of dyke in 6 days. After 2 days, 4 more people join. How many days early do they finish?

Solution

Each person builds $360 : 12 : 6 = 5$ m a day.

In 2 days they build $12 \times 5 \times 2 = 120$ m; 240 m remain.

16 people build 80 m a day, needing $240 : 80 = 3$ days.

Total 5 days: 1 day early.

Answer: 1 day.

Problem 50 ★★

The same money buys 8 pens or 12 notebooks. A pen costs 2 000 dong more than a notebook. How much money is it?

Solution

Buying 8 pens instead of 8 notebooks costs $8 \times 2\,000 = 16\,000$ dong more.

That extra is the price of $12 - 8 = 4$ notebooks, so a notebook is 4 000 dong.

Money: $4\,000 \times 12 = 48\,000$ dong.

Answer: 48 000 dong.

8 Work Problems

① Treat the whole job as one unit

- Treat **the whole job** (or **a full tank**) as 1 unit.
- If someone finishes the job in n hours, then **each hour** they do $\frac{1}{n}$ of the job.
- When several people **work together**, the work done in 1 hour is the *sum* of their separate parts.
- An **outlet pipe** empties the tank, so its part is *subtracted*.
- Time needed = (work to be done) : (work done in 1 hour).

Example 1

The first person can finish a job alone in 6 hours, the second in 3 hours. How long will it take them working together?

Guided solution.

Each hour the first person does $\frac{1}{6}$ of the job and the second $\frac{1}{3}$.

Together they do $\frac{1}{6} + \frac{1}{3} = \frac{1}{2}$ of the job each hour.

Time together: $1 : \frac{1}{2} = 2$ hours.

Answer: 2 hours.

Example 2

Two people together finish a job in 12 days. The first alone would take 20 days. How long would the second take alone?

Guided solution.

Each day both together do $\frac{1}{12}$ of the job and the first does $\frac{1}{20}$.

So the second does $\frac{1}{12} - \frac{1}{20} = \frac{5}{60} - \frac{3}{60} = \frac{1}{30}$ each day, and would take 30 days alone.

Answer: 30 days.

Example 3

Pipe I fills a tank in 4 hours, pipe II in 6 hours. The tank is empty; both pipes run for 1 hour, then pipe I is turned off. How much longer must pipe II run to fill the tank?

Guided solution.

In 1 hour both pipes fill $\frac{1}{4}+\frac{1}{6}=\frac{5}{12}$ of the tank, leaving $1-\frac{5}{12}=\frac{7}{12}$.

Pipe II needs $\frac{7}{12}:\frac{1}{6}=\frac{7}{2}$ hours $=3$ hours 30 minutes.

Answer: 3 hours 30 minutes.

Example 4

An inlet fills a tank in 3 hours; an outlet empties a full tank in 5 hours. Starting empty with both open, how long until the tank is full?

Guided solution.

Each hour the water rises by $\frac{1}{3}-\frac{1}{5}=\frac{2}{15}$ of the tank.

Time to fill: $1:\frac{2}{15}=\frac{15}{2}$ hours $=7$ hours 30 minutes.

Answer: 7 hours 30 minutes.

Practice

Problem 1

An sweeps the schoolyard in 30 minutes, Binh in 20 minutes. How long does it take them together?

Solution

Each minute together they sweep $\frac{1}{30}+\frac{1}{20}=\frac{5}{60}=\frac{1}{12}$ of the yard, so they take 12 minutes.

Answer: 12 minutes.

Problem 2

Pipe I fills a tank in 2 hours, pipe II in 3 hours. Starting empty with both open, how long until it is full?

Solution

Each hour both fill $\frac{1}{2}+\frac{1}{3}=\frac{5}{6}$ of the tank.

Time: $1:\frac{5}{6}=\frac{6}{5}$ hours $=1$ hour 12 minutes.

Answer: 1 hour 12 minutes.

Problem 3

Two workers together finish a job in 6 hours. The first alone takes 10 hours. How long does the second take alone?

Solution

Each hour the second does $\frac{1}{6} - \frac{1}{10} = \frac{1}{15}$ of the job, so they take 15 hours alone.

Answer: 15 hours.

Problem 4

Two people harvest a field together in 2 hours. The first alone takes 3 hours. What fraction of the field does the second harvest each hour?

Solution

Each hour the second harvests $\frac{1}{2} - \frac{1}{3} = \frac{1}{6}$ of the field.

Answer: $\frac{1}{6}$ of the field.

Problem 5

Three friends tidy the library. Alone, Mai would take 4 hours, Hung 6 hours and Lan 12 hours. How long does it take all three together?

Solution

Each hour the three do $\frac{1}{4} + \frac{1}{6} + \frac{1}{12} = \frac{3+2+1}{12} = \frac{1}{2}$ of the job, so they take 2 hours.

Answer: 2 hours.

Problem 6

An inlet fills a tank in 4 hours; an outlet empties a full tank in 6 hours. Starting empty with both open, how long until it is full?

Solution

Each hour the tank gains $\frac{1}{4} - \frac{1}{6} = \frac{1}{12}$, so it is full after 12 hours.

Answer: 12 hours.

Problem 7

A tank is already $\frac{1}{3}$ full. A pipe fills an empty tank in 6 hours. How long will it take the pipe to fill the rest?

Solution

Still to fill: $1 - \frac{1}{3} = \frac{2}{3}$.

Time: $\frac{2}{3} : \frac{1}{6} = 4$ hours.

Answer: 4 hours.

Problem 8

Pipe I fills a tank in 5 hours, pipe II in 10 hours. Starting empty, pipe I runs for 2 hours and is then replaced by pipe II. How long must pipe II run?

Solution

Pipe I fills $\frac{1}{5} \times 2 = \frac{2}{5}$ of the tank, leaving $\frac{3}{5}$.

Pipe II needs $\frac{3}{5} : \frac{1}{10} = 6$ hours.

Answer: 6 hours.

Problem 9

Team A finishes a job alone in 9 days, team B in 18 days. Team A works alone for 3 days, then team B joins. How many more days do they need together?

Solution

In 3 days team A does $\frac{3}{9} = \frac{1}{3}$ of the job, leaving $\frac{2}{3}$.

Together they do $\frac{1}{9} + \frac{1}{18} = \frac{1}{6}$ a day, so they need $\frac{2}{3} : \frac{1}{6} = 4$ days.

Answer: 4 days.

Problem 10

Mr Ba finishes a job alone in 10 days, Mr Tu in 15 days. They work together for 4 days, then Mr Ba stops. How long does Mr Tu need to finish?

Solution

Together they do $\frac{1}{10}+\frac{1}{15}=\frac{1}{6}$ a day, so $\frac{4}{6}=\frac{2}{3}$ in 4 days, leaving $\frac{1}{3}$.

Mr Tu needs $\frac{1}{3}:\frac{1}{15}=5$ days.

Answer: 5 days.

Problem 11

Two people together finish a job in 4 hours; the first alone would take 6 hours. They work together for 2 hours, then the first stops. How much longer must the second work?

Solution

The second does $\frac{1}{4}-\frac{1}{6}=\frac{1}{12}$ an hour.

In 2 hours together they finish $\frac{2}{4}=\frac{1}{2}$ of the job.

The second needs $\frac{1}{2}:\frac{1}{12}=6$ more hours.

Answer: 6 hours.

Problem 12

Two people do a job together and are paid 1 800 000 dong. Alone, the first would take 6 days and the second 3 days. If the pay is shared by the work each did, how much does each get?

Solution

Each day the first does $\frac{1}{6}$ and the second $\frac{1}{3}=\frac{2}{6}$ of the job, so their work is in the ratio $1:2$.

First: $1\,800\,000:3=600\,000$ dong; second: $1\,200\,000$ dong.

Answer: 600 000 dong and 1 200 000 dong.

Problem 13

One person finishes a job in 4 hours, another in 12 hours. How long do they take together?

Solution

Each hour together: $\frac{1}{4}+\frac{1}{12}=\frac{4}{12}=\frac{1}{3}$ of the job.

Together they take 3 hours.

Answer: 3 hours.

Problem 14

Pipe I fills a tank in 5 hours, pipe II in 20 hours. Starting empty with both open, how long until it is full?

Solution

Each hour: $\frac{1}{5} + \frac{1}{20} = \frac{5}{20} = \frac{1}{4}$ of the tank.

Full after 4 hours.

Answer: 4 hours.

Problem 15

Two people finish a job together in 8 days; the first alone takes 12 days. How long does the second take alone?

Solution

Each day the second does $\frac{1}{8} - \frac{1}{12} = \frac{3}{24} - \frac{2}{24} = \frac{1}{24}$ of the job.

Alone: 24 days.

Answer: 24 days.

Problem 16

Two pipes fill a tank together in 2 hours; pipe I alone takes 3 hours. How long does pipe II take alone?

Solution

Each hour pipe II fills $\frac{1}{2} - \frac{1}{3} = \frac{1}{6}$ of the tank.

Alone: 6 hours.

Answer: 6 hours.

Problem 17

Three workers alone would take 6, 10 and 15 days. How long do all three take together?

Solution

Each day: $\frac{1}{6} + \frac{1}{10} + \frac{1}{15} = \frac{5+3+2}{30} = \frac{1}{3}$ of the job.

All three: 3 days.

Answer: 3 days.

Problem 18

An inlet fills a tank in 6 hours; an outlet empties a full tank in 9 hours. Starting empty with both open, how long until it is full?

Solution

Each hour: $\frac{1}{6} - \frac{1}{9} = \frac{3}{18} - \frac{2}{18} = \frac{1}{18}$ of the tank.

Full after 18 hours.

Answer: 18 hours.

Problem 19

A tank is $\frac{1}{4}$ full. A pipe fills an empty tank in 8 hours. How long does it need to fill the rest?

Solution

Still to fill: $\frac{3}{4}$ of the tank.

Time: $\frac{3}{4} : \frac{1}{8} = 6$ hours.

Answer: 6 hours.

Problem 20

A tank is $\frac{2}{3}$ full. An outlet empties a full tank in 6 hours. How long does it take to empty this tank?

Solution

Each hour it empties $\frac{1}{6}$ of the tank.

Time: $\frac{2}{3} : \frac{1}{6} = 4$ hours.

Answer: 4 hours.

Problem 21

Team A alone takes 12 days and team B 6 days. A works alone for 3 days, then B joins. How many more days do they need?

Solution

In 3 days A does $\frac{3}{12} = \frac{1}{4}$; $\frac{3}{4}$ is left.

Together they do $\frac{1}{12} + \frac{1}{6} = \frac{1}{4}$ a day.

Time: $\frac{3}{4} : \frac{1}{4} = 3$ days.

Answer: 3 days.

Problem 22

Pipe I fills a tank in 4 hours, pipe II in 12. Starting empty, pipe I runs for 1 hour, then pipe II is opened too. How much longer until it is full?

Solution

After 1 hour pipe I has filled $\frac{1}{4}$; $\frac{3}{4}$ is left.

Both together: $\frac{1}{4} + \frac{1}{12} = \frac{1}{3}$ an hour.

Time: $\frac{3}{4} : \frac{1}{3} = \frac{9}{4}$ hours $= 2$ h 15 min.

Answer: 2 h 15 min.

Problem 23

Two people together do $\frac{1}{2}$ of a job in 3 days. The first alone would take 10 days. How long would the second take alone?

Solution

Together they do $\frac{1}{2} : 3 = \frac{1}{6}$ a day.

The second does $\frac{1}{6} - \frac{1}{10} = \frac{1}{15}$ a day.

Alone: 15 days.

Answer: 15 days.

Problem 24

An does $\frac{1}{3}$ of a job in 2 hours; Binh does $\frac{1}{2}$ of it in 6 hours. How long do they take together?

Solution

An would take 6 hours for the whole job; Binh 12 hours.

Together: $\frac{1}{6} + \frac{1}{12} = \frac{1}{4}$ an hour.

So 4 hours.
Answer: 4 hours.

Problem 25

Each hour pipe I fills $\frac{1}{6}$ of a tank and pipe II $\frac{1}{3}$. What fraction of an empty tank do they fill together in 1 hour 20 minutes?

Solution

1 h 20 min $= \frac{4}{3}$ hours.

Together: $\frac{1}{6} + \frac{1}{3} = \frac{1}{2}$ of the tank an hour.

In $\frac{4}{3}$ hours: $\frac{1}{2} \times \frac{4}{3} = \frac{2}{3}$.

Answer: $\frac{2}{3}$ of the tank.

Problem 26

Two teams repair a road together in 6 days; team I alone takes 10 days. They work together for 2 days, then team I leaves. How long does team II need to finish?

Solution

Team II does $\frac{1}{6} - \frac{1}{10} = \frac{1}{15}$ a day.

After 2 days together $\frac{2}{6} = \frac{1}{3}$ is done; $\frac{2}{3}$ is left.

Team II needs $\frac{2}{3} : \frac{1}{15} = 10$ days.

Answer: 10 days.

Problem 27

A pipe fills a tank in 3 hours, but because of a leak it takes 4 hours. How long would the leak take to empty a full tank?

Solution

Each hour the leak loses $\frac{1}{3} - \frac{1}{4} = \frac{1}{12}$ of the tank.

It empties the tank in 12 hours.

Answer: 12 hours.

Problem 28

Two workers share 2 000 000 dong for a job done together. Alone, the first would take 4 days and the second 6. Shared by the work done, how much does each get?

Solution

Each day the first does $\frac{1}{4} = \frac{3}{12}$ and the second $\frac{1}{6} = \frac{2}{12}$; the work is in the ratio $3 : 2$.

First: $2\,000\,000 : 5 \times 3 = 1\,200\,000$ dong; second: $800\,000$ dong.

Answer: 1 200 000 dong and 800 000 dong.

Problem 29

Pipe I fills a tank in 30 minutes, pipe II in 45 minutes. Together, starting empty, how many minutes do they take?

Solution

Each minute: $\frac{1}{30} + \frac{1}{45} = \frac{3}{90} + \frac{2}{90} = \frac{1}{18}$ of the tank.

Full after 18 minutes.

Answer: 18 minutes.

Problem 30

Two tractors plough a field together in 4 days; tractor I alone takes 6 days. After 2 days together, tractor I breaks down. How long does tractor II need to finish?

Solution

Tractor II ploughs $\frac{1}{4} - \frac{1}{6} = \frac{1}{12}$ of the field a day.

After 2 days together, half is done.

Tractor II needs $\frac{1}{2} : \frac{1}{12} = 6$ days.

Answer: 6 days.

Challenge – Extension

Problem 31 *

A pipe fills a sound tank in 2 hours. Because of a leak, it now takes 2 hours 30 minutes. How long would the leak take to empty a full tank?

Solution

2 h 30 min $= \frac{5}{2}$ hours.

The pipe adds $\frac{1}{2}$ a tank each hour; with the leak the tank gains only $\frac{2}{5}$ an hour.

The leak loses $\frac{1}{2} - \frac{2}{5} = \frac{1}{10}$ a tank each hour, so it empties a full tank in 10 hours.

Answer: 10 hours.

Problem 32 *

A tank is empty. Pipe I fills it in 3 hours and pipe II in 6 hours; pipe III empties a full tank in 4 hours. With all three open, how long until it is full?

Solution

Each hour the tank gains $\frac{1}{3} + \frac{1}{6} - \frac{1}{4} = \frac{4+2-3}{12} = \frac{1}{4}$, so it is full after 4 hours.

Answer: 4 hours.

Problem 33 *

Three workers: alone, A finishes the job in 6 days, B in 12 days and C in 12 days. A and B work together for 2 days, then C joins. How much longer do the three need?

Solution

A and B do $\frac{1}{6} + \frac{1}{12} = \frac{1}{4}$ a day, so half the job in 2 days.

All three do $\frac{1}{4} + \frac{1}{12} = \frac{1}{3}$ a day, so they need $\frac{1}{2} : \frac{1}{3} = \frac{3}{2}$ days, i.e. a day and a half.

Answer: 1.5 days.

Problem 34 *

The master craftsman finishes a job alone in 12 hours. He works for 3 hours, then his assistant works for 4 hours, and together they have done $\frac{7}{12}$ of the job. How long would they take working together from the start?

Solution

In 3 hours the master does $\frac{3}{12}$.

The assistant's 4 hours give $\frac{7}{12} - \frac{3}{12} = \frac{4}{12}$, so the assistant does $\frac{1}{12}$ an hour.

Together: $\frac{1}{12}+\frac{1}{12}=\frac{1}{6}$ an hour, so 6 hours.

Answer: 6 hours.

Problem 35 ★★

A and B together finish a job in 12 days; B and C in 15 days; A and C in 20 days.

a) How long do all three take together? **b)** How long does each take alone?

Solution

a) Add the three pairs: $\frac{1}{12}+\frac{1}{15}+\frac{1}{20}=\frac{5+4+3}{60}=\frac{1}{5}$.

Each person is counted twice, so all three do $\frac{1}{5}:2=\frac{1}{10}$ a day and take 10 days.

b) A: $\frac{1}{10}-\frac{1}{15}=\frac{1}{30}$ (30 days); B: $\frac{1}{10}-\frac{1}{20}=\frac{1}{20}$ (20 days); C: $\frac{1}{10}-\frac{1}{12}=\frac{1}{60}$ (60 days).

Answer: a) 10 days; **b)** A: 30, B: 20, C: 60 days.

Problem 36 ★★

Two pipes fill an empty tank. Both together for 2 hours fill $\frac{1}{2}$ of it. Pipe I for 3 hours and pipe II for 1 hour fill $\frac{7}{12}$ of it. How long does each pipe take alone?

Solution

In 1 hour both fill $\frac{1}{2}:2=\frac{1}{4}$.

The second case is "both for 1 hour" plus "pipe I for 2 more hours".

So pipe I fills $\frac{7}{12}-\frac{1}{4}=\frac{1}{3}$ in 2 hours, i.e.

$\frac{1}{6}$ an hour.

Pipe II: $\frac{1}{4}-\frac{1}{6}=\frac{1}{12}$ an hour.

Answer: Pipe I: 6 hours; pipe II: 12 hours.

Problem 37 ★

A tank is half full. Pipe I fills an empty tank in 6 hours; pipe II empties a full one in 8 hours. With both open, how long until it is full?

Solution

Each hour: $\frac{1}{6} - \frac{1}{8} = \frac{1}{24}$ of the tank.

Time: $\frac{1}{2} : \frac{1}{24} = 12$ hours.

Answer: 12 hours.

Problem 38 *

A working 3 days and B 4 days do $\frac{5}{12}$ of a job; A working 6 days and B 2 days do $\frac{7}{12}$. How long would each take alone?

Solution

Doubling the first case: A for 6 days and B for 8 days do $\frac{10}{12}$.

Compared with the second case, 6 days of B do $\frac{10}{12} - \frac{7}{12} = \frac{1}{4}$, so B does $\frac{1}{24}$ a day.

A's 3 days give $\frac{5}{12} - \frac{4}{24} = \frac{1}{4}$, so A does $\frac{1}{12}$ a day.

A: 12 days; B: 24 days.

Answer: A 12, B 24 days.

Problem 39 *

Two people finish a job together in 12 hours. If the first works 4 hours and the second 6 hours, they do $\frac{2}{5}$ of it. How long would each take alone?

Solution

Working together for 4 hours gives $\frac{4}{12} = \frac{1}{3}$.

The second's extra 2 hours give $\frac{2}{5} - \frac{1}{3} = \frac{1}{15}$, i.e.

$\frac{1}{30}$ an hour.

The first: $\frac{1}{12} - \frac{1}{30} = \frac{1}{20}$ an hour.

First: 20 h; second: 30 h.

Answer: 20 h and 30 h.

Problem 40 *

An inlet fills a tank in 4 hours; an outlet empties a full tank in 6 hours. Starting empty, the inlet runs for 2 hours, then the outlet is opened too. How long after the start is the tank full?

Solution

After 2 hours the tank is $\frac{2}{4} = \frac{1}{2}$ full.

With both open it gains $\frac{1}{4} - \frac{1}{6} = \frac{1}{12}$ an hour, needing $\frac{1}{2} : \frac{1}{12} = 6$ more hours.

In all: $2 + 6 = 8$ hours.

Answer: 8 hours.

Problem 41 *

A and B together finish a job in 6 days; A alone takes 10 days. C alone takes twice as long as B. How long do all three take together?

Solution

B does $\frac{1}{6} - \frac{1}{10} = \frac{1}{15}$ a day, so B takes 15 days and C 30 days.

All three: $\frac{1}{10} + \frac{1}{15} + \frac{1}{30} = \frac{6}{30} = \frac{1}{5}$ a day.

Together: 5 days.

Answer: 5 days.

Problem 42 *

Three people finish a job together and are paid 3 600 000 dong. Alone, A would take 6 days, B 12 and C 4. Shared by the work done, how much does each get?

Solution

Each day A does $\frac{2}{12}$, B $\frac{1}{12}$ and C $\frac{3}{12}$; the ratio is $2 : 1 : 3$.

One part: $3\,600\,000 : 6 = 600\,000$ dong.

A: 1 200 000; B: 600 000; C: 1 800 000 dong.

Answer: A 1 200 000; B 600 000; C 1 800 000 dong.

Problem 43 *

Running pipe I for 2 hours and then pipe II for 3 hours fills a tank. Both together fill it in 2 hours 24 minutes. How long does each take alone?

Solution

2 h 24 min $= \frac{12}{5}$ hours, so together they fill $\frac{5}{12}$ an hour.

Both for 2 hours give $\frac{10}{12}$; pipe II's extra hour gives the rest, $1 - \frac{10}{12} = \frac{1}{6}$.

Pipe II: 6 h; pipe I fills $\frac{5}{12} - \frac{1}{6} = \frac{1}{4}$ an hour: 4 h.

Answer: Pipe I 4 h, pipe II 6 h.

Problem 44 *

A alone takes 10 days and B alone 15. They start together, but A takes the first 5 days off while B works throughout. How many days does the job take?

Solution

In the first 5 days B does $\frac{5}{15} = \frac{1}{3}$; $\frac{2}{3}$ is left.

Then both do $\frac{1}{10} + \frac{1}{15} = \frac{1}{6}$ a day, needing $\frac{2}{3} : \frac{1}{6} = 4$ days.

In all: $5 + 4 = 9$ days.

Answer: 9 days.

Problem 45 *

A tank has two inlets and one outlet. Inlet I fills it in 3 hours, inlet II in 4; the outlet empties it in 6. Starting empty with all three open, how long until it is full?

Solution

Each hour: $\frac{1}{3} + \frac{1}{4} - \frac{1}{6} = \frac{4+3-2}{12} = \frac{5}{12}$ of the tank.

Time: $1 : \frac{5}{12} = \frac{12}{5}$ hours $= 2$ h 24 min.

Answer: 2 h 24 min.

Problem 46 *

Class 5A cleans the schoolyard in 3 hours, class 5B in 6. 5A works for 1 hour, then 5B joins. How much longer do they need?

Solution

After 1 hour 5A has done $\frac{1}{3}$; $\frac{2}{3}$ is left.

Together: $\frac{1}{3}+\frac{1}{6}=\frac{1}{2}$ an hour.

Time: $\frac{2}{3}:\frac{1}{2}=\frac{4}{3}$ hours $=1$ h 20 min.

Answer: 1 h 20 min.

Problem 47 *

One pump empties a pond in 10 hours. After 4 hours a second pump joins, and 2 hours later the pond is empty. How long would the second pump take alone?

Solution

In 4 hours pump I removes $\frac{4}{10}=\frac{2}{5}$; $\frac{3}{5}$ is left.

The two remove $\frac{3}{5}$ in 2 hours: $\frac{3}{10}$ an hour.

Pump II: $\frac{3}{10}-\frac{1}{10}=\frac{1}{5}$ an hour, so 5 hours alone.

Answer: 5 hours.

Problem 48 **

Pipes I and II together fill a tank in 2 hours; pipes II and III in 3 hours; pipe I alone in 3 hours. How long do all three take together?

Solution

Pipe II: $\frac{1}{2}-\frac{1}{3}=\frac{1}{6}$ an hour.

Pipe III: $\frac{1}{3}-\frac{1}{6}=\frac{1}{6}$ an hour.

All three: $\frac{1}{3}+\frac{1}{6}+\frac{1}{6}=\frac{2}{3}$ an hour.

Time: $1:\frac{2}{3}=\frac{3}{2}$ hours $=1$ h 30 min.

Answer: 1 h 30 min.

Problem 49 **

Worker A alone is 6 hours quicker than worker B alone. Together they finish in 4 hours. How long does each take alone? (Hint: try values.)

Solution

Each alone must take more than 4 hours.

Try A 5 h, B 11 h: $\frac{1}{5}+\frac{1}{11}=\frac{16}{55}\neq\frac{1}{4}$.

Try A 6 h, B 12 h: $\frac{1}{6}+\frac{1}{12}=\frac{3}{12}=\frac{1}{4}$ (correct).

A: 6 h; B: 12 h.

Answer: A 6 h, B 12 h.

Problem 50 ★★

A master craftsman working 8 hours a day finishes a job in 12 days. If an assistant who works at half his rate joins him (8 hours a day), how many days will the job take?

Solution

Each day the master does $\frac{1}{12}$ and the assistant $\frac{1}{24}$ of the job.

Together: $\frac{1}{12}+\frac{1}{24}=\frac{1}{8}$ a day.

So 8 days.

Answer: 8 days.

9 (*) The Method of Two Differences

① Compare two ways of sharing

Many problems describe *the same amount* shared out (or arranged) in **two different ways**. Compare:

- **The difference per share:** how much more each share (each child, each bench, each hour,...) gets in one way than in the other.
- **The total difference:** how much more is needed in one way than in the other. If one way has some *left over* and the other is *short*, the total difference is *left over* + *short*; if both are left over (or both short), subtract the smaller from the larger.

Number of shares = (total difference) : (difference per share). From it, find the amount asked for.

The *method of supposition* rests on the same idea: suppose for a moment that everything is of one kind, then compare with what really happens.

Example 1

A teacher shares sweets among a group. Giving 5 to each child leaves 8 over; giving 7 to each child is 4 short. How many children and how many sweets are there?

Guided solution.

The second time each child gets $7 - 5 = 2$ more.

The second way needs $8 + 4 = 12$ more sweets (the 8 left over, plus 4 more).

Children: $12 : 2 = 6$.

Sweets: $5 \times 6 + 8 = 38$. (*Check:* $7 \times 6 - 4 = 38$.)

Answer: 6 children, 38 sweets.

Example 2

Class 5B sets out benches for a meeting. With 3 pupils per bench, 6 pupils have no seat. With 4 per bench, 2 benches are left empty. How many benches and how many pupils are there?

Guided solution.

Two empty benches in the second way means $4 \times 2 = 8$ more pupils would be needed to fill every bench.

Difference per bench: $4 - 3 = 1$ pupil.

Total difference: $6 + 8 = 14$ pupils.

Benches: $14 : 1 = 14$.
Pupils: $3 \times 14 + 6 = 48$.
Answer: 14 benches, 48 pupils.

Example 3

Some chickens and dogs have 36 heads and 100 legs altogether. How many chickens and how many dogs are there?

Guided solution.

Suppose all 36 animals were chickens: there would be $36 \times 2 = 72$ legs, which is $100 - 72 = 28$ too few.
Swapping one chicken for one dog adds $4 - 2 = 2$ legs.
Dogs: $28 : 2 = 14$; chickens: $36 - 14 = 22$.
Answer: 22 chickens, 14 dogs.

Example 4

Someone travels from home to town. Going 12 km each hour, they arrive 1 hour later than planned; going 15 km each hour, 1 hour early. Find the distance and the planned time.

Guided solution.

In exactly the planned time, at 12 km an hour they would be 12 km short (the late hour); at 15 km an hour they would go 15 km too far.
Difference per hour: $15 - 12 = 3$ km; total difference: $12 + 15 = 27$ km.
Planned time: $27 : 3 = 9$ hours.
Distance: $12 \times (9 + 1) = 120$ km.
Answer: 120 km; 9 hours.

Practice

Problem 1

Sharing notebooks: 4 each leaves 5 over; 5 each is 3 short. Find the number of pupils and of notebooks.

Solution

Difference per pupil: 1; total difference: $5 + 3 = 8$.
Pupils: 8; notebooks: $4 \times 8 + 5 = 37$.
Answer: 8 pupils, 37 notebooks.

Problem 2

Sharing cakes: 3 each leaves 16 over; 5 each still leaves 4 over. How many children and cakes are there?

Solution

Both ways leave some over: total difference $16 - 4 = 12$; per child 2.
Children: 6; cakes: $3 \times 6 + 16 = 34$.
Answer: 6 children, 34 cakes.

Problem 3

Sharing oranges: 6 each is 2 short; 8 each is 14 short. Find the number of children and of oranges.

Solution

Both ways are short: total difference $14 - 2 = 12$; per child 2.
Children: 6; oranges: $6 \times 6 - 2 = 34$.
Answer: 6 children, 34 oranges.

Problem 4

A school hires buses for a trip. With 40 pupils per bus, 10 have no seat; with 45 per bus, 15 seats are spare. How many buses and pupils are there?

Solution

Per bus: 5 seats; in total: $10 + 15 = 25$.
Buses: $25 : 5 = 5$.
Pupils: $40 \times 5 + 10 = 210$.
Answer: 5 buses, 210 pupils.

Problem 5

Mum takes some money to the market. 5 kg of apples would be 20 000 dong too much; 4 kg would leave 10 000 dong over. Find the price of 1 kg and how much money she has.

Solution

One extra kilogram makes the difference $20\,000 + 10\,000 = 30\,000$ dong, so 1 kg costs 30 000.
Money: $30\,000 \times 4 + 10\,000 = 130\,000$ dong.
Answer: 30 000 dong; 130 000 dong.

Problem 6

With 5 people per bench, 4 must stand; with 6 per bench, 1 bench is empty. How many benches and people are there?

Solution

The empty bench means 6 people short.
Total difference: $4 + 6 = 10$; per bench 1.
Benches: 10; people: $5 \times 10 + 4 = 54$.
Answer: 10 benches, 54 people.

Problem 7

A group arrives at a guesthouse. With 4 per room, 6 guests have no room; with 6 per room, 2 rooms are empty. How many rooms and guests are there?

Solution

Two empty rooms mean 12 guests short.
Total difference $6 + 12 = 18$; per room 2.
Rooms: 9; guests: $4 \times 9 + 6 = 42$.
Answer: 9 rooms, 42 guests.

Problem 8

Find a number such that 7 times it is 13 more than 50, and 5 times it is 5 less than 50.

Solution

The two products differ by $13 + 5 = 18$, and the multipliers by $7 - 5 = 2$.
The number is $18 : 2 = 9$.
Answer: 9.

Problem 9

3 pens and 5 notebooks cost 49 000 dong; 3 pens and 8 notebooks of the same kind cost 64 000 dong. Find the price of a pen and of a notebook.

Solution

The two purchases differ by 3 notebooks and $64\,000 - 49\,000 = 15\,000$ dong, so a notebook costs 5 000 dong.
3 pens cost $49\,000 - 25\,000 = 24\,000$ dong, so a pen costs 8 000 dong.
Answer: Pen: 8 000 dong; notebook: 5 000 dong.

Problem 10

4 pencils and 3 rubbers cost 22 000 dong; 2 pencils and 3 rubbers cost 14 000 dong. Find the price of each.

Solution

2 pencils cost $22\,000 - 14\,000 = 8\,000$, so one costs 4 000 dong.
3 rubbers cost $14\,000 - 8\,000 = 6\,000$, so one costs 2 000 dong.
Answer: Pencil: 4 000 dong; rubber: 2 000 dong.

Problem 11

The 30 pupils of class 5C buy 100 notebooks; each boy buys 4 and each girl buys 3. How many boys and girls are there?

Solution

If all 30 were girls they would buy 90, which is 10 too few.
Each boy buys 1 more than a girl, so there are 10 boys and 20 girls.
Answer: 10 boys, 20 girls.

Problem 12

A cyclist rides from A to B. At 10 km an hour they arrive 1 hour late; at 15 km an hour, 1 hour early. Find the distance AB.

Solution

In the planned time, at 10 km an hour they would be 10 km short; at 15, 15 km too far.
Planned time: $(10 + 15) : (15 - 10) = 5$ hours.
Distance: $10 \times (5 + 1) = 60$ km.
Answer: 60 km.

Problem 13

Sharing sweets in a group: 4 each leaves 6 over; 6 each is 4 short. How many children and how many sweets are there?

Solution

The two sharings differ by $6 + 4 = 10$ sweets.
Each child differs by $6 - 4 = 2$.
Children: $10 : 2 = 5$.
Sweets: $4 \times 5 + 6 = 26$.
Answer: 5 children; 26 sweets.

Problem 14

Sharing notebooks: 3 each leaves 12 over; 5 each still leaves 2 over. Find the number of pupils and of notebooks.

Solution

The leftovers differ by $12 - 2 = 10$.
Each pupil differs by $5 - 3 = 2$.
Pupils: $10 : 2 = 5$.
Notebooks: $3 \times 5 + 12 = 27$.
Answer: 5 pupils; 27 notebooks.

Problem 15

Sharing tangerines: 7 each is 3 short; 9 each is 15 short. Find the number of children and of tangerines.

Solution

The shortfalls differ by $15 - 3 = 12$.
Each child differs by $9 - 7 = 2$.
Children: $12 : 2 = 6$.
Tangerines: $7 \times 6 - 3 = 39$.
Answer: 6 children; 39 tangerines.

Problem 16

There are 30 chickens and rabbits with 84 legs in all. How many of each are there?

Solution

Suppose all 30 are chickens: $2 \times 30 = 60$ legs.
Missing legs: $84 - 60 = 24$.
Each rabbit has 2 more legs, so rabbits: $24 : 2 = 12$.
Chickens: $30 - 12 = 18$.
Answer: 18 chickens, 12 rabbits.

Problem 17

A car park holds 25 bicycles (2 wheels) and pedicabs (3 wheels), with 62 wheels in all. How many of each are there?

Solution

Suppose all 25 are bicycles: $2 \times 25 = 50$ wheels.
Missing: $62 - 50 = 12$ wheels.

Each pedicab has 1 more wheel, so pedicabs: 12.
Bicycles: $25 - 12 = 13$.
Answer: 13 bicycles, 12 pedicabs.

Problem 18

Lan has 20 notes of 2 000 and 5 000 dong, worth 64 000 dong in all. How many of each are there?

Solution

Suppose all are 2 000 notes: $2\,000 \times 20 = 40\,000$ dong.
Missing: $64\,000 - 40\,000 = 24\,000$ dong.
A 5 000 note is worth 3 000 more.
5 000 notes: $24\,000 : 3\,000 = 8$.
2 000 notes: $20 - 8 = 12$.
Answer: 12 notes of 2 000, 8 notes of 5 000.

Problem 19

2 pens and 3 notebooks cost 34 000 dong; 2 pens and 5 notebooks cost 46 000 dong. Find the price of each.

Solution

The second buys 2 more notebooks for 12 000 dong more.
One notebook: $12\,000 : 2 = 6\,000$ dong.
2 pens: $34\,000 - 6\,000 \times 3 = 16\,000$ dong.
One pen: 8 000 dong.
Answer: Pen 8 000 dong, notebook 6 000 dong.

Problem 20

3 kg of oranges and 2 kg of apples cost 150 000 dong; 5 kg of oranges and 2 kg of apples cost 200 000 dong. Find the price per kg of each.

Solution

2 kg of oranges cost $200\,000 - 150\,000 = 50\,000$ dong.
1 kg of oranges: 25 000 dong.
2 kg of apples: $150\,000 - 25\,000 \times 3 = 75\,000$ dong.
1 kg of apples: 37 500 dong.
Answer: Oranges 25 000 dong/kg, apples 37 500 dong/kg.

Problem 21

Find a number such that 6 times it is 14 more than 40, and 4 times it is 4 less than 40.

Solution

The two products differ by $14 + 4 = 18$.
The multipliers differ by $6 - 4 = 2$.
The number: $18 : 2 = 9$.
Check: $9 \times 6 = 54 = 40 + 14$; $9 \times 4 = 36 = 40 - 4$.
Answer: 9.

Problem 22

With 4 people on each bench, 3 must stand; with 5 on each bench, 2 benches are empty. How many benches and people are there?

Solution

The 2 empty benches could seat $5 \times 2 = 10$.
The two arrangements differ by $3 + 10 = 13$ people.
Each bench differs by $5 - 4 = 1$.
Benches: 13.
People: $4 \times 13 + 3 = 55$.
Answer: 13 benches, 55 people.

Problem 23

Pupils stand in rows. With 6 per row, 4 are left over; with 7 per row (same number of rows), 3 are missing. How many rows and pupils are there?

Solution

The two ways differ by $4 + 3 = 7$ pupils.
Each row differs by 1.
Rows: 7.
Pupils: $6 \times 7 + 4 = 46$.
Answer: 7 rows, 46 pupils.

Problem 24

A rider goes from A to B. At 15 km per hour he arrives 2 hours late; at 20 km per hour he arrives 1 hour early. Find the planned time and the distance AB.

Solution

At 15 km/h in the planned time he is $15 \times 2 = 30$ km short.
At 20 km/h in the planned time he goes $20 \times 1 = 20$ km too far.
The difference is $30 + 20 = 50$ km, and $20 - 15 = 5$ km per hour.
Planned time: $50 : 5 = 10$ hours.
AB: $15 \times (10 + 2) = 180$ km.
Answer: 10 hours; 180 km.

Problem 25

A test has 10 questions: 10 points for a right answer, 5 points off for a wrong one. Minh answers all and scores 70. How many are right?

Solution

All right would give $10 \times 10 = 100$ points.
He is $100 - 70 = 30$ points down.
Each wrong answer costs $10 + 5 = 15$ points.
Wrong: $30 : 15 = 2$; right: $10 - 2 = 8$.
Answer: 8 questions.

Problem 26

20 spiders (8 legs) and beetles (6 legs) have 136 legs in all. How many of each are there?

Solution

Suppose all are beetles: $6 \times 20 = 120$ legs.
Missing: $136 - 120 = 16$.
Each spider has 2 more, so spiders: $16 : 2 = 8$.
Beetles: $20 - 8 = 12$.
Answer: 8 spiders, 12 beetles.

Problem 27

10 boats, large ones carrying 6 and small ones carrying 4, carry exactly 48 people. How many of each are there?

Solution

Suppose all are small: $4 \times 10 = 40$ people.
Missing: $48 - 40 = 8$.
A large boat carries 2 more, so large boats: $8 : 2 = 4$.
Small boats: $10 - 4 = 6$.

Answer: 4 large, 6 small.

Problem 28

Binh buys 16 stamps at 3 000 and 5 000 dong, paying 62 000 dong. How many of each does he buy?

Solution

Suppose all cost 3 000: $3\,000 \times 16 = 48\,000$ dong.
Missing: $62\,000 - 48\,000 = 14\,000$ dong.
A 5 000 stamp costs 2 000 more.
5 000 stamps: $14\,000 : 2\,000 = 7$.
3 000 stamps: $16 - 7 = 9$.
Answer: 9 at 3 000, 7 at 5 000.

Problem 29

4 notebooks and 2 pens cost 40 000 dong; 2 notebooks and 3 pens cost 36 000 dong. Find the price of each.

Solution

Double the second purchase: 4 notebooks and 6 pens cost 72 000 dong.
Compared with the first, that is 4 more pens for 32 000 dong more.
One pen: $32\,000 : 4 = 8\,000$ dong.
One notebook: $(40\,000 - 8\,000 \times 2) : 4 = 6\,000$ dong.
Answer: Notebook 6 000 dong, pen 8 000 dong.

Problem 30

Sharing balls among classes: 5 each leaves 3 over; 7 each is 5 short. Find the number of classes and of balls.

Solution

The two ways differ by $3 + 5 = 8$ balls.
Each class differs by 2.
Classes: $8 : 2 = 4$.
Balls: $5 \times 4 + 3 = 23$.
Answer: 4 classes, 23 balls.

Challenge – Extension

Problem 31 *

Sharing sweets: 3 each leaves 2 over; 4 each means one child gets only 2. How many children and sweets are there?

Solution

The second way is $4 - 2 = 2$ short.
Total difference $2 + 2 = 4$; per child 1.
Children: 4; sweets: $3 \times 4 + 2 = 14$.
Answer: 4 children, 14 sweets.

Problem 32 *

Pupils line up in rows. With 8 per row, 3 are left over; with 9 per row (same number of rows), the last row is 5 short. How many pupils are there?

Solution

Total difference $3 + 5 = 8$; per row 1.
Rows: 8; pupils: $8 \times 8 + 3 = 67$.
Answer: 67 pupils.

Problem 33 *

A team planned to make 30 items a day. In fact it made 36 a day, finished 2 days early and made 12 extra items. How many items were planned?

Solution

Working all the planned days at 36 a day would make $36 \times 2 + 12 = 84$ more than planned.
Each day gives 6 extra, so the plan was $84 : 6 = 14$ days, i.e.
$30 \times 14 = 420$ items.
Answer: 420 items.

Problem 34 *

An buys 5 blue pens and 3 red pens for 33 000 dong. Binh buys 3 blue and 5 red pens of the same kinds for 31 000 dong. Find the price of each kind.

Solution

Together: 8 blue and 8 red cost 64 000, so 1 blue + 1 red cost 8 000 dong.
Compared with Binh, An has 2 more blue and 2 fewer red and pays 2 000 more, so a blue pen costs 1 000 more than a red one.
Blue: $(8\,000 + 1\,000) : 2 = 4\,500$ dong; red: 3 500 dong.
Answer: Blue: 4 500 dong; red: 3 500 dong.

Problem 35 ★★

A worker is hired for 30 days. Each working day pays 200 000 dong and each day off costs a 50 000 dong deduction. After 30 days the worker receives 4 750 000 dong. How many days off were taken?

Solution

Working all 30 days would pay 6 000 000, which is 1 250 000 more than received.
Each day off lowers the pay by $200\,000 + 50\,000 = 250\,000$, so the days off are $1\,250\,000 : 250\,000 = 5$.
Answer: 5 days.

Problem 36 ★★

A contest has 20 problems. Each correct answer scores 5 points; each wrong or skipped one loses 2 points. Minh scores 72. How many did Minh get right?

Solution

All 20 correct would give 100, which is 28 more.
Each wrong answer costs $5 + 2 = 7$, so $28 : 7 = 4$ are wrong and 16 right.
Answer: 16 problems.

Problem 37 ★

Sharing sweets: 5 each leaves 7 over; 6 each means one child gets only 2. Find the number of children and of sweets.

Solution

With 6 each, the last child is $6 - 2 = 4$ short.
The two ways differ by $7 + 4 = 11$; each child differs by 1.
Children: 11.
Sweets: $5 \times 11 + 7 = 62$.
Answer: 11 children, 62 sweets.

Problem 38 ★

There are 38 chickens and dogs. The chickens have 16 more legs than the dogs. How many of each are there?

Solution

If all 38 were chickens, the difference would be $2 \times 38 = 76$ legs.
That is $76 - 16 = 60$ too many.
Swapping a chicken for a dog lowers the difference by $2 + 4 = 6$.
Dogs: $60 : 6 = 10$; chickens: $38 - 10 = 28$.
Check: $56 - 40 = 16$.

Answer: 28 chickens, 10 dogs.

Problem 39 *

A worker is hired for 26 days: paid 300 000 dong for each working day and fined 100 000 dong for each day off. He receives 6 200 000 dong. How many days off did he take?

Solution

Working all 26 days: $300\,000 \times 26 = 7\,800\,000$ dong.
Shortfall: $7\,800\,000 - 6\,200\,000 = 1\,600\,000$ dong.
Each day off costs $300\,000 + 100\,000 = 400\,000$ dong.
Days off: $1\,600\,000 : 400\,000 = 4$.
Answer: 4 days.

Problem 40 *

A car leaves A. At 40 km/h it reaches B at 11 o'clock; at 50 km/h at 10 o'clock. Find AB and the departure time.

Solution

At 40 km/h the trip takes $11 - 10 = 1$ hour longer.
If the fast car drove on for that hour after B, it would pass B by $50 \times 1 = 50$ km.
So in the same time the fast car covers 50 km more, gaining $50 - 40 = 10$ km each hour.
Time at 40 km/h: $50 : 10 = 5$ hours.
AB: $40 \times 5 = 200$ km.
Departure: $11 - 5 = 6$ o'clock.
Answer: 200 km; leaves at 6 o'clock.

Problem 41 *

Pupils sit at desks. With 2 per desk, 8 have no seat; with 3 per desk, 1 desk is empty and one desk has only 1 pupil. Find the number of desks and pupils.

Solution

With 3 per desk, the empty desk is 3 short and the other 2 short: 5 short in all.
The two ways differ by $8 + 5 = 13$ pupils; each desk differs by 1.
Desks: 13.
Pupils: $2 \times 13 + 8 = 34$.
Answer: 13 desks, 34 pupils.

Problem 42 *

A test has 25 questions: 4 points for each right answer, 1 off for each wrong one. Hoa answers all and scores 70. How many are wrong?

Solution

All right gives $4 \times 25 = 100$.
Shortfall: $100 - 70 = 30$.
Each wrong answer costs $4 + 1 = 5$.
Wrong: $30 : 5 = 6$.
Answer: 6 questions.

Problem 43 *

2 shirts and 3 trousers cost 1 100 000 dong; 3 shirts and 2 trousers cost 1 150 000 dong. Find each price.

Solution

Together: 5 shirts and 5 trousers cost 2 250 000 dong.
One of each: $2\,250\,000 : 5 = 450\,000$ dong.
The second buys one more shirt and one fewer trousers for 50 000 more, so a shirt costs 50 000 more.
Shirt: $(450\,000 + 50\,000) : 2 = 250\,000$ dong.
Trousers: $450\,000 - 250\,000 = 200\,000$ dong.
Answer: Shirt 250 000 dong, trousers 200 000 dong.

Problem 44 *

Mum buys some oranges. Selling at 30 000 dong/kg she gains 40 000 dong; at 25 000 dong/kg she loses 10 000 dong. Find the amount of oranges and the cost.

Solution

The two prices give totals differing by $40\,000 + 10\,000 = 50\,000$ dong.
Per kg: $30\,000 - 25\,000 = 5\,000$ dong.
Oranges: $50\,000 : 5\,000 = 10$ kg.
Cost: $30\,000 \times 10 - 40\,000 = 260\,000$ dong.
Answer: 10 kg; cost 260 000 dong.

Problem 45 *

18 spiders (8 legs, no wings) and dragonflies (6 legs, 2 pairs of wings) have 124 legs. How many pairs of wings are there?

Solution

If all were dragonflies: $6 \times 18 = 108$ legs.
Missing $124 - 108 = 16$; a spider has 2 more.
Spiders: 8; dragonflies: 10.
Pairs of wings: $2 \times 10 = 20$.
Answer: 20 pairs.

Problem 46 *

A shop has 45 boxes of cakes, holding 6 or 10 each, 350 cakes in all. How many boxes of each are there?

Solution

If all held 6: $6 \times 45 = 270$.
Missing 80; a box of 10 has 4 more.
Boxes of 10: $80 : 4 = 20$.
Boxes of 6: 25.
Answer: 25 boxes of 6, 20 boxes of 10.

Problem 47 **

An cycles to school. At 12 km/h he is 5 minutes late; at 15 km/h he is 5 minutes early. How far is the school?

Solution

5 minutes $= \frac{1}{12}$ hour.

At 12 km/h in the set time he is $12 \times \frac{1}{12} = 1$ km short.

At 15 km/h he goes $15 \times \frac{1}{12} = 1.25$ km too far.

Difference: 2.25 km, at 3 km per hour.
Set time: $2.25 : 3 = 0.75$ hour.
Distance: $12 \times 0.75 + 1 = 10$ km.
Answer: 10 km.

Problem 48 *

A car park has 50 vehicles: cars (4 wheels) and motorbikes (2 wheels). The cars have 20 more wheels than the motorbikes. How many of each are there?

Solution

If all 50 were cars, the difference would be $4 \times 50 = 200$.
That is 180 too many.
Swapping a car for a motorbike lowers it by $4 + 2 = 6$.
Motorbikes: $180 : 6 = 30$; cars: 20.
Check: $80 - 60 = 20$.
Answer: 20 cars, 30 motorbikes.

Problem 49 ★★

A test has 30 questions: 5 points for right, 2 off for wrong, 0 for skipped. Binh scores 112, and has twice as many wrong as skipped. How many are right?

Solution

Group 1 skipped with 2 wrong into a block of 3.
All right: $5 \times 30 = 150$; shortfall: $150 - 112 = 38$.
Each block costs $5 + (5 + 2) \times 2 = 19$.
Blocks: $38 : 19 = 2$, i.e.
2 skipped and 4 wrong.
Right: $30 - 2 - 4 = 24$.
Answer: 24 questions.

Problem 50 ★

Hung takes some money to buy notebooks. Buying 8 leaves 4 000 dong; buying 10 is 6 000 dong short. How much money does he have?

Solution

The two purchases differ by $4\,000 + 6\,000 = 10\,000$ dong for 2 notebooks.
One notebook: 5 000 dong.
Money: $5\,000 \times 8 + 4\,000 = 44\,000$ dong.
Answer: 44 000 dong.

10 Working Backwards

① Start from the final result

When a problem gives the **final result** of a chain of operations (or a chain of changes), start from that result and **go backwards** one step at a time, using the *inverse operation* at each step:

add ⟷ subtract multiply ⟷ divide

- Draw a **flow diagram** of the steps so that none is missed or taken out of order.
- For "take away a fraction, then add (or remove) a few more" problems, undo the *extra units* first and then the *fraction* at each backward step.
- For problems where amounts pass between several people (or jars), make a **table** of the amounts after each transfer and work from the bottom row up.

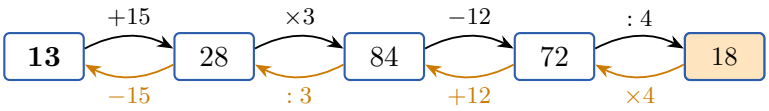
Always **check** by working forwards again.

Example 1

Find a number such that adding 15, multiplying by 3, subtracting 12 and finally dividing by 4 gives 18.

Guided solution.

The diagram (forwards on top, backwards below):



Backwards: $18 \times 4 = 72$;
$72 + 12 = 84$;
$84 : 3 = 28$;
$28 - 15 = 13$.
Check: $(13 + 15) \times 3 = 84$;
$84 - 12 = 72$;
$72 : 4 = 18$.
Answer: 13.

Example 2

A woman takes eggs to market. First she sells half of them and 1 more. Then she sells half of what is left and 1 more. She has 10 eggs left. How many did she bring?

Guided solution.

Undo the second sale: without the extra egg she would have $10 + 1 = 11$ left, which is half of what she had before.
So before the second sale: $11 \times 2 = 22$ eggs.
Undo the first sale: $(22 + 1) \times 2 = 46$ eggs.
Answer: 46 eggs.

Example 3

On day one, $\frac{1}{3}$ of the water in a tank is used; on day two, $\frac{2}{5}$ of what is left. Now 120 litres remain. How much water was in the tank at first?

Guided solution.

After day two, $1 - \frac{2}{5} = \frac{3}{5}$ of the day-one remainder is left.

That remainder was $120 : \frac{3}{5} = 200$ litres.

200 litres is $1 - \frac{1}{3} = \frac{2}{3}$ of the original.

At first: $200 : \frac{2}{3} = 300$ litres.

Answer: 300 litres.

Example 4

Three drums hold 240 litres of oil. First, drum I pours into drum II as much as drum II holds. Then drum II pours into drum III as much as drum III holds. Then drum III pours into drum I as much as drum I holds. Now all three drums hold the same. How much did each hold at first?

Guided solution.

At the end each drum holds $240 : 3 = 80$ litres.
A drum that "receives as much as it holds" doubles; going back, we *halve* the receiving drum and *give* that amount back to the pouring drum.

	Drum I	Drum II	Drum III
At the end	80	80	80
Before step 3	40	80	120
Before step 2	40	140	60
At first	**110**	**70**	**60**

Answer: Drum I: 110 litres; drum II: 70 litres; drum III: 60 litres.

Practice

Problem 1

Find a number such that multiplying it by 4, subtracting 18 and dividing by 2 gives 25.

Solution

$25 \times 2 = 50$;
$50 + 18 = 68$;
$68 : 4 = 17$.
Answer: 17.

Problem 2

Nam thinks of a number. He multiplies it by 6, adds 8, divides by 2 and subtracts 14, getting 20. What was his number?

Solution

$20 + 14 = 34$;
$34 \times 2 = 68$;
$68 - 8 = 60$;
$60 : 6 = 10$.
Answer: 10.

Problem 3

Find a number such that adding $\frac{1}{2}$ and then multiplying by $\frac{2}{3}$ gives 3.

Solution

$3 : \frac{2}{3} = \frac{9}{2}$;
$\frac{9}{2} - \frac{1}{2} = 4$.
Answer: 4.

Problem 4

If An's age is doubled, 5 is added and the result is divided by 3, the answer is 9. How old is An?

Solution

$9 \times 3 = 27$;
$27 - 5 = 22$;
$22 : 2 = 11$.

Answer: 11 years old.

Problem 5

Lan spends $\frac{1}{2}$ of her money on books and $\frac{1}{3}$ of the rest on pens, leaving 20 000 dong. How much did she have at first?

Solution

20 000 is $\frac{2}{3}$ of what was left after the books, i.e. 30 000.

At first: $30\,000 \times 2 = 60\,000$ dong.

Answer: 60 000 dong.

Problem 6

Hoa reads $\frac{1}{4}$ of a book on day one, $\frac{1}{3}$ of the remaining pages on day two, and the last 60 pages on day three. How many pages does the book have?

Solution

60 pages are $\frac{2}{3}$ of what was left after day one: $60 : \frac{2}{3} = 90$.

And 90 pages are $\frac{3}{4}$ of the book: $90 : \frac{3}{4} = 120$.

Answer: 120 pages.

Problem 7

Mr Ba sells half his hens and 2 more, then half of the rest and 2 more, and has 8 left. How many hens did he have?

Solution

Before the second sale: $(8 + 2) \times 2 = 20$.

At first: $(20 + 2) \times 2 = 44$.

Answer: 44 hens.

Problem 8

At the market Mum first spends half her money and 5 000 dong more, then half the rest and 5 000 more, leaving 10 000 dong. How much did she take?

Solution

Before the second purchase: $(10\,000 + 5\,000) \times 2 = 30\,000$.

At first: $(30\,000 + 5\,000) \times 2 = 70\,000$ dong.

Answer: 70 000 dong.

Problem 9

15 litres are poured from tank A into tank B, then 5 litres from B back into A. Now each holds 40 litres. How much did each hold at first?

Solution

Tank A: $40 - 5 + 15 = 50$ litres; tank B: $40 + 5 - 15 = 30$ litres.

Answer: A: 50 litres; B: 30 litres.

Problem 10

6 oranges move from basket I to basket II, then 2 move from II back to I. Now basket I has 20 and basket II has 30. How many were in each at first?

Solution

Basket I: $20 - 2 + 6 = 24$; basket II: $30 + 2 - 6 = 26$.

Answer: 24 and 26 oranges.

Problem 11

Jar I pours half its water into jar II; then jar II pours $\frac{1}{3}$ of what it holds into jar I. Now jar I has 8 litres and jar II 12. How much did each have at first?

Solution

Before the second pour, jar II kept $\frac{2}{3}$, so it had $12 : \frac{2}{3} = 18$ and gave 6; jar I had $8 - 6 = 2$.

At first jar I had $2 \times 2 = 4$ and gave 2; jar II had $18 - 2 = 16$.

Answer: Jar I: 4 litres; jar II: 16 litres.

Problem 12

Find a number: add 7, multiply by 5, then subtract 10 to get 65.

Solution

Before subtracting 10: $65 + 10 = 75$.
Before multiplying by 5: $75 : 5 = 15$.
Before adding 7: $15 - 7 = 8$.
Answer: 8.

Problem 13

Find a number: multiply by 3, add 12, then divide by 5 to get 9.

Solution

Before dividing by 5: $9 \times 5 = 45$.
Before adding 12: $45 - 12 = 33$.
The number: $33 : 3 = 11$.
Answer: 11.

Problem 14

Find a number: subtract 4, multiply by 6, then add 3 to get 51.

Solution

Before adding 3: $51 - 3 = 48$.
Before multiplying by 6: $48 : 6 = 8$.
The number: $8 + 4 = 12$.
Answer: 12.

Problem 15

Find a number: divide by 2, add 15, then multiply by 4 to get 100.

Solution

Before multiplying by 4: $100 : 4 = 25$.
Before adding 15: $25 - 15 = 10$.
The number: $10 \times 2 = 20$.
Answer: 20.

Problem 16

Find x: multiply by 2.5, then subtract 1.5 to get 8.5.

Solution

Before subtracting 1.5: $8.5 + 1.5 = 10$.
$x = 10 : 2.5 = 4$.
Answer: $x = 4$.

Problem 17

Find a decimal: add 2.4, then multiply by 3 to get 12.6.

Solution

Before multiplying by 3: $12.6 : 3 = 4.2$.
The number: $4.2 - 2.4 = 1.8$.
Answer: 1.8.

Problem 18

Mai spends 25 000 dong on a book, gets 10 000 dong from Mum, then spends 8 000 dong on a pen and has 17 000 dong left. How much did she start with?

Solution

Before the pen: $17\,000 + 8\,000 = 25\,000$ dong.
Before Mum's gift: $25\,000 - 10\,000 = 15\,000$ dong.
At first: $15\,000 + 25\,000 = 40\,000$ dong.
Answer: 40 000 dong.

Problem 19

A bus leaves the depot. At the first stop 6 get off and 4 get on; at the second 10 get off and 3 get on. Now 21 passengers are aboard. How many were there at the start?

Solution

Before the second stop: $21 - 3 + 10 = 28$.
Before the first stop: $28 - 4 + 6 = 30$.
Answer: 30 passengers.

Problem 20

Dad's age divided by 4, plus 3, gives 13. How old is Dad?

Solution

Before adding 3: $13 - 3 = 10$.
Dad's age: $10 \times 4 = 40$.
Answer: 40 years old.

Problem 21

Lan spends $\frac{1}{3}$ of her money, then $\frac{1}{2}$ of the rest, and has 25 000 dong left. How much did she have?

Solution

25 000 dong is $\frac{1}{2}$ of what was left after the first spend.

So $25\,000 \times 2 = 50\,000$ dong was left.

That is $\frac{2}{3}$ of the original.

Original: $50\,000 : \frac{2}{3} = 75\,000$ dong.

Answer: 75 000 dong.

Problem 22

On day one $\frac{1}{4}$ of a tank's water is used; on day two $\frac{1}{3}$ of the rest. 300 litres remain. How much was there?

Solution

300 litres is $\frac{2}{3}$ of what was left after day one.

After day one: $300 : \frac{2}{3} = 450$ litres.

That is $\frac{3}{4}$ of the original.

Original: $450 : \frac{3}{4} = 600$ litres.

Answer: 600 litres.

Problem 23

A seller first sells half her eggs plus 2, then half the rest plus 1, leaving 9. How many eggs were there?

Solution

Before the extra 1: $9 + 1 = 10$, which is half of what was left after the first sale.
After the first sale: $10 \times 2 = 20$.
Likewise $20 + 2 = 22$ is half the original.
Original: $22 \times 2 = 44$.
Answer: 44 eggs.

Problem 24

Move 8 pomelos from basket A to B, then 3 from B to A. Now A has 25 and B has 35. How many did each have at first?

Solution

Before the second move: A $25 - 3 = 22$, B $35 + 3 = 38$.
At first: A $22 + 8 = 30$, B $38 - 8 = 30$.
Answer: 30 in each.

Problem 25

A shop sells $\frac{2}{5}$ of its rice on day one and half the rest on day two, leaving 45 kg. How much rice was there?

Solution

After day one: $45 \times 2 = 90$ kg.
That is $\frac{3}{5}$ of the original.
Original: $90 : \frac{3}{5} = 150$ kg.
Answer: 150 kg.

Problem 26

Find x if $\left(x - \frac{1}{3}\right) \times \frac{3}{4} = \frac{1}{2}$.

Solution

$x - \frac{1}{3} = \frac{1}{2} : \frac{3}{4} = \frac{2}{3}$.
$x = \frac{2}{3} + \frac{1}{3} = 1$.
Answer: $x = 1$.

Problem 27

An gives Binh $\frac{1}{3}$ of his marbles, then gives Chi 4, and has 12 left. How many did An have?

Solution

Before giving Chi: $12 + 4 = 16$.

That is $\frac{2}{3}$ of the original.

Original: $16 : \frac{2}{3} = 24$.

Answer: 24 marbles.

Problem 28

Find a number: multiply by 4, add 0.6, then divide by 2 to get 5.3.

Solution

Before dividing by 2: $5.3 \times 2 = 10.6$.

Before adding 0.6: 10.

The number: $10 : 4 = 2.5$.

Answer: 2.5.

Problem 29

From a roll of cloth, first $\frac{1}{4}$ is cut, then 5 m, then half of what remains, leaving 8 m. How long was the roll?

Solution

Before the third cut: $8 \times 2 = 16$ m.

Before the second: $16 + 5 = 21$ m.

That is $\frac{3}{4}$ of the roll.

Roll: $21 : \frac{3}{4} = 28$ m.

Answer: 28 m.

Problem 30

Uncle Tu sells $\frac{1}{3}$ of his ducks, then buys 15 more and has 75. How many ducks did he have?

Solution

Before buying: $75 - 15 = 60$.

That is $\frac{2}{3}$ of the flock.

Original: $60 : \frac{2}{3} = 90$.

Answer: 90 ducks.

Challenge – Extension

Problem 31 *

A shop sells $\frac{1}{3}$ of a roll of cloth and 2 m more, then half of the rest and 3 m more, leaving 12 m. How long was the roll?

Solution

Before the second sale: $(12 + 3) \times 2 = 30$ m.

Then $30 + 2 = 32$ m is $\frac{2}{3}$ of the roll, so the roll is $32 : \frac{2}{3} = 48$ m.

Answer: 48 m.

Problem 32 *

A monkey picks some peaches. Each day it eats half of what it has and 1 more. On the morning of day five it has just 1 left. How many peaches did it pick?

Solution

The monkey ate for 4 days.

Undo each day with (number left $+1$)$\times 2$.

$1 \to 4 \to 10 \to 22 \to 46$.

Answer: 46 peaches.

Problem 33 *

An, Binh and Cuong have 96 marbles. An gives Binh as many as Binh has; then Binh gives Cuong as many as Cuong has; finally Cuong gives An as many as An has. Now they all have the same. How many did each have at first?

Solution

At the end each has 32.
Work backwards (as in Example 4):
Before step 3: An 16, Binh 32, Cuong 48.
Before step 2: An 16, Binh 56, Cuong 24.
At first: An 44, Binh 28, Cuong 24.
Answer: An: 44; Binh: 28; Cuong: 24 marbles.

Problem 34 ★★

A, B and C play three rounds of a game. After each round the loser must give each of the others as many marbles as they already have. A loses round 1, B round 2 and C round 3. Afterwards each has 24 marbles. How many did each have at first?

Solution

Total: 72 marbles.
Going back, in each round the two winners halve and the loser gets those marbles back.
Before round 3: A 12, B 12, C 48.
Before round 2: A 6, B 42, C 24.
At first: A 39, B 21, C 12.
Answer: A: 39; B: 21; C: 12 marbles.

Problem 35 ★

Each day a monkey eats half its peaches plus 2 more. After 3 days it has 3 left. How many did it start with?

Solution

Start of day three: $(3 + 2) \times 2 = 10$.
Start of day two: $(10 + 2) \times 2 = 24$.
At first: $(24 + 2) \times 2 = 52$.
Answer: 52 peaches.

Problem 36 ★

Pour from jar A into jar B as much as B holds; then from B into A as much as A holds. Now each has 24 litres. How much did each have at first?

Solution

The second pour doubled A, so before it A had $24 : 2 = 12$ and B $24 + 12 = 36$.
The first pour doubled B, so before it B had $36 : 2 = 18$ and A $12 + 18 = 30$.

Answer: A 30 litres, B 18 litres.

Problem 37

A shop sells half its rice plus 5 kg, then $\frac{1}{3}$ of the rest plus 4 kg, then $\frac{1}{4}$ of the rest plus 3 kg, leaving 15 kg. How much was there?

Solution

Before the third sale: $15 + 3 = 18$ kg is $\frac{3}{4}$, so $18 : \frac{3}{4} = 24$ kg.

Before the second: $24 + 4 = 28$ kg is $\frac{2}{3}$, so 42 kg.

At first: $(42 + 5) \times 2 = 94$ kg.

Answer: 94 kg.

Problem 38

Nam thinks of a number. Three times over he doubles it and subtracts 3, ending with 19. What was the number?

Solution

Before the third round: $(19 + 3) : 2 = 11$.

Before the second: $(11 + 3) : 2 = 7$.

The number: $(7 + 3) : 2 = 5$.

Answer: 5.

Problem 39

A team repairs a road: in week one $\frac{1}{4}$ of it plus 3 km; in week two $\frac{1}{3}$ of the rest plus 1 km; in week three the last 9 km. How long is the road?

Solution

After week one, $9 + 1 = 10$ km is $\frac{2}{3}$ of what was left, so 15 km was left.

$15 + 3 = 18$ km is $\frac{3}{4}$ of the road.

Road: $18 : \frac{3}{4} = 24$ km.

Answer: 24 km.

Problem 40

Ms Hanh spends $\frac{2}{5}$ of her pay on food, $\frac{1}{3}$ of the rest on schooling and saves half of what remains, keeping 1 200 000 dong. What is her pay?

Solution

Before saving: $1\,200\,000 \times 2 = 2\,400\,000$ dong.

Before schooling: $2\,400\,000 : \frac{2}{3} = 3\,600\,000$ dong.

Pay: $3\,600\,000 : \frac{3}{5} = 6\,000\,000$ dong.

Answer: 6 000 000 dong.

Problem 41

Grandpa's age minus 4, divided by 7, plus 2, times 5 is 60. How old is he?

Solution

Before $\times 5$: 12.
Before $+2$: 10.
Before $: 7$: 70.
Age: $70 + 4 = 74$.
Answer: 74 years old.

Problem 42

Duckweed doubles its area every day and covers the whole pond at the end of day 10. On which day did it cover $\frac{1}{4}$ of the pond?

Solution

At the end of day 9 it covered $\frac{1}{2}$ (doubling gives the whole).

At the end of day 8 it covered $\frac{1}{4}$.

Answer: Day 8.

Problem 43

Three drums hold 180 litres. Pour 10 l from I to II, then 20 l from II to III, then 5 l from III to I. Now all are equal. How much did each hold?

Solution

Finally each has $180 : 3 = 60$ l.
Before the third pour: I 55, III 65.
Before the second: II 80, III 45.
At first: I 65, II 70, III 45.
Answer: I 65 l, II 70 l, III 45 l.

Problem 44 *

Find x if $(x \times 1.5 - 2.25) : 0.5 = 7.5$.

Solution

$x \times 1.5 - 2.25 = 7.5 \times 0.5 = 3.75$.
$x \times 1.5 = 6$.
$x = 4$.
Answer: $x = 4$.

Problem 45 *

Grandma spends half her money plus 3 000 dong, then half the rest plus 2 000, then half the rest plus 1 000, and has nothing left. How much did she take?

Solution

Before the third spend: $(0 + 1\,000) \times 2 = 2\,000$ dong.
Before the second: $(2\,000 + 2\,000) \times 2 = 8\,000$ dong.
At first: $(8\,000 + 3\,000) \times 2 = 22\,000$ dong.
Answer: 22 000 dong.

Problem 46 *

Two baskets hold 60 oranges. Move $\frac{1}{3}$ of basket I into II, then $\frac{1}{4}$ of basket II (as it is then) into I. Now they are equal. How many were in each?

Solution

Finally each has 30.

Before the second move, II had $30 : \frac{3}{4} = 40$, so I had 20.

Before the first, I had $20 : \frac{2}{3} = 30$, so II had 30.

Answer: 30 in each.

Problem 47 *

Ha does maths for 45 minutes, rests 15 minutes, then reads for 1 hour 20 minutes, finishing at 21:05. When did she start?

Solution

Reading began at $21{:}05 - 1{:}20 = 19{:}45$.
Rest began at 19:30.
Maths began at 18:45.
Answer: 18:45.

Problem 48 *

Think of any number, multiply by 9, add 18, divide by 9, then subtract the number you thought of. Show the result is always 2.

Solution

Call the number a.
$a \times 9 + 18 = (a + 2) \times 9$.
Divide by 9: $a + 2$.
Subtract a: 2, whatever a is.
Answer: Always 2.

Problem 49 **

A snail at the bottom of a 10 m well climbs 3 m each day and slips 2 m each night. On which day does it reach the top?

Solution

Work back from the last day: it climbs 3 m and reaches the top, so it started that day at $10 - 3 = 7$ m.
Each day and night it gains $3 - 2 = 1$ m.
Reaching 7 m takes 7 days and nights.
So it gets out on day 8.
Answer: Day 8.

Problem 50 **

Grandma has some apples. The eldest takes half plus half an apple; the second takes half the rest plus half an apple; the youngest takes half the rest plus half an apple, and none are left. No apple is cut. How many were there?

Solution

Before the youngest: $\left(0+\frac{1}{2}\right)\times 2=1$.

Before the second: $\left(1+\frac{1}{2}\right)\times 2=3$.

At first: $\left(3+\frac{1}{2}\right)\times 2=7$.

Check: they take 4, 2 and 1 whole apples.

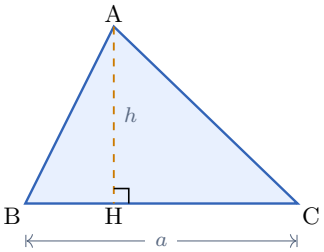
Answer: 7 apples.

11 Triangles and Their Area

① Base, height and area

A **triangle** has 3 sides, 3 vertices and 3 angles. By its angles it is *acute* (three acute angles), *obtuse* (one obtuse angle) or *right-angled* (one right angle).

The segment drawn from a vertex **perpendicular** to the opposite side is a **height**; that side is the **base**. Every triangle has 3 heights.



The **area of a triangle** is the base times the height (in the same unit), divided by 2:

$$S = \frac{a \times h}{2} \qquad h = \frac{S \times 2}{a} \qquad a = \frac{S \times 2}{h}$$

In a right-angled triangle, the two sides at the right angle are each other's base and height.

Example 1

Find the area of a triangle with:

a) base 8 cm, height 5 cm;

b) base 2.5 dm, height 12 cm.

Guided solution.

a) $S = 8 \times 5 : 2 = 20$ cm^2.

b) Use the same unit first: 2.5 dm = 25 cm.

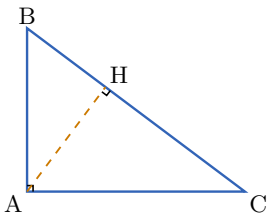
$S = 25 \times 12 : 2 = 150$ cm^2.

Answer: a) 20 cm^2;

b) 150 cm^2.

Example 2

Triangle ABC has a right angle at A, with $AB = 6$ cm, $AC = 8$ cm and $BC = 10$ cm. Find its area and the length of the height AH.



Guided solution.

Take AC as the base; then AB is the height: $S = 6 \times 8 : 2 = 24$ cm^2.

Take BC as the base; then AH is the height: $AH = 24 \times 2 : 10 = 4.8$ cm.

Answer: 24 cm^2;

$AH = 4.8$ cm.

Example 3

A triangle has area 36 cm^2 and base 9 cm. Find its height.

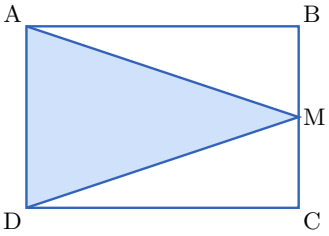
Guided solution.

$h = S \times 2 : a = 36 \times 2 : 9 = 8$ cm.

Answer: 8 cm.

Example 4

Rectangle ABCD has $AB = 12$ cm and $BC = 8$ cm; M is the midpoint of BC. Find the areas of triangles ABM, MCD and AMD.



Guided solution.

$BM = MC = 8 : 2 = 4$ cm.

$S_{ABM} = 12 \times 4 : 2 = 24$ cm^2;

$S_{MCD} = 12 \times 4 : 2 = 24$ cm^2.

$S_{ABCD} = 12 \times 8 = 96$ cm^2, so $S_{AMD} = 96 - 24 - 24 = 48$ cm^2.

Remark: triangle AMD has base AD and height equal to AB, so its area is half the rectangle.

Answer: 24 cm^2; 24 cm^2; 48 cm^2.

Practice

Problem 1

Find the area of a triangle with base a and height h:

a) $a = 15$ cm, $h = 8$ cm;

b) $a = 4.5$ m, $h = 2$ m;

c) $a = \frac{3}{4}$ m, $h = \frac{2}{5}$ m;

d) $a = 1.2$ dm, $h = 15$ cm.

Solution

a) $15 \times 8 : 2 = 60$ cm^2.

b) $4.5 \times 2 : 2 = 4.5$ m^2.

c) $\frac{3}{4} \times \frac{2}{5} : 2 = \frac{3}{20}$ m^2.

d) 1.2 dm $= 12$ cm; $12 \times 15 : 2 = 90$ cm^2.

Answer: 60 cm^2; 4.5 m^2; $\frac{3}{20}$ m^2; 90 cm^2.

Problem 2

Find the area of a right-angled triangle whose two shorter sides are 3.5 dm and 2 dm.

Solution

$S = 3.5 \times 2 : 2 = 3.5$ dm^2.

Answer: 3.5 dm^2.

Problem 3

A triangle has area 45 cm^2 and height 6 cm. Find its base.

Solution

$a = 45 \times 2 : 6 = 15$ cm.

Answer: 15 cm.

Problem 4

A triangular plot has base 40 m and height $\frac{3}{4}$ of the base. Find its area.

Solution

Height: $40 \times \frac{3}{4} = 30$ m.

Area: $40 \times 30 : 2 = 600$ m^2.

Answer: 600 m^2.

Problem 5

A triangular card has base 2.4 dm and height $\frac{2}{3}$ of the base. Find its area.

Solution

Height: $2.4 \times \frac{2}{3} = 1.6$ dm.

Area: $2.4 \times 1.6 : 2 = 1.92$ dm^2.

Answer: 1.92 dm^2.

Problem 6

Triangle ABC has a right angle at A, $AB = 9$ cm, $AC = 12$ cm and $BC = 15$ cm. Find the height AH.

Solution

$S = 9 \times 12 : 2 = 54$ cm^2.

$AH = 54 \times 2 : 15 = 7.2$ cm.

Answer: 7.2 cm.

Problem 7

A triangle has the same area as a square of side 6 cm. Its base is 9 cm. Find its height.

Solution

Area: $6 \times 6 = 36$ cm^2.

Height: $36 \times 2 : 9 = 8$ cm.

Answer: 8 cm.

Problem 8

A triangle has base 18 cm. Lengthening the base by 4 cm (same height) adds 20 cm^2 to the area. Find the original area.

Solution

The extra part is a triangle with base 4 cm and area 20 cm^2; its height is $20 \times 2 : 4 = 10$ cm.

Original area: $18 \times 10 : 2 = 90$ cm^2.

Answer: 90 cm^2.

Problem 9

Square ABCD has side 10 cm; M is the midpoint of BC. Find the areas of triangle ABM and quadrilateral AMCD.

Solution

$S_{ABM} = 10 \times 5 : 2 = 25$ cm^2.

$S_{AMCD} = 10 \times 10 - 25 = 75$ cm^2.

Answer: 25 cm^2; 75 cm^2.

Problem 10

Triangle ABC has area 60 cm^2 and $BC = 15$ cm. Point M lies on BC with $BM = 5$ cm. Find the area of triangle ABM.

Solution

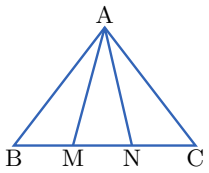
The height from A to BC is $60 \times 2 : 15 = 8$ cm.

Triangle ABM has the same height: $S_{ABM} = 5 \times 8 : 2 = 20$ cm^2.

Answer: 20 cm^2.

Problem 11

How many triangles are there in the figure? Name them.



Solution

Each triangle has vertex A and a base joining two of the four points B, M, N, C.

There are $3 + 2 + 1 = 6$ such segments.

Triangles: ABM, AMN, ANC, ABN, AMC, ABC.

Answer: 6 triangles.

Problem 12

Find the area of a triangle with base 14 cm and height 9 cm.

Solution

$S = 14 \times 9 : 2 = 63$ cm^2.

Answer: 63 cm^2.

Problem 13

Find the area of a triangle with base 2.5 m and height 1.6 m.

Solution

$S = 2.5 \times 1.6 : 2 = 2$ m^2.

Answer: 2 m^2.

Problem 14

Find the area of a triangle with base $\frac{3}{4}$ m and height $\frac{2}{5}$ m.

Solution

$S = \frac{3}{4} \times \frac{2}{5} : 2 = \frac{3}{20}$ m^2.

Answer: $\frac{3}{20}$ m^2.

Problem 15

Find the area of a triangle with base 3 dm and height 25 cm.

Solution

3 dm = 30 cm.

$S = 30 \times 25 : 2 = 375$ cm^2.

Answer: 375 cm^2.

Problem 16

Find the area of a right triangle whose legs are 7 cm and 10 cm.

Solution

One leg is the base and the other the height.

$S = 7 \times 10 : 2 = 35$ cm^2.

Answer: 35 cm^2.

Problem 17

A triangle has area 54 cm^2 and base 12 cm. Find its height.

Solution

Height = area ×2 : base.

$h = 54 \times 2 : 12 = 9$ cm.

Answer: 9 cm.

Problem 18

A triangle has area 7.5 dm^2 and height 2.5 dm. Find its base.

Solution

Base = area ×2 : height.

$a = 7.5 \times 2 : 2.5 = 6$ dm.

Answer: 6 dm.

Problem 19

A triangular plot has base 50 m and height 24 m. Each square metre yields 0.8 kg of vegetables. What is the total yield?

Solution

Area: $50 \times 24 : 2 = 600$ m^2.

Yield: $0.8 \times 600 = 480$ kg.

Answer: 480 kg.

Problem 20

A triangle has base 16 cm and height $\frac{3}{4}$ of the base. Find its area.

Solution

Height: $16 \times \frac{3}{4} = 12$ cm.

Area: $16 \times 12 : 2 = 96$ cm^2.

Answer: 96 cm^2.

Problem 21

The base and height of a triangle add up to 30 cm, and the base is 6 cm longer. Find the area.

Solution

Base: $(30 + 6) : 2 = 18$ cm.

Height: 12 cm.

Area: $18 \times 12 : 2 = 108$ cm^2.
Answer: 108 cm^2.

Problem 22

Triangle ABC has a right angle at A, $AB = 15$ cm, $AC = 20$ cm, $BC = 25$ cm. Find its area and the height AH.

Solution

Area: $15 \times 20 : 2 = 150$ cm^2.
With BC as base, AH is the height.
$AH = 150 \times 2 : 25 = 12$ cm.
Answer: 150 cm^2; $AH = 12$ cm.

Problem 23

Triangle one has base 12 cm and height 8 cm. Triangle two has the same area and base 16 cm. Find its height.

Solution

Area: $12 \times 8 : 2 = 48$ cm^2.
Height of triangle two: $48 \times 2 : 16 = 6$ cm.
Answer: 6 cm.

Problem 24

Each triangular flag has base 30 cm and height 20 cm. How many square decimetres of cloth do 20 flags need? (Ignore seams.)

Solution

One flag: $30 \times 20 : 2 = 300$ cm^2.
20 flags: $6\,000$ cm^2 $= 60$ dm^2.
Answer: 60 dm^2.

Problem 25

A right triangle has area 24 cm^2 and one leg 6 cm. Find the other leg.

Solution

The other leg is the height for the base 6 cm.
Length: $24 \times 2 : 6 = 8$ cm.
Answer: 8 cm.

Problem 26

Rectangle ABCD is 12 cm by 8 cm. Diagonal AC cuts it into two triangles. Find the area of each.

Solution

Triangle ABC has a right angle at B with legs 12 cm and 8 cm.
$S_{ABC} = 12 \times 8 : 2 = 48$ cm^2.
Likewise $S_{ACD} = 48$ cm^2 (half the rectangle).
Answer: 48 cm^2 each.

Problem 27

A triangle has area 150 cm^2. Raising its height by 3 cm (same base) adds 30 cm^2. Find the base and the height.

Solution

The extra part is a triangle with the same base, height 3 cm and area 30 cm^2.
Base: $30 \times 2 : 3 = 20$ cm.
Height: $150 \times 2 : 20 = 15$ cm.
Answer: Base 20 cm, height 15 cm.

Problem 28

A triangle with base 8 cm has the same area as a square of side 8 cm. Find its height.

Solution

Square: $8 \times 8 = 64$ cm^2.
Height: $64 \times 2 : 8 = 16$ cm.
Answer: 16 cm.

Problem 29

A triangular field has base 120 m and height 80 m. Give its area in hectares.

Solution

Area: $120 \times 80 : 2 = 4\,800$ m^2.
1 ha $= 10\,000$ m^2, so this is 0.48 ha.
Answer: 0.48 ha.

Problem 30

Triangle ABC has area 48 cm^2; M is the midpoint of BC. Find the area of triangle ABM.

Solution

ABM and ABC share the height from A.
$BM = \frac{1}{2}BC$, so $S_{ABM} = \frac{1}{2}S_{ABC}$.
$S_{ABM} = 48 : 2 = 24$ cm^2.
Answer: 24 cm^2.

Challenge – Extension

Problem 31 *

A triangle has area 120 cm^2. Shortening the base by 3 cm (same height) takes 18 cm^2 off the area. Find the base.

Solution

Height: $18 \times 2 : 3 = 12$ cm.
Base: $120 \times 2 : 12 = 20$ cm.
Answer: 20 cm.

Problem 32 *

The two shorter sides of a right-angled triangle add up to 14 cm, and one is 2 cm longer than the other. Find its area.

Solution

The sides are $(14 + 2) : 2 = 8$ cm and 6 cm.
Area: $8 \times 6 : 2 = 24$ cm^2.
Answer: 24 cm^2.

Problem 33 *

Rectangle ABCD has $AB = 16$ cm and $AD = 10$ cm. Point E is on AB with $AE = 6$ cm. Find the areas of triangles ADE, EBC and ECD.

Solution

$S_{ADE} = 6 \times 10 : 2 = 30$ cm^2;

$EB = 10$ cm, so $S_{EBC} = 10 \times 10 : 2 = 50$ cm^2.

$S_{ECD} = 16 \times 10 : 2 = 80$ cm^2 (half the rectangle).

Check: $30 + 50 + 80 = 160 = 16 \times 10$.

Answer: 30 cm^2; 50 cm^2; 80 cm^2.

Problem 34 **

Triangle ABC has a right angle at A, $AB = 6$ cm and $AC = 8$ cm. M is on AB with $AM = 2$ cm, and N is on AC with $AN = 4$ cm. Find the area of quadrilateral BMNC.

Solution

$S_{ABC} = 6 \times 8 : 2 = 24$ cm^2;

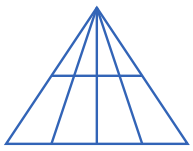
$S_{AMN} = 2 \times 4 : 2 = 4$ cm^2.

$S_{BMNC} = 24 - 4 = 20$ cm^2.

Answer: 20 cm^2.

Problem 35 **

How many triangles are there in the figure?



Solution

Ignoring the middle crossbar: the base has 5 points, giving $4+3+2+1 = 10$ triangles with vertex A.

The crossbar also has 5 points, giving 10 more triangles with vertex A.

In all: 20 triangles.

Answer: 20 triangles.

Problem 36 *

In triangle ABC, $BC = 20$ cm. M lies on BC with $BM = 8$ cm, and triangle ABM has area 40 cm^2. Find the area of ABC.

Solution

The common height from A: $40 \times 2 : 8 = 10$ cm.

$S_{ABC} = 20 \times 10 : 2 = 100$ cm^2.

Answer: 100 cm^2.

Problem 37

Rectangle ABCD has $AB = 20$ cm, $AD = 12$ cm. M is the midpoint of AB and N the midpoint of BC. Find the area of triangle DMN.

Solution

Rectangle: $20 \times 12 = 240$ cm^2.

$S_{AMD} = 60$, $S_{MBN} = 30$, $S_{NCD} = 60$ cm^2.

$S_{DMN} = 240 - 60 - 30 - 60 = 90$ cm^2.

Answer: 90 cm^2.

Problem 38

Square ABCD has side 12 cm, and E is any point on AB. Find the area of triangle ECD and the total area of triangles AED and EBC.

Solution

Triangle ECD has base $CD = 12$ cm and height 12 cm.

$S_{ECD} = 72$ cm^2.

The other two total $144 - 72 = 72$ cm^2, wherever E is.

Answer: Both are 72 cm^2.

Problem 39

A triangle has area 40 cm^2. If its base is tripled and its height halved, what is the new area?

Solution

Tripling the base triples the area.

Halving the height halves it.

New area: $40 \times 3 : 2 = 60$ cm^2.

Answer: 60 cm^2.

Problem 40

The legs of a right triangle add up to 23 cm and differ by 7 cm. Find its area.

Solution

Longer leg: $(23 + 7) : 2 = 15$ cm.

Shorter: 8 cm.

Area: $15 \times 8 : 2 = 60$ cm^2.

Answer: 60 cm^2.

Problem 41

Triangle ABC has area 90 cm^2. M on BC makes MC twice BM. Find the areas of ABM and AMC.

Solution

$BM = \frac{1}{3}BC$ and $MC = \frac{2}{3}BC$; both share the height from A.

$S_{ABM} = 90 : 3 = 30$ cm^2.

$S_{AMC} = 60$ cm^2.

Answer: $S_{ABM} = 30$ cm^2, $S_{AMC} = 60$ cm^2.

Problem 42

Triangle ABC has a right angle at A, $AB = 12$ cm, $AC = 16$ cm. M is the midpoint of AC. Compare the areas of ABM and BMC.

Solution

$AM = MC = 8$ cm; both have height $AB = 12$ cm.

$S_{ABM} = 48$ cm^2.

$S_{BMC} = 48$ cm^2.

They are equal.

Answer: Equal, 48 cm^2 each.

Problem 43

Triangle ABC has $BC = 24$ cm and height $AH = 10$ cm. BC is extended beyond C by $CD = 6$ cm. Find the areas of ACD and ABD.

Solution

ABC, ACD and ABD all have height $AH = 10$ cm.

$S_{ACD} = 6 \times 10 : 2 = 30$ cm^2.

$S_{ABD} = 30 \times 10 : 2 = 150$ cm^2.

Answer: 30 cm^2 and 150 cm^2.

Problem 44

Two triangles have equal heights. The base of the first is $\frac{2}{3}$ of the second's, and the second's area is 24 cm^2 larger. Find each area.

Solution

With equal heights, areas are in the ratio of the bases: $S_1 = \frac{2}{3}S_2$.

The difference of $3 - 2 = 1$ part is 24 cm^2.

$S_1 = 48$ cm^2; $S_2 = 72$ cm^2.

Answer: 48 cm^2 and 72 cm^2.

Problem 45

A garden is 40 m by 25 m. Flowers grow on a triangle whose base is one long side and whose apex is on the opposite side; grass covers the rest. Find both areas.

Solution

The triangle has base 40 m and height 25 m.

Flowers: 500 m^2.

Garden: 1 000 m^2.

Grass: 500 m^2.

Answer: Flowers 500 m^2, grass 500 m^2.

Problem 46

Triangle ABC has area 72 cm^2. D is on AB with $AD = \frac{1}{3}AB$; E is the midpoint of AC. Find the area of ADE.

Solution

Join D to C.

ADC and ABC share the height from C, and $AD = \frac{1}{3}AB$.

$S_{ADC} = 72 : 3 = 24$ cm^2.

ADE and ADC share the height from D, and $AE = \frac{1}{2}AC$.

$S_{ADE} = 12$ cm^2.

Answer: 12 cm^2.

Problem 47 *

Square ABCD has side 10 cm. E is the midpoint of BC and F the midpoint of CD. Find the area of triangle AEF.

Solution

$S_{ABE} = 25$ cm^2; $S_{ADF} = 25$ cm^2.

$S_{ECF} = 5 \times 5 : 2 = 12.5$ cm^2.

$S_{AEF} = 100 - 25 - 25 - 12.5 = 37.5$ cm^2.

Answer: 37.5 cm^2.

Problem 48 **

A square has a diagonal of 8 cm. Find its area.

Solution

A diagonal splits the square into two equal triangles on the base 8 cm.

The diagonals are equal, perpendicular and bisect each other, so each triangle's height is $8 : 2 = 4$ cm.

Each triangle: $8 \times 4 : 2 = 16$ cm^2.

Square: 32 cm^2.

Answer: 32 cm^2.

Problem 49 **

Triangle ABC has area 80 cm^2. M is the midpoint of BC and N the midpoint of AM. Find the area of BNC.

Solution

$S_{ABM} = S_{ACM} = 40$ cm^2.

BNM and BAM share the height from B and $NM = \frac{1}{2}AM$, so $S_{BNM} = 20$ cm^2.

Likewise $S_{CNM} = 20$ cm^2.

$S_{BNC} = 40$ cm^2.

Answer: 40 cm^2.

Problem 50 **

Triangle ABC has area 120 cm^2. M is on BC with $BM = \frac{1}{4}BC$; N is on AM with $AN = \frac{2}{3}AM$.

Find the areas of ABN and NBM.

Solution

$S_{ABM} = 120 : 4 = 30$ cm^2 (same height from A).

ABN and ABM share the height from B, with $AN = \frac{2}{3}AM$.

$S_{ABN} = 20$ cm^2.

$S_{NBM} = 10$ cm^2.

Answer: $S_{ABN} = 20$ cm^2, $S_{NBM} = 10$ cm^2.

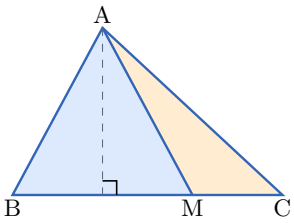
12 (*) Comparing Areas of Triangles

① Same height, compare the bases

- Two triangles with **the same height** have areas in the same ratio as **their bases**. For example, if M is on BC then

$$\frac{S_{ABM}}{S_{AMC}} = \frac{BM}{MC}.$$

- Two triangles with **the same base** have areas in the same ratio as **their heights**.
- The segment from a vertex to the **midpoint** of the opposite side splits a triangle into two triangles of equal area.



To compare two triangles, it often helps to pass through a "middle" triangle that shares a height with each of them.

Example 1

Triangle ABC has area 45 cm^2. Point M is on BC with BM twice MC. Find the areas of triangles ABM and AMC.

Guided solution.

Triangles ABM and AMC share the height from A, so their areas are in the ratio $BM : MC = 2 : 1$.

If S_{AMC} is 1 part, S_{ABM} is 2 parts and S_{ABC} is 3 parts.

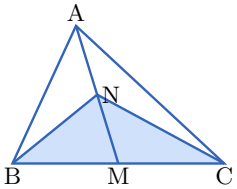
$S_{AMC} = 45 : 3 = 15$ cm^2;

$S_{ABM} = 15 \times 2 = 30$ cm^2.

Answer: 30 cm^2 and 15 cm^2.

Example 2

Triangle ABC has area 60 cm^2; M is the midpoint of BC and N is the midpoint of AM. Find the areas of triangles ABN and BNC.



Guided solution.

M is the midpoint of BC, so $S_{ABM} = S_{AMC} = 60 : 2 = 30$ cm^2.
N is the midpoint of AM, so $S_{ABN} = S_{NBM} = 30 : 2 = 15$ cm^2; likewise $S_{NMC} = 15$ cm^2.
$S_{BNC} = S_{NBM} + S_{NMC} = 15 + 15 = 30$ cm^2.
Answer: $S_{ABN} = 15$ cm^2;
$S_{BNC} = 30$ cm^2.

Example 3

Triangle ABC has area 72 cm^2. M is on AB with $AM = \frac{1}{3}AB$, and N is on AC with $AN = \frac{1}{2}AC$. Find the areas of triangle AMN and quadrilateral BMNC.

Guided solution.

Join M to C (the "middle" triangle AMC).

AMC and ABC share the height from C, and $AM = \frac{1}{3}AB$, so $S_{AMC} = \frac{1}{3} \times 72 = 24$ cm^2.

AMN and AMC share the height from M, and $AN = \frac{1}{2}AC$, so $S_{AMN} = \frac{1}{2} \times 24 = 12$ cm^2.

$S_{BMNC} = 72 - 12 = 60$ cm^2.
Answer: 12 cm^2; 60 cm^2.

Example 4

Square ABCD has side 12 cm; M is the midpoint of AB and N the midpoint of BC. Find the area of triangle DMN.

Guided solution.

Take the square and remove the three right-angled corner triangles:
$S_{AMD} = 6 \times 12 : 2 = 36$;
$S_{MBN} = 6 \times 6 : 2 = 18$;
$S_{NCD} = 6 \times 12 : 2 = 36$ cm^2.
$S_{DMN} = 12 \times 12 - 36 - 18 - 36 = 54$ cm^2.
Answer: 54 cm^2.

Practice

Problem 1

Triangle ABC has area 40 cm^2. M is on BC with $BM = 3$ cm and $MC = 5$ cm. Find the areas of ABM and AMC.

Solution

The areas are in the ratio 3 : 5, 8 parts in all.

$S_{ABM} = 40 : 8 \times 3 = 15$ cm^2; $S_{AMC} = 25$ cm^2.

Answer: 15 cm^2 and 25 cm^2.

Problem 2

In triangle ABC, M is the midpoint of BC and $S_{ABM} = 18$ cm^2. Find S_{ABC}.

Solution

$S_{AMC} = S_{ABM} = 18$ cm^2, so $S_{ABC} = 36$ cm^2.

Answer: 36 cm^2.

Problem 3

Triangle ABC has area 90 cm^2. D is on AC with $AD = \frac{2}{3}AC$. Find the areas of ABD and BDC.

Solution

They share the height from B: $S_{ABD} = \frac{2}{3} \times 90 = 60$ cm^2 and $S_{BDC} = 30$ cm^2.

Answer: 60 cm^2 and 30 cm^2.

Problem 4

Points M and N divide side BC of triangle ABC into three equal parts $BM = MN = NC$. If $S_{AMN} = 15$ cm^2, find S_{ABC}.

Solution

ABM, AMN and ANC share a height and have equal bases, so their areas are equal.

$S_{ABC} = 15 \times 3 = 45$ cm^2.

Answer: 45 cm^2.

Problem 5

Triangle ABC has area 20 cm^2. Extend BC beyond C to D with $CD = BC$. Find the areas of ACD and ABD.

Solution

ACD and ABC share the height from A and have equal bases: $S_{ACD} = 20$ cm^2.

$S_{ABD} = 20 + 20 = 40$ cm^2.

Answer: 20 cm^2; 40 cm^2.

Problem 6

Triangle ABC has area 60 cm^2; M is the midpoint of BC; N is on AC with $AN = \frac{1}{3}AC$. Find S_{MNC}.

Solution

$S_{AMC} = 60 : 2 = 30$ cm^2.

MNC and AMC share the height from M and $NC = \frac{2}{3}AC$, so $S_{MNC} = \frac{2}{3} \times 30 = 20$ cm^2.

Answer: 20 cm^2.

Problem 7

Triangle ABC has area 54 cm^2. M is on BC with $BM = \frac{1}{3}BC$; N is the midpoint of AM. Find the areas of BNM and ANC.

Solution

$S_{ABM} = \frac{1}{3} \times 54 = 18$ and $S_{AMC} = 36$ cm^2.

N is the midpoint of AM, so $S_{BNM} = 18 : 2 = 9$ cm^2 and $S_{ANC} = 36 : 2 = 18$ cm^2.

Answer: 9 cm^2; 18 cm^2.

Problem 8

Triangle ABC has area 48 cm^2. M is on AB with $AM = \frac{2}{3}AB$; N is on AC with $AN = \frac{3}{4}AC$. Find the areas of AMN and BMNC.

Solution

$S_{AMC} = \frac{2}{3} \times 48 = 32$ cm^2;

$S_{AMN} = \frac{3}{4} \times 32 = 24$ cm^2;

$S_{BMNC} = 48 - 24 = 24$ cm^2.

Answer: 24 cm^2; 24 cm^2.

Problem 9

In triangle ABC, D is on BC with $BD = \frac{1}{4}BC$, and $S_{ABD} = 12$ cm^2. Find S_{ADC}.

Solution

$DC = 3 \times BD$ and they share the height from A, so $S_{ADC} = 12 \times 3 = 36$ cm^2.

Answer: 36 cm^2.

Problem 10

Triangle ABC has area 60 cm^2. M on BC divides it so that $BM : MC = 1 : 2$. Find the areas of ABM and AMC.

Solution

ABM, AMC and ABC share the height from A.

$BM = \frac{1}{3}BC$, so $S_{ABM} = 60 \times \frac{1}{3} = 20$ cm^2.

$S_{AMC} = 60 - 20 = 40$ cm^2.

Answer: $S_{ABM} = 20$ cm^2, $S_{AMC} = 40$ cm^2.

Problem 11

Triangle ABC has area 84 cm^2. M on BC divides it so that $BM : MC = 3 : 4$. Find the areas of ABM and AMC.

Solution

ABM, AMC and ABC share the height from A.

$BM = \frac{3}{7}BC$, so $S_{ABM} = 84 \times \frac{3}{7} = 36$ cm^2.

$S_{AMC} = 84 - 36 = 48$ cm^2.

Answer: $S_{ABM} = 36$ cm^2, $S_{AMC} = 48$ cm^2.

Problem 12

Triangle ABC has area 100 cm^2. M on BC divides it so that $BM : MC = 2 : 3$. Find the areas of ABM and AMC.

Solution

ABM, AMC and ABC share the height from A.

$BM = \frac{2}{5}BC$, so $S_{ABM} = 100 \times \frac{2}{5} = 40$ cm^2.

$S_{AMC} = 100 - 40 = 60$ cm^2.
Answer: $S_{ABM} = 40$ cm^2, $S_{AMC} = 60$ cm^2.

Problem 13

Triangle ABC has area 72 cm^2. M on BC divides it so that $BM : MC = 5 : 1$. Find the areas of ABM and AMC.

Solution

ABM, AMC and ABC share the height from A.
$BM = \frac{5}{6}BC$, so $S_{ABM} = 72 \times \frac{5}{6} = 60$ cm^2.
$S_{AMC} = 72 - 60 = 12$ cm^2.
Answer: $S_{ABM} = 60$ cm^2, $S_{AMC} = 12$ cm^2.

Problem 14

Triangle ABC has area 45 cm^2. M on BC divides it so that $BM : MC = 4 : 5$. Find the areas of ABM and AMC.

Solution

ABM, AMC and ABC share the height from A.
$BM = \frac{4}{9}BC$, so $S_{ABM} = 45 \times \frac{4}{9} = 20$ cm^2.
$S_{AMC} = 45 - 20 = 25$ cm^2.
Answer: $S_{ABM} = 20$ cm^2, $S_{AMC} = 25$ cm^2.

Problem 15

In triangle ABC, M on BC has $BM = \frac{1}{3}BC$. Triangle ABM has area 15 cm^2. Find the areas of ABC and AMC.

Solution

ABM and ABC share the height from A, so $S_{ABM} = \frac{1}{3}S_{ABC}$.
$S_{ABC} = 15 : \frac{1}{3} = 45$ cm^2.
$S_{AMC} = 45 - 15 = 30$ cm^2.
Answer: $S_{ABC} = 45$ cm^2, $S_{AMC} = 30$ cm^2.

Problem 16

In triangle ABC, M on BC has $BM = \frac{2}{5}BC$. Triangle ABM has area 24 cm^2. Find the areas of ABC and AMC.

Solution

ABM and ABC share the height from A, so $S_{ABM} = \frac{2}{5}S_{ABC}$.

$S_{ABC} = 24 : \frac{2}{5} = 60$ cm^2.

$S_{AMC} = 60 - 24 = 36$ cm^2.

Answer: $S_{ABC} = 60$ cm^2, $S_{AMC} = 36$ cm^2.

Problem 17

In triangle ABC, M on BC has $BM = \frac{3}{4}BC$. Triangle ABM has area 21 cm^2. Find the areas of ABC and AMC.

Solution

ABM and ABC share the height from A, so $S_{ABM} = \frac{3}{4}S_{ABC}$.

$S_{ABC} = 21 : \frac{3}{4} = 28$ cm^2.

$S_{AMC} = 28 - 21 = 7$ cm^2.

Answer: $S_{ABC} = 28$ cm^2, $S_{AMC} = 7$ cm^2.

Problem 18

In triangle ABC, M on BC has $BM = \frac{1}{6}BC$. Triangle ABM has area 12 cm^2. Find the areas of ABC and AMC.

Solution

ABM and ABC share the height from A, so $S_{ABM} = \frac{1}{6}S_{ABC}$.

$S_{ABC} = 12 : \frac{1}{6} = 72$ cm^2.

$S_{AMC} = 72 - 12 = 60$ cm^2.

Answer: $S_{ABC} = 72$ cm^2, $S_{AMC} = 60$ cm^2.

Problem 19

Triangle ABC has area 80 cm^2 and $BC = 16$ cm. Point M on BC makes triangle ABM have area 30 cm^2. Find BM.

Solution

ABM and ABC share the height from A, so their areas are in the ratio of the bases.

$\frac{BM}{BC} = \frac{30}{80} = \frac{3}{8}$.

$BM = 16 \times \frac{3}{8} = 6$ cm.

Answer: $BM = 6$ cm.

Problem 20

Triangle ABC has area 96 cm^2 and $BC = 12$ cm. Point M on BC makes triangle ABM have area 36 cm^2. Find BM.

Solution

ABM and ABC share the height from A, so their areas are in the ratio of the bases.

$\frac{BM}{BC} = \frac{36}{96} = \frac{3}{8}$.

$BM = 12 \times \frac{3}{8} = 4.5$ cm.

Answer: $BM = 4.5$ cm.

Problem 21

Triangle ABC has area 75 cm^2 and $BC = 10$ cm. Point M on BC makes triangle ABM have area 45 cm^2. Find BM.

Solution

ABM and ABC share the height from A, so their areas are in the ratio of the bases.

$\frac{BM}{BC} = \frac{45}{75} = \frac{3}{5}$.

$BM = 10 \times \frac{3}{5} = 6$ cm.

Answer: $BM = 6$ cm.

Problem 22

Triangle ABC has area 36 cm^2. Extend BC beyond C by $CD = \frac{1}{2}BC$. Find the areas of ACD and ABD.

Solution

ABC, ACD and ABD share the height from A.

$S_{ACD} = 36 \times \frac{1}{2} = 18$ cm^2.

$S_{ABD} = 36 + 18 = 54$ cm^2.

Answer: $S_{ACD} = 18$ cm^2, $S_{ABD} = 54$ cm^2.

Problem 23

Triangle ABC has area 50 cm^2. Extend BC beyond C by $CD = \frac{2}{5}BC$. Find the areas of ACD and ABD.

Solution

ABC, ACD and ABD share the height from A.

$S_{ACD} = 50 \times \frac{2}{5} = 20$ cm^2.

$S_{ABD} = 50 + 20 = 70$ cm^2.

Answer: $S_{ACD} = 20$ cm^2, $S_{ABD} = 70$ cm^2.

Problem 24

Triangle ABC has area 28 cm^2. Extend BC beyond C by $CD = 2 \times BC$. Find the areas of ACD and ABD.

Solution

ABC, ACD and ABD share the height from A.

$S_{ACD} = 28 \times 2 = 56$ cm^2.

$S_{ABD} = 28 + 56 = 84$ cm^2.

Answer: $S_{ACD} = 56$ cm^2, $S_{ABD} = 84$ cm^2.

Problem 25

Triangle ABC has area 60 cm^2. M on AB has $AM = \frac{1}{2}AB$; N on AC has $AN = \frac{1}{3}AC$. Find the areas of AMN and of quadrilateral BMNC.

Solution

Join M to C.

AMC and ABC share the height from C: $S_{AMC} = 60 \times \frac{1}{2} = 30$ cm^2.

AMN and AMC share the height from M: $S_{AMN} = 30 \times \frac{1}{3} = 10$ cm^2.

$S_{BMNC} = 60 - 10 = 50$ cm^2.

Answer: $S_{AMN} = 10$ cm^2, $S_{BMNC} = 50$ cm^2.

Problem 26

Triangle ABC has area 90 cm^2. M on AB has $AM = \frac{2}{3}AB$; N on AC has $AN = \frac{1}{2}AC$. Find the areas of AMN and of quadrilateral BMNC.

Solution

Join M to C.

AMC and ABC share the height from C: $S_{AMC} = 90 \times \frac{2}{3} = 60$ cm^2.

AMN and AMC share the height from M: $S_{AMN} = 60 \times \frac{1}{2} = 30$ cm^2.

$S_{BMNC} = 90 - 30 = 60$ cm^2.

Answer: $S_{AMN} = 30$ cm^2, $S_{BMNC} = 60$ cm^2.

Problem 27

Triangle ABC has area 84 cm^2. M on AB has $AM = \frac{3}{4}AB$; N on AC has $AN = \frac{2}{3}AC$. Find the areas of AMN and of quadrilateral BMNC.

Solution

Join M to C.

AMC and ABC share the height from C: $S_{AMC} = 84 \times \frac{3}{4} = 63$ cm^2.

AMN and AMC share the height from M: $S_{AMN} = 63 \times \frac{2}{3} = 42$ cm^2.

$S_{BMNC} = 84 - 42 = 42$ cm^2.

Answer: $S_{AMN} = 42$ cm^2, $S_{BMNC} = 42$ cm^2.

Problem 28

Triangles ABC and DBC share the base BC. The height from A is 8 cm and from D is 6 cm. If $S_{ABC} = 40$ cm^2, find S_{DBC}.

Solution

With a shared base, areas are in the ratio of the heights: $\frac{S_{DBC}}{S_{ABC}} = \frac{6}{8}$.

$S_{DBC} = 40 \times 6 : 8 = 30$ cm^2.

Answer: 30 cm^2.

Problem 29

Triangles ABC and DBC share the base BC. The height from A is 9 cm and from D is 12 cm. If $S_{ABC} = 54$ cm^2, find S_{DBC}.

Solution

With a shared base, areas are in the ratio of the heights: $\frac{S_{DBC}}{S_{ABC}} = \frac{12}{9}$.

$S_{DBC} = 54 \times 12 : 9 = 72$ cm^2.

Answer: 72 cm^2.

Problem 30

Triangles ABC and DBC share the base BC. The height from A is 10 cm and from D is 4 cm. If $S_{ABC} = 35$ cm^2, find S_{DBC}.

Solution

With a shared base, areas are in the ratio of the heights: $\frac{S_{DBC}}{S_{ABC}} = \frac{4}{10}$.

$S_{DBC} = 35 \times 4 : 10 = 14$ cm^2.

Answer: 14 cm^2.

Challenge – Extension

Problem 31 *

Triangle ABC has area 64 cm^2; M and N are the midpoints of AB and AC. Find the areas of AMN and BMNC.

Solution

$S_{AMC} = 64 : 2 = 32$ cm^2; $S_{AMN} = 32 : 2 = 16$ cm^2.

$S_{BMNC} = 64 - 16 = 48$ cm^2.

The area of AMN is $\frac{1}{4}$ *of the area of ABC.*

Answer: 16 cm^2; 48 cm^2.

Problem 32 *

Triangle ABC has area 30 cm^2. Extend AB beyond B to D with $BD = AB$, and AC beyond C to E with $CE = AC$. Find S_{ADE}.

Solution

$AD = 2 \times AB$, so $S_{ADC} = 2 \times 30 = 60$ cm^2.

$AE = 2 \times AC$, so $S_{ADE} = 2 \times 60 = 120$ cm^2.

Answer: 120 cm^2.

Problem 33 *

Triangle ABC has area 120 cm^2. D is on BC with $BD = 2 \times DC$; E is the midpoint of AD. Find S_{BEC}.

Solution

$S_{ABD} = 80$ and $S_{ADC} = 40$ cm^2.

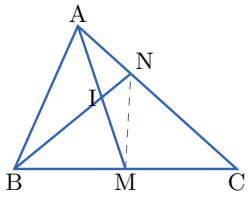
E is the midpoint of AD, so $S_{BED} = 40$ and $S_{EDC} = 20$ cm^2.

$S_{BEC} = 40 + 20 = 60$ cm^2.

Answer: 60 cm^2.

Problem 34 **

In triangle ABC, M is the midpoint of BC and I is the midpoint of AM. Line BI meets AC at N. Show that $AN = \frac{1}{3}AC$.



Solution

Join M to N.

Since $AI = IM$, $S_{ABI} = S_{MBI}$ and $S_{AIN} = S_{MIN}$.

Adding: $S_{ABN} = S_{MBN}$.

Since $BM = MC$, $S_{MBN} = S_{MNC}$.

Hence $S_{BNC} = 2 \times S_{ABN}$.

ABN and CBN share the height from B, so $NC = 2 \times AN$, i.e.

$AN = \frac{1}{3}AC$.

Answer: Shown as required.

Problem 35 **

Triangle ABC has area 90 cm^2. M is on BC with $BM = \frac{1}{3}BC$, N on CA with $CN = \frac{1}{3}CA$, and P on AB with $AP = \frac{1}{3}AB$. Find S_{MNP}.

Solution

Triangle APN: $S_{APC} = \frac{1}{3} \times 90 = 30$; $AN = \frac{2}{3}AC$, so $S_{APN} = \frac{2}{3} \times 30 = 20$ cm^2.

Likewise $S_{BMP} = 20$ and $S_{CNM} = 20$ cm^2.

So $S_{MNP} = 90 - 3 \times 20 = 30$ cm^2.

Answer: 30 cm^2.

Problem 36 *

Triangle ABC has area 60 cm^2. M on BC has $BM = \frac{1}{3}BC$; N on AM has $AN = \frac{1}{2}AM$. Find the areas of ABN, NBM and ANC.

Solution

$S_{ABM} = 60 \times \frac{1}{3} = 20$ cm^2; $S_{AMC} = 40$ cm^2.

ABN and ABM share the height from B: $S_{ABN} = 20 \times \frac{1}{2} = 10$ cm^2.

$S_{NBM} = 20 - 10 = 10$ cm^2.

ANC and AMC share the height from C: $S_{ANC} = 40 \times \frac{1}{2} = 20$ cm^2.

Answer: $S_{ABN} = 10$, $S_{NBM} = 10$, $S_{ANC} = 20$ (cm^2).

Problem 37 *

Triangle ABC has area 96 cm^2. M on BC has $BM = \frac{1}{4}BC$; N on AM has $AN = \frac{2}{3}AM$. Find the areas of ABN, NBM and ANC.

Solution

$S_{ABM} = 96 \times \frac{1}{4} = 24$ cm^2; $S_{AMC} = 72$ cm^2.

ABN and ABM share the height from B: $S_{ABN} = 24 \times \frac{2}{3} = 16$ cm^2.

$S_{NBM} = 24 - 16 = 8$ cm^2.

ANC and AMC share the height from C: $S_{ANC} = 72 \times \frac{2}{3} = 48$ cm^2.

Answer: $S_{ABN} = 16$, $S_{NBM} = 8$, $S_{ANC} = 48$ (cm^2).

Problem 38

Triangle ABC has area 150 cm^2. M on BC has $BM = \frac{2}{5}BC$; N on AM has $AN = \frac{3}{5}AM$. Find the areas of ABN, NBM and ANC.

Solution

$S_{ABM} = 150 \times \frac{2}{5} = 60$ cm^2; $S_{AMC} = 90$ cm^2.

ABN and ABM share the height from B: $S_{ABN} = 60 \times \frac{3}{5} = 36$ cm^2.

$S_{NBM} = 60 - 36 = 24$ cm^2.

ANC and AMC share the height from C: $S_{ANC} = 90 \times \frac{3}{5} = 54$ cm^2.

Answer: $S_{ABN} = 36$, $S_{NBM} = 24$, $S_{ANC} = 54$ (cm^2).

Problem 39

M on AB has $AM = \frac{1}{2}AB$ and N on AC has $AN = \frac{2}{5}AC$. Triangle AMN has area 10 cm^2. Find the area of ABC.

Solution

Join M to C.

AMN and AMC share the height from M: $S_{AMC} = 10 : \frac{2}{5} = 25$ cm^2.

AMC and ABC share the height from C: $S_{ABC} = 25 : \frac{1}{2} = 50$ cm^2.

Answer: 50 cm^2.

Problem 40

M on AB has $AM = \frac{3}{4}AB$ and N on AC has $AN = \frac{2}{3}AC$. Triangle AMN has area 18 cm^2. Find the area of ABC.

Solution

Join M to C.

AMN and AMC share the height from M: $S_{AMC} = 18 : \frac{2}{3} = 27$ cm^2.

AMC and ABC share the height from C: $S_{ABC} = 27 : \frac{3}{4} = 36$ cm^2.

Answer: 36 cm^2.

Problem 41 *

Triangle ABC has area 12 cm^2. Extend AB beyond B by $BD = AB$ and AC beyond C by $CE = 2 \times AC$. Find the area of ADE.

Solution

$AD = 2 \times AB$, $AE = 3 \times AC$.
Join D to C.
ADC and ABC share the height from C: $S_{ADC} = 12 \times 2 = 24$ cm^2.
ADE and ADC share the height from D: $S_{ADE} = 24 \times 3 = 72$ cm^2.
Answer: 72 cm^2.

Problem 42 *

Triangle ABC has area 15 cm^2. Extend AB beyond B by $BD = 2 \times AB$ and AC beyond C by $CE = 2 \times AC$. Find the area of ADE.

Solution

$AD = 3 \times AB$, $AE = 3 \times AC$.
Join D to C.
ADC and ABC share the height from C: $S_{ADC} = 15 \times 3 = 45$ cm^2.
ADE and ADC share the height from D: $S_{ADE} = 45 \times 3 = 135$ cm^2.
Answer: 135 cm^2.

Problem 43 **

Triangle ABC has area 80 cm^2. M, N, P lie on AB, BC, CA with $AM = \frac{1}{4}AB$, $BN = \frac{1}{4}BC$, $CP = \frac{1}{4}CA$. Find the area of MNP.

Solution

In triangle AMP: $AM = \frac{1}{4}AB$, $AP = \frac{3}{4}AC$.

As before (join M to C): $S_{AMP} = 80 \times \frac{1}{4} \times \frac{3}{4} = 15$ cm^2.

Likewise $S_{BNM} = S_{CPN} = 15$ cm^2.

$S_{MNP} = 80 - 3 \times 15 = 35$ cm^2.

Answer: 35 cm^2.

Problem 44 **

Triangle ABC has area 72 cm^2. M, N, P lie on AB, BC, CA with $AM = \frac{1}{3}AB$, $BN = \frac{1}{3}BC$, $CP = \frac{1}{3}CA$. Find the area of MNP.

Solution

In triangle AMP: $AM = \frac{1}{3}AB$, $AP = \frac{2}{3}AC$.

As before (join M to C): $S_{AMP} = 72 \times \frac{1}{3} \times \frac{2}{3} = 16$ cm^2.

Likewise $S_{BNM} = S_{CPN} = 16$ cm^2.

$S_{MNP} = 72 - 3 \times 16 = 24$ cm^2.

Answer: 24 cm^2.

Problem 45 *

In triangle ABC, M on BC has $BM : MC = 3 : 1$. Triangle ABM is 20 cm^2 larger than AMC. Find the area of ABC.

Solution

ABM and AMC share the height from A, so $S_{ABM} : S_{AMC} = 3 : 1$.

The difference of 2 parts is 20 cm^2; one part is 10 cm^2.

$S_{ABC} = 10 \times 4 = 40$ cm^2.

Answer: 40 cm^2.

Problem 46 *

In triangle ABC, M on BC has $BM : MC = 5 : 2$. Triangle ABM is 24 cm^2 larger than AMC. Find the area of ABC.

Solution

ABM and AMC share the height from A, so $S_{ABM} : S_{AMC} = 5 : 2$.

The difference of 3 parts is 24 cm^2; one part is 8 cm^2.

$S_{ABC} = 8 \times 7 = 56$ cm^2.
Answer: 56 cm^2.

Problem 47 *

Rectangle ABCD has $AB = 24$ cm, $BC = 16$ cm. M on AB has $AM = 8$ cm; N on BC has $BN = 6$ cm. Find the area of triangle DMN.

Solution

$S_{ABCD} = 24 \times 16 = 384$ cm^2.
$S_{AMD} = 64$, $S_{MBN} = 48$, $S_{NCD} = 120$ cm^2.
$S_{DMN} = 384 - 64 - 48 - 120 = 152$ cm^2.
Answer: 152 cm^2.

Problem 48 *

Rectangle ABCD has $AB = 30$ cm, $BC = 20$ cm. M on AB has $AM = 12$ cm; N on BC has $BN = 10$ cm. Find the area of triangle DMN.

Solution

$S_{ABCD} = 30 \times 20 = 600$ cm^2.
$S_{AMD} = 120$, $S_{MBN} = 90$, $S_{NCD} = 150$ cm^2.
$S_{DMN} = 600 - 120 - 90 - 150 = 240$ cm^2.
Answer: 240 cm^2.

Problem 49 *

Rectangle ABCD has $AB = 18$ cm, $BC = 12$ cm. M on AB has $AM = 6$ cm; N on BC has $BN = 4$ cm. Find the area of triangle DMN.

Solution

$S_{ABCD} = 18 \times 12 = 216$ cm^2.
$S_{AMD} = 36$, $S_{MBN} = 24$, $S_{NCD} = 72$ cm^2.
$S_{DMN} = 216 - 36 - 24 - 72 = 84$ cm^2.
Answer: 84 cm^2.

Problem 50 *

Triangle ABC has $BC = 16$ cm. Extending BC beyond C by 5 cm gives a triangle 25 cm^2 larger. Find the area of ABC.

Solution

The extra part is a triangle with base 5 cm and the same height from A.

Height: $25 \times 2 : 5 = 10$ cm.

$S_{ABC} = 16 \times 10 : 2 = 80$ cm^2.

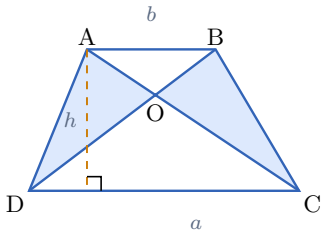
Answer: 80 cm^2.

13 Trapezia and Their Area

① Two parallel bases

A **trapezium** is a quadrilateral with *one pair of parallel opposite sides*: the **longer base** and the **shorter base**. The other two sides are the **legs**. A trapezium with one leg perpendicular to the bases is a **right trapezium**.

The **height** is the distance between the bases.



$$S = \frac{(a+b)\times h}{2} \qquad h = \frac{S\times 2}{a+b} \qquad a+b = \frac{S\times 2}{h}$$

The diagonals. In trapezium ABCD (AB parallel to CD), triangles ACD and BCD share base CD and have equal heights, so $S_{ACD} = S_{BCD}$; removing the common part gives $S_{AOD} = S_{BOC}$.

Example 1

Find the area of a trapezium with bases 12 cm and 8 cm and height 5 cm.

Guided solution.

$S = (12+8)\times 5 : 2 = 50$ cm^2.

Answer: 50 cm^2.

Example 2

A trapezium-shaped field has a longer base of 120 m; the shorter base is $\frac{2}{3}$ of the longer, and the height is the mean of the two bases. Every 100 m^2 yields 64 kg of paddy. How much paddy does the field yield?

Guided solution.

Shorter base: $120 \times \frac{2}{3} = 80$ m.

Height: $(120+80) : 2 = 100$ m.

Area: $(120+80)\times 100 : 2 = 10\,000$ m^2.

Paddy: $64 \times (10\,000 : 100) = 6\,400$ kg.

Answer: 6 400 kg.

Example 3

A trapezium has area 60 cm^2 and height 6 cm; the longer base is 4 cm longer than the shorter. Find both bases.

Guided solution.

Sum of the bases: $60 \times 2 : 6 = 20$ cm.
This is a sum-and-difference problem: longer base $(20 + 4) : 2 = 12$ cm, shorter 8 cm.

Answer: 12 cm and 8 cm.

Example 4

Trapezium ABCD has shorter base $AB = 6$ cm, longer base $CD = 9$ cm and height 4 cm. Find the areas of triangles ABD and BCD, and show that $S_{AOD} = S_{BOC}$ (O is where the diagonals cross).

Guided solution.

Both triangles have the trapezium's height: $S_{ABD} = 6 \times 4 : 2 = 12$ cm^2 and $S_{BCD} = 9 \times 4 : 2 = 18$ cm^2.
$S_{ACD} = S_{BCD} = 18$ cm^2 (same base CD, same height).
Removing their common part, triangle OCD, gives $S_{AOD} = S_{BOC}$.

Answer: 12 cm^2; 18 cm^2.

Practice

Problem 1

Find the area of a trapezium with bases a, b and height h:

a) $a = 14$ cm, $b = 6$ cm, $h = 7$ cm;

b) $a = 2.8$ m, $b = 1.6$ m, $h = 0.5$ m;

c) $a = \frac{3}{4}$ dm, $b = \frac{1}{2}$ dm, $h = \frac{2}{3}$ dm;

d) $a = 1.5$ dm, $b = 9$ cm, $h = 8$ cm.

Solution

a) $(14 + 6) \times 7 : 2 = 70$ cm^2.

b) $(2.8 + 1.6) \times 0.5 : 2 = 1.1$ m^2.

c) $\left(\frac{3}{4} + \frac{1}{2}\right) \times \frac{2}{3} : 2 = \frac{5}{12}$ dm^2.

d) 1.5 dm $= 15$ cm; $(15 + 9) \times 8 : 2 = 96$ cm^2.

Answer: 70 cm^2; 1.1 m^2; $\frac{5}{12}$ dm^2; 96 cm^2.

Problem 2

A right trapezium has bases 10 cm and 7 cm, and the leg perpendicular to them is 6 cm. Find its area.

Solution

The perpendicular leg is the height: $S = (10 + 7) \times 6 : 2 = 51$ cm^2.

Answer: 51 cm^2.

Problem 3

A trapezium has area 72 cm^2 and bases 10 cm and 14 cm. Find its height.

Solution

$h = 72 \times 2 : (10 + 14) = 6$ cm.

Answer: 6 cm.

Problem 4

A trapezium has area 45 m^2, height 5 m and shorter base 7 m. Find the longer base.

Solution

Sum of bases: $45 \times 2 : 5 = 18$ m.
Longer base: $18 - 7 = 11$ m.

Answer: 11 m.

Problem 5

A trapezium-shaped field has bases 60 m and 40 m and height 30 m. Every 100 m^2 yields 70 kg of paddy. How much paddy does the field yield?

Solution

Area: $(60 + 40) \times 30 : 2 = 1\,500$ m^2.
Paddy: $70 \times 15 = 1\,050$ kg.

Answer: 1 050 kg.

Problem 6

A trapezium's bases add up to 48 cm; the shorter is $\frac{3}{5}$ of the longer, and the height is $\frac{1}{4}$ of the longer base. Find its area.

Solution

Longer base: $48 : (3 + 5) \times 5 = 30$ cm; height: $30 \times \frac{1}{4} = 7.5$ cm.

$S = 48 \times 7.5 : 2 = 180$ cm^2.

Answer: 180 cm^2.

Problem 7

A trapezium has bases 25 cm and 15 cm. Lengthening the longer base by 5 cm adds 20 cm^2 to the area. Find the original area.

Solution

The added part is a triangle with base 5 cm and the trapezium's height: $h = 20 \times 2 : 5 = 8$ cm.

$S = (25 + 15) \times 8 : 2 = 160$ cm^2.

Answer: 160 cm^2.

Problem 8

Rectangle ABCD has $AB = 20$ cm and $BC = 12$ cm. M is on AB with $AM = 8$ cm. Find the area of trapezium MBCD.

Solution

MBCD has bases $MB = 12$ cm and $CD = 20$ cm and height $BC = 12$ cm: $S = (12 + 20) \times 12 : 2 = 192$ cm^2.

Another way: $20 \times 12 - 8 \times 12 : 2 = 240 - 48 = 192$ cm^2.

Answer: 192 cm^2.

Problem 9

Trapezium ABCD has $AB = 4$ cm, $CD = 10$ cm and height 6 cm. Find the areas of triangles ABC and ACD.

Solution

$S_{ABC} = 4 \times 6 : 2 = 12$ cm^2;

$S_{ACD} = 10 \times 6 : 2 = 30$ cm^2.

Check: $12 + 30 = 42 = (4 + 10) \times 6 : 2$.

Answer: 12 cm^2; 30 cm^2.

Problem 10

Find the area of a trapezium with bases 15 cm and 9 cm and height 6 cm.

Solution

$S = (15 + 9) \times 6 : 2 = 72$ cm^2.

Answer: 72 cm^2.

Problem 11

Find the area of a trapezium with bases 6.5 cm and 3.5 cm and height 4.2 cm.

Solution

$S = (6.5 + 3.5) \times 4.2 : 2 = 21$ cm^2.

Answer: 21 cm^2.

Problem 12

Find the area of a trapezium with bases $\frac{3}{4}$ m and $\frac{1}{2}$ m and height $\frac{2}{3}$ m.

Solution

$S = \left(\frac{3}{4} + \frac{1}{2}\right) \times \frac{2}{3} : 2 = \frac{5}{4} \times \frac{2}{3} : 2 = \frac{5}{12}$ m^2.

Answer: $\frac{5}{12}$ m^2.

Problem 13

Find the area of a trapezium with bases 14 dm and 10 dm and height 7.5 dm.

Solution

$S = (14 + 10) \times 7.5 : 2 = 90$ dm^2.

Answer: 90 dm^2.

Problem 14

Find the area of a trapezium with bases 1.2 m and 80 cm and height 5 dm.

Solution

In decimetres: 1.2 m = 12 dm; 80 cm = 8 dm.

$S = (12 + 8) \times 5 : 2 = 50$ dm^2.

Answer: 50 dm^2.

Problem 15

A trapezium has area 84 cm^2 and bases 16 cm and 12 cm. Find its height.

Solution

Height = area $\times 2$: sum of bases.

$h = 84 \times 2 : (16 + 12) = 6$ cm.

Answer: 6 cm.

Problem 16

A trapezium has area 45 cm^2 and bases 8.5 cm and 6.5 cm. Find its height.

Solution

Height = area $\times 2$: sum of bases.

$h = 45 \times 2 : (8.5 + 6.5) = 6$ cm.

Answer: 6 cm.

Problem 17

A trapezium has area 150 cm^2 and bases 23 cm and 17 cm. Find its height.

Solution

Height = area $\times 2$: sum of bases.

$h = 150 \times 2 : (23 + 17) = 7.5$ cm.

Answer: 7.5 cm.

Problem 18

A trapezium has area 96 cm^2, height 8 cm and shorter base 10 cm. Find the longer base.

Solution

Sum of bases: $96 \times 2 : 8 = 24$ cm.

Longer base: $24 - 10 = 14$ cm.

Answer: 14 cm.

Problem 19

A trapezium has area 63 cm^2, height 7 cm and shorter base 7.5 cm. Find the longer base.

Solution

Sum of bases: $63 \times 2 : 7 = 18$ cm.
Longer base: $18 - 7.5 = 10.5$ cm.
Answer: 10.5 cm.

Problem 20

A trapezium has area 110 cm^2, height 10 cm and shorter base 9 cm. Find the longer base.

Solution

Sum of bases: $110 \times 2 : 10 = 22$ cm.
Longer base: $22 - 9 = 13$ cm.
Answer: 13 cm.

Problem 21

A trapezium-shaped field has bases 80 m and 50 m and height 40 m. Every 100 m^2 yields 65 kg of paddy. What is the total yield?

Solution

Area: $(80 + 50) \times 40 : 2 = 2\,600$ m^2.
Yield: $2\,600 : 100 \times 65 = 1\,690$ kg.
Answer: 1 690 kg.

Problem 22

A trapezium-shaped field has bases 150 m and 90 m and height 60 m. Every 100 m^2 yields 60 kg of paddy. What is the total yield?

Solution

Area: $(150 + 90) \times 60 : 2 = 7\,200$ m^2.
Yield: $7\,200 : 100 \times 60 = 4\,320$ kg.
Answer: 4 320 kg.

Problem 23

A trapezium's bases add up to 36 cm and differ by 8 cm; its height is 10 cm. Find each base and the area.

Solution

Longer base: $(36 + 8) : 2 = 22$ cm; shorter: 14 cm.

Area: $36 \times 10 : 2 = 180$ cm^2.

Answer: Bases 22 cm and 14 cm; 180 cm^2.

Problem 24

A trapezium's bases add up to 50 cm and differ by 14 cm; its height is 12 cm. Find each base and the area.

Solution

Longer base: $(50 + 14) : 2 = 32$ cm; shorter: 18 cm.

Area: $50 \times 12 : 2 = 300$ cm^2.

Answer: Bases 32 cm and 18 cm; 300 cm^2.

Problem 25

A trapezium-shaped plot has longer base 60 m; the shorter base is $\frac{2}{3}$ of it and the height is $\frac{1}{2}$ of it. Find the area.

Solution

Shorter base: $60 \times \frac{2}{3} = 40$ m.

Height: $60 \times \frac{1}{2} = 30$ m.

Area: $(60 + 40) \times 30 : 2 = 1\,500$ m^2.

Answer: 1 500 m^2.

Problem 26

A trapezium-shaped plot has longer base 48 m; the shorter base is $\frac{3}{4}$ of it and the height is $\frac{5}{8}$ of it. Find the area.

Solution

Shorter base: $48 \times \frac{3}{4} = 36$ m.

Height: $48 \times \frac{5}{8} = 30$ m.

Area: $(48 + 36) \times 30 : 2 = 1\,260$ m^2.

Answer: 1 260 m^2.

Problem 27

Trapezium ABCD has right angles at A and D, $AB = 9$ cm, $CD = 15$ cm and $AD = 8$ cm. Find its area.

Solution

AD is perpendicular to both bases, so it is the height.

$S = (9 + 15) \times 8 : 2 = 96$ cm^2.

Answer: 96 cm^2.

Problem 28

Trapezium ABCD has right angles at A and D, $AB = 4.5$ cm, $CD = 7.5$ cm and $AD = 6$ cm. Find its area.

Solution

AD is perpendicular to both bases, so it is the height.

$S = (4.5 + 7.5) \times 6 : 2 = 36$ cm^2.

Answer: 36 cm^2.

Problem 29

Rectangle ABCD has $AB = 25$ cm and $BC = 14$ cm. M on AB has $AM = 9$ cm. Find the areas of trapezium MBCD and triangle AMD.

Solution

MBCD has bases $MB = 16$ cm and $CD = 25$ cm and height $BC = 14$ cm.

$S_{MBCD} = (16 + 25) \times 14 : 2 = 287$ cm^2.

$S_{AMD} = 9 \times 14 : 2 = 63$ cm^2.

Answer: $S_{MBCD} = 287$ cm^2, $S_{AMD} = 63$ cm^2.

Problem 30

Rectangle ABCD has $AB = 30$ cm and $BC = 16$ cm. M on AB has $AM = 12$ cm. Find the areas of trapezium MBCD and triangle AMD.

Solution

MBCD has bases $MB = 18$ cm and $CD = 30$ cm and height $BC = 16$ cm.

$S_{MBCD} = (18 + 30) \times 16 : 2 = 384$ cm^2.

$S_{AMD} = 12 \times 16 : 2 = 96$ cm^2.

Answer: $S_{MBCD} = 384$ cm^2, $S_{AMD} = 96$ cm^2.

Challenge – Extension

Problem 31 *

A trapezium has bases 12 cm and 18 cm. Lengthening the shorter base by 6 cm turns it into a rectangle and adds 27 cm^2. Find the trapezium's area.

Solution

The added part is a triangle with side 6 cm and the trapezium's height: $h = 27 \times 2 : 6 = 9$ cm.
$S = (12 + 18) \times 9 : 2 = 135$ cm^2.
Answer: 135 cm^2.

Problem 32 *

A trapezium-shaped field has bases adding up to 140 m, the longer 20 m longer than the shorter. Lengthening each base by 10 m adds 500 m^2. Find the original area and both bases.

Solution

The bases grow by 20 m in total, so $h = 500 \times 2 : 20 = 50$ m.
$S = 140 \times 50 : 2 = 3\,500$ m^2.
Longer base: $(140 + 20) : 2 = 80$ m; shorter: 60 m.
Answer: 3 500 m^2; 80 m and 60 m.

Problem 33 *

In trapezium ABCD, the shorter base AB is $\frac{1}{3}$ of the longer base CD, and $S_{ABD} = 15$ cm^2. Find the area of the trapezium.

Solution

ABD and BCD both have the trapezium's height, and $CD = 3 \times AB$, so $S_{BCD} = 15 \times 3 = 45$ cm^2.
$S_{ABCD} = 15 + 45 = 60$ cm^2.
Answer: 60 cm^2.

Problem 34 **

Right trapezium ABCD has right angles at A and D, $AB = 6$ cm, $CD = 12$ cm and $AD = 8$ cm. M is the midpoint of AD. Find S_{MBC}.

Solution

$S_{ABCD} = (6 + 12) \times 8 : 2 = 72$;
$S_{ABM} = 6 \times 4 : 2 = 12$;
$S_{MDC} = 12 \times 4 : 2 = 24$ cm^2.
$S_{MBC} = 72 - 12 - 24 = 36$ cm^2, exactly half the trapezium.
Answer: 36 cm^2.

Problem 35 ★★

In trapezium ABCD (AB parallel to CD), the diagonals cross at O. $S_{AOB} = 4$ cm^2 and $S_{BOC} = 8$ cm^2. Find the area of the trapezium.

Solution

AOB and BOC share the height from B, so $OC = 2 \times OA$.
By the trapezium property, $S_{AOD} = S_{BOC} = 8$ cm^2.
COD and AOD share the height from D and $OC = 2 \times OA$, so $S_{COD} = 16$ cm^2.
$S_{ABCD} = 4 + 8 + 8 + 16 = 36$ cm^2.
Answer: 36 cm^2.

Problem 36 ★

A trapezium has bases 30 cm and 18 cm. Lengthening the longer base by 6 cm (same height) adds 27 cm^2. Find the original area.

Solution

The extra part is a triangle with base 6 cm and the trapezium's height.
Height: $27 \times 2 : 6 = 9$ cm.
Area: $(30 + 18) \times 9 : 2 = 216$ cm^2.
Answer: 216 cm^2.

Problem 37 ★

A trapezium has bases 22 cm and 14 cm. Lengthening the longer base by 4 cm (same height) adds 18 cm^2. Find the original area.

Solution

The extra part is a triangle with base 4 cm and the trapezium's height.
Height: $18 \times 2 : 4 = 9$ cm.
Area: $(22 + 14) \times 9 : 2 = 162$ cm^2.
Answer: 162 cm^2.

Problem 38

A trapezium-shaped garden has bases summing to 90 m, differing by 30 m. Adding 5 m to each base (same height) adds 200 m^2. Find the original area.

Solution

The extra part is a trapezium with bases 5 m and 5 m and the same height.

Height: $200 \times 2 : (5 + 5) = 40$ m.

Original area: $90 \times 40 : 2 = 1\,800$ m^2.

Answer: 1 800 m^2.

Problem 39

A trapezium-shaped garden has bases summing to 120 m, differing by 40 m. Adding 8 m to each base (same height) adds 320 m^2. Find the original area.

Solution

The extra part is a trapezium with bases 8 m and 8 m and the same height.

Height: $320 \times 2 : (8 + 8) = 40$ m.

Original area: $120 \times 40 : 2 = 2\,400$ m^2.

Answer: 2 400 m^2.

Problem 40

In trapezium ABCD the shorter base AB is $\frac{1}{2}$ of the longer base CD. Triangle ABD has area 12 cm^2. Find the area of the trapezium.

Solution

ABD and BCD have the same height (the trapezium's height).

So $S_{ABD} = \frac{1}{2} S_{BCD}$.

$S_{BCD} = 12 : \frac{1}{2} = 24$ cm^2.

$S_{ABCD} = 36$ cm^2.

Answer: 36 cm^2.

Problem 41

In trapezium ABCD the shorter base AB is $\frac{2}{3}$ of the longer base CD. Triangle ABD has area 18 cm^2. Find the area of the trapezium.

Solution

ABD and BCD have the same height (the trapezium's height).

So $S_{ABD} = \frac{2}{3} S_{BCD}$.

$S_{BCD} = 18 : \frac{2}{3} = 27$ cm^2.

$S_{ABCD} = 45$ cm^2.

Answer: 45 cm^2.

Problem 42 ★★

Trapezium ABCD (AB parallel to CD) has diagonals meeting at O. $S_{AOB} = 4$ cm^2 and $S_{BOC} = 6$ cm^2. Find the area of the trapezium.

Solution

$S_{ABD} = S_{ABC}$ (same base AB, same height); removing AOB gives $S_{AOD} = S_{BOC} = 6$ cm^2.

AOB and BOC share the height from B, so $\frac{OA}{OC} = \frac{4}{6}$.

AOD and COD share the height from D, so $S_{COD} = 6 \times \frac{6}{4} = 9$ cm^2.

$S_{ABCD} = 4 + 6 + 6 + 9 = 25$ cm^2.

Answer: 25 cm^2.

Problem 43 ★★

Trapezium ABCD (AB parallel to CD) has diagonals meeting at O. $S_{AOB} = 9$ cm^2 and $S_{BOC} = 12$ cm^2. Find the area of the trapezium.

Solution

$S_{ABD} = S_{ABC}$ (same base AB, same height); removing AOB gives $S_{AOD} = S_{BOC} = 12$ cm^2.

AOB and BOC share the height from B, so $\frac{OA}{OC} = \frac{9}{12}$.

AOD and COD share the height from D, so $S_{COD} = 12 \times \frac{12}{9} = 16$ cm^2.

$S_{ABCD} = 9 + 12 + 12 + 16 = 49$ cm^2.

Answer: 49 cm^2.

Problem 44 ★★

Trapezium ABCD (AB parallel to CD) has diagonals meeting at O. $S_{AOB} = 2$ cm^2 and $S_{BOC} = 6$ cm^2. Find the area of the trapezium.

Solution

$S_{ABD} = S_{ABC}$ (same base AB, same height); removing AOB gives $S_{AOD} = S_{BOC} = 6$ cm^2.

AOB and BOC share the height from B, so $\frac{OA}{OC} = \frac{2}{6}$.

AOD and COD share the height from D, so $S_{COD} = 6 \times \frac{6}{2} = 18$ cm^2.

$S_{ABCD} = 2 + 6 + 6 + 18 = 32$ cm^2.

Answer: 32 cm^2.

Problem 45

Trapezium ABCD has right angles at A and D, $AB = 8$ cm, $CD = 14$ cm, $AD = 10$ cm. M is the midpoint of AD. Find the area of triangle MBC.

Solution

$S_{ABCD} = (8 + 14) \times 10 : 2 = 110$ cm^2.

$AM = MD = 5$ cm.

$S_{MAB} = 20$, $S_{MDC} = 35$ cm^2.

$S_{MBC} = 110 - 20 - 35 = 55$ cm^2.

Answer: 55 cm^2.

Problem 46

Trapezium ABCD has right angles at A and D, $AB = 5$ cm, $CD = 11$ cm, $AD = 12$ cm. M is the midpoint of AD. Find the area of triangle MBC.

Solution

$S_{ABCD} = (5 + 11) \times 12 : 2 = 96$ cm^2.

$AM = MD = 6$ cm.

$S_{MAB} = 15$, $S_{MDC} = 33$ cm^2.

$S_{MBC} = 96 - 15 - 33 = 48$ cm^2.

Answer: 48 cm^2.

Problem 47

A right trapezium has bases 10 cm and 16 cm. Lengthening the shorter base by 6 cm makes a rectangle 24 cm^2 larger. Find the trapezium's area.

Solution

The added piece is a right triangle with legs 6 cm and the height.

Height: $24 \times 2 : 6 = 4$ cm.

Area: $(10 + 16) \times 4 : 2 = 52$ cm^2.
Answer: 52 cm^2.

Problem 48

A right trapezium has bases 14 cm and 20 cm. Lengthening the shorter base by 6 cm makes a rectangle 27 cm^2 larger. Find the trapezium's area.

Solution

The added piece is a right triangle with legs 6 cm and the height.
Height: $27 \times 2 : 6 = 4.5$ cm.
Area: $(14 + 20) \times 4.5 : 2 = 76.5$ cm^2.
Answer: 76.5 cm^2.

Problem 49

A trapezium with bases 20 cm and 16 cm has the same area as a square of side 12 cm. Find its height.

Solution

Square: $12 \times 12 = 144$ cm^2.
Height: $144 \times 2 : (20 + 16) = 8$ cm.
Answer: 8 cm.

Problem 50

Diagonal AC splits trapezium ABCD (AB parallel to CD) into triangles with $S_{ABC} = 24$ cm^2 and $S_{ACD} = 40$ cm^2. The height is 8 cm. Find both bases.

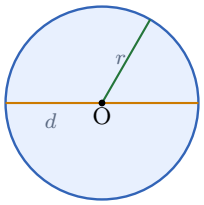
Solution

Both triangles have the trapezium's height, 8 cm.
$AB = 24 \times 2 : 8 = 6$ cm.
$CD = 40 \times 2 : 8 = 10$ cm.
Answer: $AB = 6$ cm, $CD = 10$ cm.

14 Circles: Circumference and Area

① Centre, radius, diameter

Every point on a **circle** is the same distance from the **centre** O. A segment from the centre to the circle is a **radius** r; a segment through the centre joining two points of the circle is a **diameter** d, and $d = r \times 2$.



We use the number 3.14 (called "pi") to calculate:

$$C = d \times 3.14 = r \times 2 \times 3.14 \qquad S = r \times r \times 3.14$$

- Given the circumference: $d = C : 3.14$.
- Given the area: find $r \times r = S : 3.14$, then the number r.
- If the radius is multiplied by k, the circumference is multiplied by k and the area by $k \times k$.

Example 1

Find the circumference and area of a circle of radius 5 cm.

Guided solution.

$C = 5 \times 2 \times 3.14 = 31.4$ cm.

$S = 5 \times 5 \times 3.14 = 78.5$ cm^2.

Answer: 31.4 cm; 78.5 cm^2.

Example 2

A circle has circumference 18.84 dm. Find its area.

Guided solution.

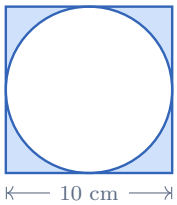
Diameter: $18.84 : 3.14 = 6$ dm; radius: 3 dm.

$S = 3 \times 3 \times 3.14 = 28.26$ dm^2.

Answer: 28.26 dm^2.

Example 3

A square has side 10 cm. A circle inside it touches all four sides. Find the shaded area (the part of the square outside the circle).



Guided solution.

The circle's diameter equals the side of the square: $d = 10$ cm, $r = 5$ cm.

$S_{\text{square}} = 10 \times 10 = 100$ cm^2;

$S_{\text{circle}} = 5 \times 5 \times 3.14 = 78.5$ cm^2.

Shaded part: $100 - 78.5 = 21.5$ cm^2.

Answer: 21.5 cm^2.

Example 4

A cart wheel has a diameter of 0.6 m. How far does the cart travel when the wheel turns 100 times?

Guided solution.

Each turn moves the cart by the wheel's circumference: $0.6 \times 3.14 = 1.884$ m.

100 turns: $1.884 \times 100 = 188.4$ m.

Answer: 188.4 m.

Practice

Problem 1

Find the circumference and area of a circle with:

a) $r = 3$ cm; **b)** $r = 0.5$ dm; **c)** $d = 8$ m.

Solution

a) $C = 18.84$ cm; $S = 28.26$ cm^2.

b) $C = 3.14$ dm; $S = 0.785$ dm^2.

c) $C = 25.12$ m; $r = 4$ m, $S = 50.24$ m^2.

Answer: See above.

Problem 2

A circle has circumference 12.56 cm. Find its radius and area.

Solution

$d = 12.56 : 3.14 = 4$ cm; $r = 2$ cm.
$S = 2 \times 2 \times 3.14 = 12.56$ cm^2.
Answer: $r = 2$ cm; $S = 12.56$ cm^2.

Problem 3

A bicycle wheel has a diameter of 65 cm. How many kilometres does the bicycle travel in 1 000 turns of the wheel?

Solution

Each turn: $65 \times 3.14 = 204.1$ cm.
1 000 turns: 204 100 cm = 2.041 km.
Answer: 2.041 km.

Problem 4

A car wheel travels 1 099 m in 500 turns. Find its diameter.

Solution

Circumference: $1\,099 : 500 = 2.198$ m.
Diameter: $2.198 : 3.14 = 0.7$ m.
Answer: 0.7 m.

Problem 5

The minute hand of a clock is 10 cm long. How far does its tip travel in 1 hour? In 15 minutes?

Solution

In 1 hour the tip goes once round: $10 \times 2 \times 3.14 = 62.8$ cm.
In 15 minutes, a quarter turn: $62.8 : 4 = 15.7$ cm.
Answer: 62.8 cm; 15.7 cm.

Problem 6

Find the area and perimeter of a semicircle with diameter 8 cm.

Solution

$S = 4 \times 4 \times 3.14 : 2 = 25.12$ cm^2.
The perimeter is half the circle plus the diameter: $8 \times 3.14 : 2 + 8 = 20.56$ cm.
Answer: 25.12 cm^2; 20.56 cm.

Problem 7

A round flower bed has a radius of 10 m. A path 1 m wide is built all round it. Find the area of the path.

Solution

The outer circle has radius 11 m.
Path: $11 \times 11 \times 3.14 - 10 \times 10 \times 3.14 = (121 - 100) \times 3.14 = 65.94$ m^2.

Answer: 65.94 m^2.

Problem 8

A square sits inside a circle of radius 5 cm with its four corners on the circle. Find the area of the part of the circle outside the square.

Solution

The square's diagonals are diameters, 10 cm long, and split it into 4 right-angled triangles with sides 5 cm: $S_{\text{square}} = 4 \times 5 \times 5 : 2 = 50$ cm^2.

$S_{\text{circle}} = 78.5$ cm^2.

Required part: $78.5 - 50 = 28.5$ cm^2.

Answer: 28.5 cm^2.

Problem 9

If the radius of a circle is multiplied by 3, by how many times are its circumference and area multiplied? Check with $r = 1$ cm.

Solution

The circumference is multiplied by 3 and the area by $3 \times 3 = 9$.
Example: $r = 1$: $C = 6.28$, $S = 3.14$; $r = 3$: $C = 18.84 = 6.28 \times 3$, $S = 28.26 = 3.14 \times 9$.

Answer: 3 times; 9 times.

Problem 10

Find the circumference and area of a circle of radius 3 cm.

Solution

$C = 3 \times 2 \times 3.14 = 18.84$ cm.

$S = 3 \times 3 \times 3.14 = 28.26$ cm^2.

Answer: $C = 18.84$ cm; $S = 28.26$ cm^2.

Problem 11

Find the circumference and area of a circle of radius 2.5 cm.

Solution

$C = 2.5 \times 2 \times 3.14 = 15.7$ cm.
$S = 2.5 \times 2.5 \times 3.14 = 19.625$ cm^2.
Answer: $C = 15.7$ cm; $S = 19.625$ cm^2.

Problem 12

Find the circumference and area of a circle of radius 0.8 m.

Solution

$C = 0.8 \times 2 \times 3.14 = 5.024$ m.
$S = 0.8 \times 0.8 \times 3.14 = 2.0096$ m^2.
Answer: $C = 5.024$ m; $S = 2.0096$ m^2.

Problem 13

Find the circumference and area of a circle of radius 12 dm.

Solution

$C = 12 \times 2 \times 3.14 = 75.36$ dm.
$S = 12 \times 12 \times 3.14 = 452.16$ dm^2.
Answer: $C = 75.36$ dm; $S = 452.16$ dm^2.

Problem 14

Find the circumference and area of a circle of diameter 10 cm.

Solution

$C = 10 \times 3.14 = 31.4$ cm.
Radius: $10 : 2 = 5$ cm.
$S = 5 \times 5 \times 3.14 = 78.5$ cm^2.
Answer: $C = 31.4$ cm; $S = 78.5$ cm^2.

Problem 15

Find the circumference and area of a circle of diameter 4.2 cm.

Solution

$C = 4.2 \times 3.14 = 13.188$ cm.
Radius: $4.2 : 2 = 2.1$ cm.
$S = 2.1 \times 2.1 \times 3.14 = 13.8474$ cm^2.
Answer: $C = 13.188$ cm; $S = 13.8474$ cm^2.

Problem 16

Find the circumference and area of a circle of diameter 2.4 m.

Solution

$C = 2.4 \times 3.14 = 7.536$ m.
Radius: $2.4 : 2 = 1.2$ m.
$S = 1.2 \times 1.2 \times 3.14 = 4.5216$ m^2.
Answer: $C = 7.536$ m; $S = 4.5216$ m^2.

Problem 17

A circle has circumference 43.96 cm. Find its radius and area.

Solution

Diameter: $43.96 : 3.14 = 14$ cm.
Radius: $14 : 2 = 7$ cm.
$S = 7 \times 7 \times 3.14 = 153.86$ cm^2.
Answer: $r = 7$ cm; $S = 153.86$ cm^2.

Problem 18

A circle has circumference 21.98 cm. Find its radius and area.

Solution

Diameter: $21.98 : 3.14 = 7$ cm.
Radius: $7 : 2 = 3.5$ cm.
$S = 3.5 \times 3.5 \times 3.14 = 38.465$ cm^2.
Answer: $r = 3.5$ cm; $S = 38.465$ cm^2.

Problem 19

A circle has circumference 6.28 m. Find its radius and area.

Solution

Diameter: $6.28 : 3.14 = 2$ m.
Radius: $2 : 2 = 1$ m.
$S = 1 \times 1 \times 3.14 = 3.14$ m^2.
Answer: $r = 1$ m; $S = 3.14$ m^2.

Problem 20

A circle has area 113.04 cm^2. Find its radius and circumference.

Solution

$r \times r = 113.04 : 3.14 = 36$.
Since $6 \times 6 = 36$, $r = 6$ cm.
$C = 6 \times 2 \times 3.14 = 37.68$ cm.
Answer: $r = 6$ cm; $C = 37.68$ cm.

Problem 21

A circle has area 12.56 m^2. Find its radius and circumference.

Solution

$r \times r = 12.56 : 3.14 = 4$.
Since $2 \times 2 = 4$, $r = 2$ m.
$C = 2 \times 2 \times 3.14 = 12.56$ m.
Answer: $r = 2$ m; $C = 12.56$ m.

Problem 22

A wheel has diameter 0.5 m. How far does it travel in 200 turns?

Solution

One turn: $0.5 \times 3.14 = 1.57$ m.
200 turns: $1.57 \times 200 = 314$ m.
Answer: 314 m.

Problem 23

A wheel has diameter 0.7 m. How far does it travel in 1 000 turns?

Solution

One turn: $0.7 \times 3.14 = 2.198$ m.
1 000 turns: $2.198 \times 1\,000 = 2\,198$ m.
Answer: 2 198 m.

Problem 24

A bicycle wheel travels 816.4 m in 400 turns. Find its diameter.

Solution

One turn: $816.4 : 400 = 2.041$ m.
Diameter: $2.041 : 3.14 = 0.65$ m.
Answer: 0.65 m.

Problem 25

Find the area and perimeter of a semicircle with diameter 10 cm.

Solution

Area: $5 \times 5 \times 3.14 : 2 = 39.25$ cm^2.
Arc: $10 \times 3.14 : 2 = 15.7$ cm.
Perimeter (with the diameter): $15.7 + 10 = 25.7$ cm.
Answer: $S = 39.25$ cm^2; perimeter 25.7 cm.

Problem 26

Find the area and perimeter of a semicircle with diameter 6 cm.

Solution

Area: $3 \times 3 \times 3.14 : 2 = 14.13$ cm^2.
Arc: $6 \times 3.14 : 2 = 9.42$ cm.
Perimeter (with the diameter): $9.42 + 6 = 15.42$ cm.
Answer: $S = 14.13$ cm^2; perimeter 15.42 cm.

Problem 27

A ceiling-fan blade is 0.6 m long from the centre. Find the area of the circle its tip sweeps out.

Solution

Radius $r = 0.6$ m.
$S = 0.6 \times 0.6 \times 3.14 = 1.1304$ m^2.
Answer: 1.1304 m^2.

Problem 28

A round pond has diameter 20 m. Fencing costs 50 000 dong per metre. How much does it cost to fence the pond?

Solution

Circumference: $20 \times 3.14 = 62.8$ m.
Cost: $50\,000 \times 62.8 = 3\,140\,000$ dong.
Answer: 3 140 000 dong.

Problem 29

Find the area of a quarter circle of radius 4 cm.

Solution

Whole circle: $4 \times 4 \times 3.14 = 50.24$ cm^2.
Quarter: $50.24 : 4 = 12.56$ cm^2.
Answer: 12.56 cm^2.

Problem 30

Find the area of a quarter circle of radius 10 cm.

Solution

Whole circle: $10 \times 10 \times 3.14 = 314$ cm^2.
Quarter: $314 : 4 = 78.5$ cm^2.
Answer: 78.5 cm^2.

Challenge – Extension

Problem 31 *

A circle has area 50.24 cm^2. Find its circumference.

Solution

$r \times r = 50.24 : 3.14 = 16$, so $r = 4$ cm.
$C = 4 \times 2 \times 3.14 = 25.12$ cm.
Answer: 25.12 cm.

Problem 32 *

If a circle's radius grows by 2 cm, by how much does its circumference grow? Does the answer depend on the original radius?

Solution

The increase is $(r+2) \times 2 \times 3.14 - r \times 2 \times 3.14 = 2 \times 2 \times 3.14 = 12.56$ cm, whatever the original radius.

Answer: 12.56 cm; it does not depend on it.

Problem 33 *

A large semicircle has diameter $AB = 12$ cm. M is the midpoint of AB, and two small semicircles are drawn on AM and MB. Compare the length of the large arc with the total length of the two small arcs.

Solution

Large arc: $12 \times 3.14 : 2 = 18.84$ cm.

Each small arc: $6 \times 3.14 : 2 = 9.42$ cm; together 18.84 cm.

They are equal.

Answer: Equal (18.84 cm).

Problem 34 *

Two circles share centre O, with radii 3 cm and 5 cm. Find the area between them (the ring).

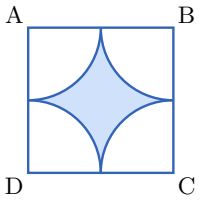
Solution

$S = 5 \times 5 \times 3.14 - 3 \times 3 \times 3.14 = (25 - 9) \times 3.14 = 50.24$ cm^2.

Answer: 50.24 cm^2.

Problem 35 **

Square ABCD has side 8 cm. Four quarter circles of radius 4 cm are drawn with centres at the four corners. Find the shaded area.



Solution

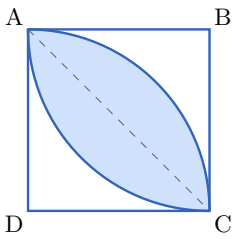
The four quarter circles of radius 4 cm make one whole circle: $4 \times 4 \times 3.14 = 50.24$ cm^2.

Shaded part: $8 \times 8 - 50.24 = 13.76$ cm^2.

Answer: 13.76 cm^2.

Problem 36

Square ABCD has side 10 cm. Two quarter circles of radius 10 cm are drawn with centres B and D; they overlap in the shaded "leaf". Find the area of the leaf.



Solution

Diagonal AC splits the leaf into two equal halves.

Each half is a quarter circle minus a right-angled triangle:

$10 \times 10 \times 3.14 : 4 - 10 \times 10 : 2 = 78.5 - 50 = 28.5$ cm^2.

Leaf: $28.5 \times 2 = 57$ cm^2.

Answer: 57 cm^2.

Problem 37

Two circles with the same centre have radii 4 m and 6 m. Find the area of the ring between them.

Solution

Large circle: $6 \times 6 \times 3.14 = 113.04$ m^2.

Small circle: $4 \times 4 \times 3.14 = 50.24$ m^2.

Required area: $113.04 - 50.24 = 62.8$ m^2.

Answer: 62.8 m^2.

Problem 38

A round flower bed has radius 8 m, with a path 2 m wide around it. Find the area of the path.

Solution

Large circle: $10 \times 10 \times 3.14 = 314$ m^2.

Small circle: $8 \times 8 \times 3.14 = 200.96$ m^2.

Required area: $314 - 200.96 = 113.04$ m^2.

Answer: 113.04 m^2.

Problem 39

A square has its four corners on a circle of radius 6 cm. Find the area of the circle outside the square.

Solution

The square's diagonals are diameters, 12 cm long and perpendicular.
It is 4 right triangles with legs 6 cm: $6 \times 6 : 2 \times 4 = 72$ cm^2.
Circle: $6 \times 6 \times 3.14 = 113.04$ cm^2.
Required: $113.04 - 72 = 41.04$ cm^2.
Answer: 41.04 cm^2.

Problem 40 *

A small wheel of radius 0.3 m makes 150 turns. A large wheel covers the same distance in 90 turns. Find its radius.

Solution

Distance: $0.3 \times 2 \times 3.14 \times 150 = 282.6$ m.
One large turn: $282.6 : 90 = 3.14$ m.
Radius: $3.14 : 3.14 : 2 = 0.5$ m.
Answer: 0.5 m.

Problem 41 *

A circle has the same perimeter as a square of side 15.7 cm. Which has the larger area, and by how much?

Solution

Square perimeter: 62.8 cm.
Circle radius: $62.8 : 3.14 : 2 = 10$ cm; area 314 cm^2.
Square area: $15.7 \times 15.7 = 246.49$ cm^2.
The circle is larger by 67.51 cm^2.
Answer: The circle, by 67.51 cm^2.

Problem 42 *

A circle has area 12.56 cm^2. If its radius is doubled, what is the new area?

Solution

Doubling r multiplies $r \times r$ by $2 \times 2 = 4$.
New area: $12.56 \times 4 = 50.24$ cm^2.
Answer: 50.24 cm^2.

Problem 43 *

A circle's radius grows by 1.5 cm. By how much does its circumference grow?

Solution

$C = r \times 2 \times 3.14$, so C grows by $1.5 \times 2 \times 3.14 = 9.42$ cm.
This does not depend on the original radius.
Answer: 9.42 cm.

Problem 44 *

An old bicycle has a front wheel of diameter 0.5 m and a back wheel of diameter 1 m. When the back wheel turns 100 times, how many times does the front wheel turn?

Solution

Distance: $1 \times 3.14 \times 100 = 314$ m.
One front turn: $0.5 \times 3.14 = 1.57$ m.
Front turns: $314 : 1.57 = 200$.
Answer: 200 turns.

Problem 45 *

A running track has two straights of 100 m and two semicircular ends of diameter 60 m. Find the length of one lap and the area inside.

Solution

Two semicircles make one circle: $60 \times 3.14 = 188.4$ m.
One lap: $100 \times 2 + 188.4 = 388.4$ m.
Rectangle: $6\,000$ m^2; circle: $30 \times 30 \times 3.14 = 2\,826$ m^2.
Inside area: $8\,826$ m^2.
Answer: 388.4 m; $8\,826$ m^2.

Problem 46 **

A quarter (bounded by two perpendicular radii) is cut from a circular card of radius 10 cm. Find the area and perimeter of what is left.

Solution

Circle: 314 cm^2; three quarters: 235.5 cm^2.
Circumference 62.8 cm; the remaining arc: 47.1 cm.
Perimeter: arc plus two radii, $47.1 + 20 = 67.1$ cm.
Answer: 235.5 cm^2; 67.1 cm.

Problem 47 ⁎⁎

A goat is tied to an outside corner of a square pen of side 6 m with a 4 m rope. It cannot enter the pen. Find the area it can graze.

Solution

The rope is shorter than the side, so the goat only swings round that corner.

The pen blocks $\frac{1}{4}$ of the circle, leaving $\frac{3}{4}$ of a circle of radius 4 m.

Area: $4 \times 4 \times 3.14 \times \frac{3}{4} = 37.68$ m^2.

Answer: 37.68 m^2.

Problem 48 ⁎

Two circles' circumferences differ by 12.56 cm, and the larger radius is 3 times the smaller. Find the area of the larger circle.

Solution

Difference of radii: $12.56 : 2 : 3.14 = 2$ cm.

$3 - 1 = 2$ parts make 2 cm, so the radii are 1 cm and 3 cm.

Larger area: $3 \times 3 \times 3.14 = 28.26$ cm^2.

Answer: 28.26 cm^2.

Problem 49 ⁎

A minute hand is 12 cm long. What area does it sweep in 20 minutes?

Solution

20 minutes is $\frac{1}{3}$ of a turn.

Circle: $12 \times 12 \times 3.14 = 452.16$ cm^2.

Swept area: $452.16 : 3 = 150.72$ cm^2.

Answer: 150.72 cm^2.

Problem 50 ⁎

A tractor wheel of diameter 0.7 m makes 120 turns a minute. How far does the tractor go in 10 minutes?

Solution

One turn: $0.7 \times 3.14 = 2.198$ m.

Turns in 10 minutes: 1 200.

Distance: $2.198 \times 1\,200 = 2\,637.6$ m.
Answer: 2 637.6 m.

15 Mixed Practice on Area

① Split, add on, cut away

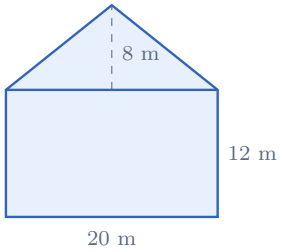
To find the area of a composite shape, reduce it to shapes whose formulas we know:

Rectangle	$a \times b$	Triangle	$a \times h : 2$
Square	$a \times a$	Trapezium	$(a + b) \times h : 2$
Circle	$r \times r \times 3.14$	Semicircle	$r \times r \times 3.14 : 2$

- **Split:** cut the shape into familiar pieces and *add* their areas.
- **Complete and subtract:** fill it out to a large familiar shape and *subtract* what was added.
- Always change measurements to **the same unit** first; $1\ \text{m}^2 = 100\ \text{dm}^2 = 10\,000\ \text{cm}^2$.

Example 1

A plot of land is shaped as shown: a rectangle 20 m by 12 m with a triangle on top whose base is the rectangle's length and whose height is 8 m. Find the area of the plot.



Guided solution.

Split it into a rectangle and a triangle:

$S_{\text{rectangle}} = 20 \times 12 = 240\ \text{m}^2$;

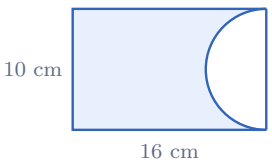
$S_{\text{triangle}} = 20 \times 8 : 2 = 80\ \text{m}^2$.

Plot: $240 + 80 = 320\ \text{m}^2$.

Answer: $320\ \text{m}^2$.

Example 2

A semicircle whose diameter is the width of a 16 cm by 10 cm wooden board is cut out of it (see the figure). Find the area of the wood that is left.



Guided solution.

$S_{\text{board}} = 16 \times 10 = 160$ cm^2.

The semicircle has $r = 5$ cm: $5 \times 5 \times 3.14 : 2 = 39.25$ cm^2.

Wood left: $160 - 39.25 = 120.75$ cm^2.

Answer: 120.75 cm^2.

Example 3

On a map with scale 1 : 1 000, a trapezium-shaped garden has bases 5 cm and 3 cm and height 4 cm. Find the real area of the garden.

Guided solution.

Real lengths: 5 cm $\rightarrow$ 5 000 cm = 50 m;
3 cm $\rightarrow$ 30 m;
4 cm $\rightarrow$ 40 m.
Real area: $(50 + 30) \times 40 : 2 = 1\,600$ m^2.

Note: change to real lengths *first*, then compute the area.

Answer: 1 600 m^2.

Example 4

A rectangular floor is 8 m by 6 m. It is tiled with square tiles of side 40 cm costing 25 000 dong each. How many tiles are needed, and what do they cost? (Ignore the grout.)

Guided solution.

Floor: $8 \times 6 = 48$ m^2 $= 480\,000$ cm^2.

One tile: $40 \times 40 = 1\,600$ cm^2.

Tiles: $480\,000 : 1\,600 = 300$.

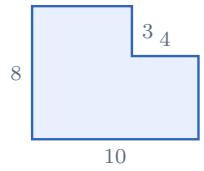
Cost: $25\,000 \times 300 = 7\,500\,000$ dong.

Answer: 300 tiles; 7 500 000 dong.

Practice

Problem 1

Find the area of the L-shape (measurements in centimetres).



Solution

Complete and subtract: $10 \times 8 - 4 \times 3 = 80 - 12 = 68$ cm^2.

Answer: 68 cm^2.

Problem 2

A rectangular plot is 25 m long, and its width is $\frac{3}{5}$ of its length. A house with a square base of side 10 m is built on it. Find the area left.

Solution

Width: $25 \times \frac{3}{5} = 15$ m.

Left: $25 \times 15 - 10 \times 10 = 375 - 100 = 275$ m^2.

Answer: 275 m^2.

Problem 3

A rectangular schoolyard is 50 m by 30 m. A round flower bed of radius 3 m is in the middle. Find the rest of the yard's area.

Solution

$50 \times 30 - 3 \times 3 \times 3.14 = 1\,500 - 28.26 = 1\,471.74$ m^2.

Answer: 1 471.74 m^2.

Problem 4

A trapezium-shaped garden has bases 30 m and 20 m and height 16 m. A quarter of it is dug out for a pond and the rest is planted. Find the planted area.

Solution

$S = (30 + 20) \times 16 : 2 = 400$ m^2.

Planted: $400 \times \frac{3}{4} = 300$ m^2.

Answer: 300 m^2.

Problem 5

On a 1 : 500 map a rectangular pitch is 6 cm by 4 cm. Find its real area.

Solution

Real size: $6 \times 500 = 3\,000$ cm $= 30$ m;
$4 \times 500 = 2\,000$ cm $= 20$ m.
Area: $30 \times 20 = 600$ m^2.
Answer: 600 m^2.

Problem 6

A classroom 6 m by 4 m is tiled with square tiles of side 50 cm. How many tiles are needed?

Solution

Room: 24 m^2 $= 240\,000$ cm^2; one tile: $2\,500$ cm^2.
Tiles: $240\,000 : 2\,500 = 96$.
Answer: 96 tiles.

Problem 7

Four equal squares of side 0.2 m are cut from the corners of a 2 m by 1.5 m metal sheet. Find the area of metal left.

Solution

$2 \times 1.5 - 4 \times 0.2 \times 0.2 = 3 - 0.16 = 2.84$ m^2.
Answer: 2.84 m^2.

Problem 8

A shape is made of a square of side 6 cm and a semicircle drawn outwards on one side of the square. Find its area and perimeter.

Solution

Area: $6 \times 6 + 3 \times 3 \times 3.14 : 2 = 36 + 14.13 = 50.13$ cm^2.
The perimeter is three sides of the square plus the half circle: $6 \times 3 + 6 \times 3.14 : 2 = 18 + 9.42 = 27.42$ cm.
Answer: 50.13 cm^2; 27.42 cm.

Problem 9

The diagonals of rectangle ABCD ($AB = 12$ cm, $BC = 8$ cm) cross at O. Show that triangles OAB, OBC, OCD and ODA have equal areas, and find each area.

Solution

O is the midpoint of each diagonal.
In triangle ABC, BO joins B to the midpoint of AC, so $S_{OAB} = S_{OBC}$; likewise $S_{OBC} = S_{OCD}$ and $S_{OCD} = S_{ODA}$.
Each triangle: $12 \times 8 : 4 = 24$ cm^2.
Answer: 24 cm^2.

Problem 10

On a map at scale 1 : 1 000, a rectangular plot is 6 cm by 4 cm. Find its real area.

Solution

Real length: $6 \times 1\,000 = 6\,000$ cm $= 60$ m.
Real width: $4 \times 1\,000 = 4\,000$ cm $= 40$ m.
Area: $60 \times 40 = 2\,400$ m^2.
Answer: 2 400 m^2.

Problem 11

On a map at scale 1 : 2 000, a rectangular plot is 5 cm by 3 cm. Find its real area.

Solution

Real length: $5 \times 2\,000 = 10\,000$ cm $= 100$ m.
Real width: $3 \times 2\,000 = 6\,000$ cm $= 60$ m.
Area: $100 \times 60 = 6\,000$ m^2.
Answer: 6 000 m^2.

Problem 12

On a map at scale 1 : 500, a rectangular plot is 8 cm by 5 cm. Find its real area.

Solution

Real length: $8 \times 500 = 4\,000$ cm $= 40$ m.
Real width: $5 \times 500 = 2\,500$ cm $= 25$ m.
Area: $40 \times 25 = 1\,000$ m^2.
Answer: 1 000 m^2.

Problem 13

A room floor is 6 m by 4 m, tiled with square tiles of side 40 cm costing 20 000 dong each. Find the number of tiles and their cost (ignore the joints).

Solution

Floor: $6 \times 4 = 24$ m^2.
One tile: $40 \times 40 = 1600$ cm^2 $= 0.16$ m^2.
Tiles: $24 : 0.16 = 150$.
Cost: $20\,000 \times 150 = 3\,000\,000$ dong.
Answer: 150 tiles; 3 000 000 dong.

Problem 14

A room floor is 8 m by 5 m, tiled with square tiles of side 50 cm costing 35 000 dong each. Find the number of tiles and their cost (ignore the joints).

Solution

Floor: $8 \times 5 = 40$ m^2.
One tile: $50 \times 50 = 2500$ cm^2 $= 0.25$ m^2.
Tiles: $40 : 0.25 = 160$.
Cost: $35\,000 \times 160 = 5\,600\,000$ dong.
Answer: 160 tiles; 5 600 000 dong.

Problem 15

A room floor is 4.8 m by 3.6 m, tiled with square tiles of side 60 cm costing 45 000 dong each. Find the number of tiles and their cost (ignore the joints).

Solution

Floor: $4.8 \times 3.6 = 17.28$ m^2.
One tile: $60 \times 60 = 3600$ cm^2 $= 0.36$ m^2.
Tiles: $17.28 : 0.36 = 48$.
Cost: $45\,000 \times 48 = 2\,160\,000$ dong.
Answer: 48 tiles; 2 160 000 dong.

Problem 16

A rectangular plot is 30 m long and its width is $\frac{2}{3}$ of the length. A 12 m $\times$ 8 m house is built on it. Find the remaining area.

Solution

Width: $30 \times \frac{2}{3} = 20$ m.
Plot: 600 m^2.
House: 96 m^2.

Remaining: 504 m^2.
Answer: 504 m^2.

Problem 17

A rectangular plot is 40 m long and its width is $\frac{3}{4}$ of the length. A 15 m × 10 m house is built on it. Find the remaining area.

Solution

Width: $40 \times \frac{3}{4} = 30$ m.
Plot: 1 200 m^2.
House: 150 m^2.
Remaining: 1 050 m^2.
Answer: 1 050 m^2.

Problem 18

A playground is 40 m by 25 m with a round sandpit of radius 3 m in the middle. Find the area of the rest.

Solution

Playground: 1000 m^2.
Sandpit: $3 \times 3 \times 3.14 = 28.26$ m^2.
Rest: 971.74 m^2.
Answer: 971.74 m^2.

Problem 19

A playground is 30 m by 20 m with a round sandpit of radius 5 m in the middle. Find the area of the rest.

Solution

Playground: 600 m^2.
Sandpit: $5 \times 5 \times 3.14 = 78.5$ m^2.
Rest: 521.5 m^2.
Answer: 521.5 m^2.

Problem 20

A shape is a square of side 8 cm with a semicircle on one side (diameter = that side), outward. Find its area.

Solution

Square: 64 cm^2.

Semicircle of radius 4 cm: 25.12 cm^2.
Total: 89.12 cm^2.
Answer: 89.12 cm^2.

Problem 21

A shape is a square of side 10 cm with a semicircle on one side (diameter = that side), outward. Find its area.

Solution

Square: 100 cm^2.

Semicircle of radius 5 cm: 39.25 cm^2.
Total: 139.25 cm^2.
Answer: 139.25 cm^2.

Problem 22

From a 3 dm × 2 dm card, a square of side 0.4 dm is cut from each corner. Find the area left.

Solution

Card: 6 dm^2.
Four squares: 0.64 dm^2.

Left: 5.36 dm^2.
Answer: 5.36 dm^2.

Problem 23

From a 5 dm × 4 dm card, a square of side 0.5 dm is cut from each corner. Find the area left.

Solution

Card: 20 dm^2.
Four squares: 1 dm^2.

Left: 19 dm^2.
Answer: 19 dm^2.

Problem 24

A trapezium-shaped garden has bases 40 m and 30 m and height 20 m. $\frac{1}{5}$ of it is a pond; vegetables grow on the rest at 3 kg per m^2. Find the harvest.

Solution

Garden: 700 m^2.
Pond: 140 m^2.
Vegetables: 560 m^2.
Harvest: $560 \times 3 = 1\,680$ kg.
Answer: 1 680 kg.

Problem 25

A trapezium-shaped garden has bases 60 m and 40 m and height 30 m. $\frac{1}{3}$ of it is a pond; vegetables grow on the rest at 2 kg per m^2. Find the harvest.

Solution

Garden: 1 500 m^2.
Pond: 500 m^2.
Vegetables: 1 000 m^2.
Harvest: $1\,000 \times 2 = 2\,000$ kg.
Answer: 2 000 kg.

Problem 26

An L-shaped plot is a 20 m × 15 m rectangle with a 8 m × 6 m corner removed. Find its area.

Solution

Big rectangle: 300 m^2.
Cut-out: 48 m^2.
Area: 252 m^2.
Answer: 252 m^2.

Problem 27

An L-shaped plot is a 30 m × 24 m rectangle with a 12 m × 10 m corner removed. Find its area.

Solution

Big rectangle: 720 m^2.
Cut-out: 120 m^2.

Area: 600 m^2.
Answer: 600 m^2.

Problem 28

A rectangular garden has perimeter 60 m, and its width is $\frac{2}{3}$ of its length. Find its area.

Solution

Half perimeter: 30 m.
5 parts; one part: 6 m.
Length 18 m, width 12 m.
Area: 216 m^2.
Answer: 216 m^2.

Problem 29

A rectangular garden has perimeter 84 m, and its width is $\frac{3}{4}$ of its length. Find its area.

Solution

Half perimeter: 42 m.
7 parts; one part: 6 m.
Length 24 m, width 18 m.
Area: 432 m^2.
Answer: 432 m^2.

Problem 30

A rectangular garden has perimeter 100 m, and its width is $\frac{1}{4}$ of its length. Find its area.

Solution

Half perimeter: 50 m.
5 parts; one part: 10 m.
Length 40 m, width 10 m.
Area: 400 m^2.
Answer: 400 m^2.

Challenge – Extension

Problem 31

A rectangular garden has perimeter 56 m. Adding 4 m to its width and taking 4 m from its length makes it a square. Find its area.

Solution

The length is $4 + 4 = 8$ m more than the width; half the perimeter is 28 m.

Length: $(28 + 8) : 2 = 18$ m; width: 10 m.

Area: 180 m^2.

Answer: 180 m^2.

Problem 32

Adding 5 m to the length of a rectangular garden adds 60 m^2; taking 3 m from its width removes 75 m^2. Find its area.

Solution

The added strip is 5 m by the width, so the width is $60 : 5 = 12$ m.

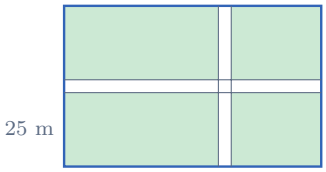
The removed strip is 3 m by the length, so the length is $75 : 3 = 25$ m.

Area: $25 \times 12 = 300$ m^2.

Answer: 300 m^2.

Problem 33

A 40 m by 25 m vegetable garden has two paths 2 m wide crossing at right angles, running the full length and width (see the figure). Find the planted area.



Solution

Pushing the planted pieces together gives a rectangle $40 - 2 = 38$ m long and $25 - 2 = 23$ m wide.

Planted area: $38 \times 23 = 874$ m^2.

Answer: 874 m^2.

Problem 34

The two largest possible non-overlapping circles are cut from a 20 cm by 10 cm card. Find the area of card left over.

Solution

Each circle has diameter 10 cm ($r = 5$ cm), and two fit side by side along the 20 cm length.

Left over: $20 \times 10 - 2 \times 5 \times 5 \times 3.14 = 200 - 157 = 43$ cm^2.

Answer: 43 cm^2.

Problem 35 ⁎⁎

Triangle ABC has a right angle at A, $AB = 12$ cm, $AC = 16$ cm and $BC = 20$ cm. Semicircles are drawn outwards on AB, AC and BC. Show that the semicircle on BC has the same area as the other two together.

Solution

Semicircle on AB ($r = 6$): $6 \times 6 \times 3.14 : 2 = 56.52$ cm^2.

Semicircle on AC ($r = 8$): $8 \times 8 \times 3.14 : 2 = 100.48$ cm^2.

Semicircle on BC ($r = 10$): $10 \times 10 \times 3.14 : 2 = 157$ cm^2 $= 56.52 + 100.48$.

Answer: $157 = 56.52 + 100.48$ (cm^2).

Problem 36 ⁎

A rectangular garden has perimeter 64 m. Adding 5 m to the width and taking 5 m off the length makes it a square. Find its area.

Solution

Then length and width are equal, so the length is $5 + 5 = 10$ m longer.

Half perimeter: 32 m.

Length: $(32 + 10) : 2 = 21$ m; width: 11 m.

Area: 231 m^2.

Answer: 231 m^2.

Problem 37 ⁎

A rectangular garden has perimeter 80 m. Adding 3 m to the width and taking 3 m off the length makes it a square. Find its area.

Solution

Then length and width are equal, so the length is $3 + 3 = 6$ m longer.

Half perimeter: 40 m.

Length: $(40 + 6) : 2 = 23$ m; width: 17 m.

Area: 391 m^2.

Answer: 391 m^2.

Problem 38

For a rectangular garden, adding 6 m to the length adds 84 m^2; taking 4 m off the width removes 72 m^2. Find its area.

Solution

The added strip is 6 m by the width, so width $= 84 : 6 = 14$ m.
The removed strip is 4 m by the length, so length $= 72 : 4 = 18$ m.
Area: $18 \times 14 = 252$ m^2.
Answer: 252 m^2.

Problem 39

For a rectangular garden, adding 4 m to the length adds 40 m^2; taking 2 m off the width removes 48 m^2. Find its area.

Solution

The added strip is 4 m by the width, so width $= 40 : 4 = 10$ m.
The removed strip is 2 m by the length, so length $= 48 : 2 = 24$ m.
Area: $24 \times 10 = 240$ m^2.
Answer: 240 m^2.

Problem 40

A garden is 30 m by 20 m. Two paths 2 m wide cross it, one parallel to each side. Find the area left for planting.

Solution

Long path: 60 m^2; short path: 40 m^2.
The $2 \times 2 = 4$ m^2 crossing is counted twice.
Paths: $60 + 40 - 4 = 96$ m^2.
Left: $600 - 96 = 504$ m^2.
Answer: 504 m^2.

Problem 41

A garden is 45 m by 32 m. Two paths 3 m wide cross it, one parallel to each side. Find the area left for planting.

Solution

Long path: 135 m^2; short path: 96 m^2.

The $3 \times 3 = 9$ m^2 crossing is counted twice.
Paths: $135 + 96 - 9 = 222$ m^2.
Left: $1440 - 222 = 1\,218$ m^2.
Answer: $1\,218$ m^2.

Problem 42 *

A 25 m by 10 m pool has a paved strip 2 m wide all round. Find the area of the strip.

Solution

Outer size: 29 m by 14 m.
Outer area: 406 m^2.
Pool: 250 m^2.
Strip: 156 m^2.
Answer: 156 m^2.

Problem 43 **

Rectangle ABCD has $AB = 12$ cm, $BC = 8$ cm. A semicircle on AB is drawn inside it. Find the area of the rectangle outside the semicircle.

Solution

Radius 6 cm $<$ 8 cm, so the semicircle fits inside.
Semicircle: $6 \times 6 \times 3.14 : 2 = 56.52$ cm^2.
Rectangle: 96 cm^2.
Required: $96 - 56.52 = 39.48$ cm^2.
Answer: 39.48 cm^2.

Problem 44 *

Square ABCD has side 10 cm. Joining the midpoints of its sides gives square MNPQ. Find its area.

Solution

Each corner is a right triangle with legs 5 cm: 12.5 cm^2.
Four of them: 50 cm^2.
$S_{MNPQ} = 100 - 50 = 50$ cm^2, half the original.
Answer: 50 cm^2.

Problem 45

A rectangle is cut into four smaller rectangles by one line parallel to each side. Three have areas 12, 18 and 20 cm^2 (12 and 18 side by side on top, 20 below the 12). Find the fourth.

Solution

The top two share a height, so their widths are in the ratio $12 : 18 = \frac{2}{3}$.

The bottom two share a height and have the same widths.

So the fourth is $20 : \frac{2}{3} = 30$ cm^2.

Answer: 30 cm^2.

Problem 46

A 40 cm by 30 cm picture has a frame 5 cm wide all round. Find the area of the frame.

Solution

Outer size: 50 cm by 40 cm.

Outer area: 2 000 cm^2.
Picture: 1 200 cm^2.
Frame: 800 cm^2.
Answer: 800 cm^2.

Problem 47

A *flower* shape is a square of side 4 cm with four outward semicircles on its sides. Find its area.

Solution

Square: 16 cm^2.

Each semicircle (radius 2 cm): 6.28 cm^2.

Four: 25.12 cm^2.
Flower: 41.12 cm^2.
Answer: 41.12 cm^2.

Problem 48

A plot has a real area of 4 800 m^2. What is its area in cm^2 on a 1 : 2 000 map?

Solution

Lengths shrink 2 000 times, so areas shrink $2\,000 \times 2\,000 = 4\,000\,000$ times.

$4\,800$ $\text{m}^2 = 48\,000\,000$ cm^2.
On the map: 12 cm^2.

Answer: 12 cm^2.

Problem 49 ★

A rectangular garden has perimeter 120 m and its length is twice its width. Vegetables grow on $\frac{3}{4}$ of it at 2 kg per m^2. How much is harvested?

Solution

Half perimeter 60 m; width $60 : 3 = 20$ m; length 40 m.

Area: 800 m^2.

Vegetables: $800 \times \frac{3}{4} = 600$ m^2.

Harvest: 1 200 kg.

Answer: 1 200 kg.

Problem 50 ★★

A right-trapezium field has bases 50 m and 30 m. Extending the shorter base by 20 m turns it into a rectangle and adds 400 m^2. Find the original area.

Solution

The added part is a right triangle with legs 20 m and the height.

Height: $400 \times 2 : 20 = 40$ m.

Original area: $(50 + 30) \times 40 : 2 = 1\,600$ m^2.

Answer: 1 600 m^2.

Mixed Review – Paper 1

Covers: Chapters 1 – 8 (whole numbers, fractions, decimals, standard word problems, proportion, work problems). **Suggested time:** 60 minutes.

Problem 1

Consider 4, 7, 10, ..., 301. **a)** How many terms are there? **b)** Find their sum.

Solution

a) $(301 - 4) : 3 + 1 = 100$ terms.

b) $(4 + 301) \times 100 : 2 = 15\,250$.

Answer: 100; 15 250.

Problem 2

Calculate cleverly: $25 \times 37 \times 4 + 63 \times 100$.

Solution

$= 37 \times 100 + 63 \times 100 = (37 + 63) \times 100 = 10\,000$.

Answer: 10 000.

Problem 3

Compare $\frac{7}{9}$ and $\frac{9}{11}$ without finding a common denominator.

Solution

$1 - \frac{7}{9} = \frac{2}{9}$ and $1 - \frac{9}{11} = \frac{2}{11}$.

Since $\frac{2}{9} > \frac{2}{11}$, $\frac{7}{9} < \frac{9}{11}$.

Answer: $\frac{7}{9} < \frac{9}{11}$.

Problem 4

Calculate: $\frac{1}{2} + \frac{1}{6} + \frac{1}{12} + \frac{1}{20} + \frac{1}{30}$.

Solution

$$= \left(1 - \frac{1}{2}\right) + \left(\frac{1}{2} - \frac{1}{3}\right) + \cdots + \left(\frac{1}{5} - \frac{1}{6}\right) = 1 - \frac{1}{6} = \frac{5}{6}.$$

Answer: $\frac{5}{6}$.

Problem 5

Calculate: $2\frac{3}{4} + 1\frac{5}{6}$.

Solution

$$= \frac{11}{4} + \frac{11}{6} = \frac{33}{12} + \frac{22}{12} = \frac{55}{12} = 4\frac{7}{12}.$$

Answer: $4\frac{7}{12}$.

Problem 6

Write as decimals: **a)** 3 m 5 cm = ... m; **b)** 2 kg 75 g = ... kg; **c)** 4 t 300 kg = ... t.

Solution

a) 3.05 m;

b) 2.075 kg;

c) 4.3 t.

Answer: 3.05; 2.075; 4.3.

Problem 7

Moving the decimal point of A one place to the right gives B. If $A + B = 69.3$, find A.

Solution

$B = 10 \times A$, so $11 \times A = 69.3$ and $A = 69.3 : 11 = 6.3$.

Answer: $A = 6.3$.

Problem 8

Class 5A has 35 pupils, and $\frac{2}{5}$ of them are girls. How many boys are there?

Solution

Girls: $35 \times \frac{2}{5} = 14$.

Boys: $35 - 14 = 21$.

Answer: 21 boys.

Problem 9

$\frac{3}{4}$ of Ha's notebooks is 36. How many notebooks does Ha have?

Solution

$36 : \frac{3}{4} = 48$.

Answer: 48 notebooks.

Problem 10

Two numbers add up to 96, and the smaller is $\frac{3}{5}$ of the larger. Find them.

Solution

8 parts in all.

Smaller: $96 : 8 \times 3 = 36$; larger: 60.

Answer: 36 and 60.

Problem 11

A mother is 28 years older than her child, whose age is $\frac{1}{5}$ of hers. Find both ages.

Solution

The difference is $5 - 1 = 4$ parts.

Child: $28 : 4 = 7$; mother: 35.

Answer: Child 7, mother 35 years.

Problem 12

A class has girls equal to $\frac{3}{4}$ of the boys. If 2 girls join, the girls are $\frac{5}{6}$ of the boys. How many boys and girls were there at first?

Solution

The boys do not change.

The 2 girls are $\frac{5}{6} - \frac{3}{4} = \frac{1}{12}$ of the boys, so there are 24 boys and $24 \times \frac{3}{4} = 18$ girls.

Answer: 24 boys, 18 girls.

Problem 13

8 people build a stretch of dyke in 15 days. How long would 12 people take? (Everyone works at the same rate.)

Solution

$8 \times 15 = 120$ worker-days;

$120 : 12 = 10$ days.

Answer: 10 days.

Problem 14

Pipe I fills a tank in 3 hours, pipe II in 6 hours. Starting empty with both open, how long until it is full?

Solution

Each hour: $\frac{1}{3} + \frac{1}{6} = \frac{1}{2}$ of the tank.

Full after 2 hours.

Answer: 2 hours.

Problem 15 *

Find a whole number that increases by 230 when the digit 5 is written on its right.

Solution

The new number is 10 times the old plus 5, so 9 times the old is $230 - 5 = 225$.

The number is $225 : 9 = 25$.

Answer: 25.

Problem 16 *

Tam alone finishes a job in 12 days. Tam and Hung work together for 4 days; then Tam stops, and Hung needs 6 more days to finish. How long would Hung take alone?

Solution

In 4 days Tam does $\frac{4}{12} = \frac{1}{3}$.

Hung works $4 + 6 = 10$ days for the other $\frac{2}{3}$, i.e.

$\frac{1}{15}$ a day.

Alone: 15 days.

Answer: 15 days.

Mixed Review – Paper 2

Covers: Chapters 9 – 15 (two differences, working backwards, triangles, trapezia, circles, composite area). **Suggested time:** 60 minutes.

Problem 1

Sharing marbles: 3 each leaves 5 over; 5 each is 7 short. Find the number of children and of marbles.

Solution

Children: $(5 + 7) : (5 - 3) = 6$.

Marbles: $3 \times 6 + 5 = 23$.

Answer: 6 children, 23 marbles.

Problem 2

Chickens and dogs: 30 heads and 84 legs. How many of each?

Solution

If all were chickens: 60 legs, 24 too few.

Dogs: $24 : 2 = 12$; chickens: 18.

Answer: 18 chickens, 12 dogs.

Problem 3

Find a number such that subtracting 7, multiplying by 3 and adding 12 gives 60.

Solution

$60 - 12 = 48$;

$48 : 3 = 16$;

$16 + 7 = 23$.

Answer: 23.

Problem 4

Lan gives her brother half her sweets and 2 more, keeping 8. How many did she have?

Solution

$(8 + 2) \times 2 = 20$.

Answer: 20 sweets.

Problem 5

Find the area of a triangle with base 2.4 dm and height 1.5 dm.

Solution

$2.4 \times 1.5 : 2 = 1.8$ dm^2.

Answer: 1.8 dm^2.

Problem 6

A triangle has area 27 cm^2 and base 9 cm. Find its height.

Solution

$27 \times 2 : 9 = 6$ cm.

Answer: 6 cm.

Problem 7

Triangle ABC has area 48 cm^2; M is on BC with $BM = \frac{1}{4}BC$. Find the areas of ABM and AMC.

Solution

$S_{ABM} = 48 : 4 = 12$ cm^2;

$S_{AMC} = 48 - 12 = 36$ cm^2.

Answer: 12 cm^2; 36 cm^2.

Problem 8

Find the area of a trapezium with bases 3.2 m and 2.4 m and height 1.5 m.

Solution

$(3.2 + 2.4) \times 1.5 : 2 = 4.2$ m^2.

Answer: 4.2 m^2.

Problem 9

A trapezium has area 98 cm^2 and height 7 cm; the shorter base is $\frac{3}{4}$ of the longer. Find both bases.

Solution

Sum of bases: $98 \times 2 : 7 = 28$ cm.
Shorter: $28 : 7 \times 3 = 12$ cm; longer: 16 cm.
Answer: 12 cm and 16 cm.

Problem 10

Find the circumference and area of a circle of radius 2.5 cm.

Solution

$C = 2.5 \times 2 \times 3.14 = 15.7$ cm;
$S = 2.5 \times 2.5 \times 3.14 = 19.625$ cm^2.
Answer: 15.7 cm; 19.625 cm^2.

Problem 11

A circle has circumference 31.4 m. Find its area.

Solution

$d = 31.4 : 3.14 = 10$ m, $r = 5$ m;
$S = 5 \times 5 \times 3.14 = 78.5$ m^2.
Answer: 78.5 m^2.

Problem 12

A 12 cm by 8 cm card has a round hole of radius 3 cm. Find the area of card left.

Solution

$12 \times 8 - 3 \times 3 \times 3.14 = 96 - 28.26 = 67.74$ cm^2.
Answer: 67.74 cm^2.

Problem 13

On a 1 : 2 000 map, a square plot has side 3 cm. Find its real area.

Solution

Real side: $3 \times 2\,000 = 6\,000$ cm $= 60$ m.
Area: $60 \times 60 = 3\,600$ m^2.
Answer: $3\,600$ m^2.

Problem 14 *

A rectangle has perimeter 48 cm, and its length is 3 times its width. Find its area.

Solution

Half the perimeter: 24 cm.
Width: $24 : 4 = 6$ cm; length: 18 cm.
Area: 108 cm^2.
Answer: 108 cm^2.

Problem 15 *

A circle sits inside a square, touching all four sides. The square's area is 64 cm^2. Find the circle's area.

Solution

The square's side is 8 cm (since $8 \times 8 = 64$), so $d = 8$ cm and $r = 4$ cm.
$S = 4 \times 4 \times 3.14 = 50.24$ cm^2.
Answer: 50.24 cm^2.

Problem 16 **

Seating pupils: with 4 per bench, 3 have no seat; with 5 per bench, 1 bench is empty and the last occupied bench has only 3 pupils. Find the number of benches and pupils.

Solution

The second way is $5 + 2 = 7$ pupils short of full.
Total difference $3 + 7 = 10$; per bench 1.
Benches: 10; pupils: $4 \times 10 + 3 = 43$.
Answer: 10 benches, 43 pupils.

End-of-Book 1 Test

Time: 60 minutes. **Total:** 10 marks. **Questions:** 19.

Part A – 0.4 *mark each.* *Part B* – 0.5 *mark each.* *Part C* – 1 *mark each.*

Part A – Quick answers (4 marks)

Problem 1

Write $\frac{36}{48}$ in its simplest form.

Solution

Divide top and bottom by 12: $\frac{3}{4}$.

Answer: $\frac{3}{4}$.

Problem 2

Work out mentally: $3.75 \times 10 = \ldots$ and $42.5 : 100 = \ldots$

Solution

Move the decimal point: 37.5 and 0.425.

Answer: 37.5; 0.425.

Problem 3

Write $4\frac{7}{8}$ as a fraction.

Solution

$\frac{4 \times 8 + 7}{8} = \frac{39}{8}$.

Answer: $\frac{39}{8}$.

Problem 4

Which is the largest of 4.503, 4.53, 4.5, 4.35?

Solution

Compare tenths, then hundredths: 4.53.

Answer: 4.53.

Problem 5

5 m^2 8 dm^2 = ... m^2.

Solution

8 dm^2 = 0.08 m^2, so the answer is 5.08 m^2.

Answer: 5.08 m^2.

Problem 6

What is $\frac{2}{3}$ of 45?

Solution

$45 \times \frac{2}{3} = 30$.

Answer: 30.

Problem 7

Find a number given that $\frac{2}{5}$ of it is 16.

Solution

$16 : \frac{2}{5} = 40$.

Answer: 40.

Problem 8

What is the circumference of a circle with diameter 4 cm?

Solution

$4 \times 3.14 = 12.56$ cm.

Answer: 12.56 cm.

Problem 9

What is the area of a triangle with base 6 cm and height 5 cm?

Solution

$6 \times 5 : 2 = 15$ cm^2.

Answer: 15 cm^2.

Problem 10

What is the area of a trapezium with bases 7 cm and 5 cm and height 4 cm?

Solution

$(7 + 5) \times 4 : 2 = 24$ cm^2.

Answer: 24 cm^2.

Part B – Working out (3 marks)

Problem 11

Calculate: $\frac{3}{4} \times \frac{8}{9} : \frac{2}{3}$.

Solution

$= \frac{2}{3} : \frac{2}{3} = 1$.

Answer: 1.

Problem 12

Find x: $x - 2\frac{1}{3} = 1\frac{3}{4}$.

Solution

$x = \frac{7}{3} + \frac{7}{4} = \frac{28}{12} + \frac{21}{12} = \frac{49}{12} = 4\frac{1}{12}$.

Answer: $x = 4\frac{1}{12}$.

Problem 13

Two numbers add up to 120 and are in the ratio $\frac{3}{5}$. Find them.

Solution

$120 : 8 = 15$; the numbers are $15 \times 3 = 45$ and $15 \times 5 = 75$.
Answer: 45 and 75.

Problem 14

6 people finish a job in 10 days. How long would 15 people take? (Same rate.)

Solution

$6 \times 10 = 60$ worker-days;
$60 : 15 = 4$ days.
Answer: 4 days.

Problem 15

Pipe I fills a tank in 6 hours, pipe II in 12 hours. Together, starting empty, how long do they take?

Solution

$\frac{1}{6} + \frac{1}{12} = \frac{1}{4}$ of the tank each hour, so 4 hours.
Answer: 4 hours.

Problem 16

Find the area of a circle of radius 6 cm.

Solution

$6 \times 6 \times 3.14 = 113.04$ cm^2.
Answer: 113.04 cm^2.

Part C – Problem solving (3 marks)

Problem 17 *

An and Binh have 54 marbles in all. If An gives Binh 6, Binh will have twice as many as An. How many does each have now?

Solution

The total is unchanged.
Afterwards An has $54 : 3 = 18$ and Binh 36.

At first: An $18 + 6 = 24$; Binh $36 - 6 = 30$.
Answer: An 24, Binh 30 marbles.

Problem 18 *

A shop sells half a roll of cloth and 3 m more, then $\frac{1}{3}$ of what is left, and 20 m remain. How long was the roll?

Solution

20 m is $\frac{2}{3}$ of what was left after the first sale, so $20 : \frac{2}{3} = 30$ m were left.
The roll was $(30 + 3) \times 2 = 66$ m.
Answer: 66 m.

Problem 19 **

Rectangle ABCD has $AB = 24$ cm and $AD = 10$ cm; M is the midpoint of CD. **a)** Find the area of triangle ABM. **b)** Find the area of trapezium ABMD.

Solution

a) ABM has base AB and height AD: $24 \times 10 : 2 = 120$ cm^2.
b) $DM = 12$ cm;
$S_{ABMD} = (24 + 12) \times 10 : 2 = 180$ cm^2.
Answer: a) 120 cm^2;
b) 180 cm^2.